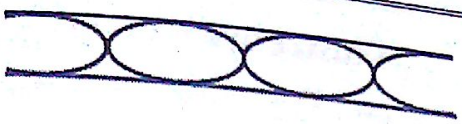
$f_3 = \frac{v}{\lambda_3} \Rightarrow \boxed{f_3 = \frac{3v}{2L}}$



$\frac{3}{2}\lambda_3 = L \Rightarrow \lambda_3 = \frac{2L}{3}$

In general, the n^{th} harmonic in a pipe open at both ends is given by:

$$\boxed{f_n = \frac{nv}{2L}} \quad \text{or} \quad \boxed{f_n = nf_1} \quad n = 1, 2, 3, \ldots$$

☞ **Remember:** Just like a fixed string, all harmonic are possible in an open pipe.

Illustrative Examples

70. What is the fundamental frequency of a 12cm closed pipe at a temperature of 30°C?

Solution: Speed of sound in air is

$$v = 331 + 0.6T = 331 + (0.6)(30) = 349 \text{ m/s}$$

$$f = \frac{v}{4L} = \frac{349}{(4)(0.12)} = 727\text{Hz}$$

71. What length of open pipe will have a frequency of 850 Hz as its first overtone? Assume the speed of sound is 340 m/s.

Solution: First overtone is equivalent to second harmonic

$$f_2 = 850 = \frac{v}{L} \Rightarrow L = \frac{v}{f_2} = \frac{340}{850}$$

$$L = 0.4\text{m}$$

72. A pipe open at both ends produces consecutive harmonic of 300Hz and 400Hz. Calculate the fundamental frequently.

Solution: Let $f_n = 300\text{Hz}$ and $f_n + 1 = 400\text{Hz}$

$\Rightarrow f_n + 1 - f_n = (n + 1)f_1 - nf_1 = 400 - 300$

$\Rightarrow f_1 = 100\text{Hz}$

Beats

Beats are fluctuations in sound intensity that occur when there is interference between two sound waves of slightly different frequency. Thus, the sound goes up and down giving a hearing sensation of loud, soft, loud, soft . . . making a destructive sound pattern called *beats*. The beat frequency is then

$$F_{beat} = |f_2 - f_1| \Rightarrow \frac{1}{f_B} = T_B$$

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Illustrative Examples

73. When two sound tones are generated, three beats per second is heard. If the frequency of one tone is 520 Hz, what is the frequency of the second tone?

Solution: Three beats per second = 3Hz = f_B

$f_B = |f_2 - f_1| \Rightarrow f_2 - f_1 = \pm f_B$, since $|x_1 - x_2| = a \Rightarrow x_1 - x_2 = \pm a$

$f_2 = f_B + f_1 = 3 + 520 = 523$Hz, or

$f_2 = -f_B + f_1 = -3 + 520 = 517$ Hz

74. Two open pipes of length 1.5m and 2.4m are sounded simultaneously. How many beats per second will be produced between the fundamentals. If the speed of sound is 340 m/s.

Solution: $f_1 = \frac{v}{2L} = \frac{340}{2(1.5)}$ and $f_2 = \frac{v}{2L} = \frac{340}{2(2.4)}$

$|f_2 - f_1| = 42.5$ Hz $\Rightarrow$ 42.5 beats per second

10.5 Sound Loudness and Intensity

The noise made by a jack hammer is much louder than the song of a bird. However, loudness is a subjective quantity. On this, we can all agree.

Loudness: Is the audible strength of a sound, which depends on the amplitude or intensity of the sound wave.

☞ In general, the louder the sound, the greater the intensity.

Intensity (I): Is the energy received by a unit area per unit time.

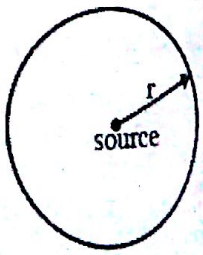
That is, intensity $= \frac{\text{Energy/time}}{\text{Area}}$

$\boxed{I = \frac{P}{A}}$ since, $\frac{\text{Energy}}{\text{time}} = \text{Power (P)}$

Its unit is w/m^2.

☞ If we assume that the sound spreads out equally in all directions, then the area covered is equal to the surface area of a sphere. Hence, the intensity at a distance r from the source is $I = \frac{P}{A}$, $A = 4\pi r^2 \rightarrow$ area of sphere

$\Rightarrow \boxed{I = \frac{P}{4\pi r^2}}$



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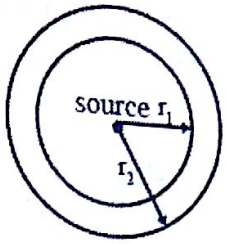
The intensities of sound at a distances of r_1 and r_2 from the source are related as follows.

$$I_1 = \frac{P}{4\pi r_1^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$$

$$I_2 = \frac{P}{4\pi r_2^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$



Taking ratio, $\boxed{\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}}$

Illustrative Example

75. The intensity of a sound source is found to be $2 \times 10^{-4} W/m^2$ at a distance of 6m. What will be the intensity at a distance of 2m from the source?

Solution: $\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} \Rightarrow I_2 = I_1\left(\frac{r_1}{r_2}\right)^2 = 2\times10^{-4}\left(\frac{6}{2}\right)^2$

$I_2 = 18 \times 10^{-4} w/m^2$

Hearing and the decibel

Hearing, like most of our senses is incredibly versatile and sensitive. And, the human ear is found to detect sounds of frequency in the range of 20Hz to 20,000Hz. Now, suppose you hear a sound of intensity I_1. Next, you listen to a second sound of intensity I_2, and the sound seems to be "twice as loud" as the first. If the two sounds are measured, it turns out that I_2 is about 10 times I_1. Our perception of sound, then, is such that increase in loudness correspond to intensities that increase by multiplicative factors. For this reason, loudness is expressed using logarithm of intensity.

Intensity level

The intensity of sound in decibel (dβ) scale is given by, $\boxed{I(d\beta) = 10\log\frac{I}{I_o}}$ where, log, indicates logarithm to the base 10.

$I_o = 1 \times 10^{-12} w/m^2$ and is called *threshold of hearing* which is the faintest sound at about 1000Hz that the ear detects.

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☞ Bel, is named in honor of Alexander Graham Bell, the inventor of the telephone. Since the 'bel' is a large unit, it is more common to measure intensity level using the $\frac{1}{10}$ of a bel. Hence, the name decibel (dβ).

Illustrative Example

76. Compute the intensity level of a sound whose intensity is $1 \times 10^{-4} w/m^2$.

Solution: $I(d\beta) = 10 \log \frac{I}{I_o} = 10 \log \frac{10^{-4}}{10^{-12}} = 80 d\beta$

77. An audio system produces an intensity level of 150dβ and another audio system produces an intensity level of 120dβ. Determine the intensity ratio of the two systems.

Solution: Recall that, $\log x - \log y = \log \left(\frac{x}{y}\right)$

$$\text{Hence, } 10 \log \frac{I_1}{I_o} - 10 \log \frac{I_2}{I_o} = 10 \log \frac{I_1}{I_2}$$

$$150 - 120 = 10 \log \frac{I_1}{I_2} \Rightarrow 30 = 10 \log \frac{I_1}{I_2}$$

$$\Rightarrow \frac{I_1}{I_1} = 10^3 = 1000$$

78. The sound level at a distance of 3m from a point source is 110dβ. At what distance is the sound level 90dβ.

Solution: $10 \log \frac{I_1}{I_0} - 10 \log \frac{I_2}{I_0} = 10 \log \frac{I_1}{I_2}$

$$110 - 90 = 10 \log \frac{I_1}{I_2} \Rightarrow 20 = 10 \log \frac{I_1}{I_2}$$

$$\Rightarrow \frac{I_1}{I_2} = 10^2 = 100$$

and we know,

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$$\frac{I_1}{I_2}=\frac{r_2^2}{r_1^2}\Rightarrow r_2^2=\left(\frac{I_1}{I_2}\right)^2(r_1^2)=(100)(3)^2$$

$$r_2=\sqrt{900}\Rightarrow r_2=30\text{m}$$

79. The sound level at a distance of 5m from a source is 120dβ. What is the sound level at a distance of 50m?

Solution: We know that,

$$\frac{I_1}{I_2}=\left(\frac{r_2}{r_1}\right)^2=\left(\frac{50}{5}\right)^2\Rightarrow I_1=100I_2$$

$$10\log\frac{I_1}{I_o}-10\log\frac{I_2}{I_o}=10\log\frac{I_1}{I_2}=10\log\frac{100I_2}{I_2}=20$$

$$120-\log\frac{I_2}{I_o}=20$$

$\Rightarrow 10\log\frac{I_2}{I_o}=100\text{dβ}$. The intensity level at 50m is 100dβ

The Speed of Sound

The speed of sound depends on the medium the sound is travelling through.

- **The speed of sound in air,** v = 331 + 0.6T, T in °C or $v=331\sqrt{1+\frac{T}{273}}$, T in °C where, T the surrounding temperature

- **The speed of sound in liquid,** $\boxed{v=\sqrt{\frac{\beta}{\rho}}}$

  Where, ρ is density of the liquid in kg/m^3.
  Bulk Modulus, β is the measure of the compressibility of the liquid. Its unit is Pascal (P).

- **The speed of sound in solid** $\boxed{v=\sqrt{\frac{\gamma}{\rho}}}$

  Where, ρ is the density of the solid in kg/m^3
  Young's Modulus, γ is measure of the stiffness of the solid. Its unit is Pascal (Pa).

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Illustrative Examples

80. Calculate the speed of sound in air at 14°C.

Solution: $v = 331 + 0.6T = 331 + 0.6(14) = 340$ m/s

Or, $v = 331\sqrt{1+\frac{T}{273}} = 331\sqrt{1+\frac{14}{273}} = 340\text{m/s}$

81. If the Bulk Modulus of water is 2×10^9 Pa. Calculate the speed of sound in water. Take $\rho = 1000$ kg/m^3.

Solution: $V = \frac{B}{\rho} = \sqrt{\frac{2\times10^9}{1000}} = 1400\text{m/s}$

82. If the Young's modulus of steel is 196×10^9 Pa, and its density is 7900 kg/m^3. Calculate the speed of sound in steel.

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Solution: $v = \sqrt{\frac{}{7900}}$

10.6 The Doppeler Effect

The apparent change in frequency due to the relative motion between a sound source and a listener is called *Doppeler* after the Austrain physicist Christian Doppeler.

Doppeler Effect: Is the apparent change in frequency of a source of sound when there is a relative motion of the source and the listener.

General Cases to Consider

A) The source and the listener move towards each other.

In this case, the frequency, perceived by the listener is *large* than the frequency of the source and is given by

$$f_L = \left(\frac{v + v_L}{v - v_S}\right) f_s \quad \leftarrow \text{increase in frequency}$$

Where: f_L = frequency perceived by the listener.
f_s = source frequency
v = speed of sound in air
v_L = speed of the listener
v_s = speed of the sound source

B) The Source and the listener move away from each other.

The frequency perceived by the listener is *less* than the frequency of the source and is given by

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$$\boxed{f_L = \left(\frac{v - v_L}{v + v_S}\right) f_s} \leftarrow \text{decrease in frequency}$$

C) **The source and the listener are both at rest or move in the same direction with the same speed**

Here, the frequency perceived by the listener is *the same* as the frequency of the source and is given by,

$$\boxed{f_L = f_s} \leftarrow \text{same frequency}$$

☞ **Note that**: In Doppeler's effect, the word "towards" is related to increase and the word "away" corresponds to decrease in frequency.

Illustrative Examples

83. A police car moving at 20m/s with its horn blowing at 1200Hz is chasing a taxi moving at 15m/s. What is the frequency heard by the taxi driver if the speed of sound is 340 m/s.

Solution:

$$f_L = \left(\frac{v - v_L}{v - v_s}\right) f_s \rightarrow \text{listener is moving away (-) and source is towards (-)}$$

$$f_L = \left(\frac{340 - 15}{340 - 20}\right)(1200) = 1219\text{Hz}$$

84. Two cars travel with the same speed away from one another. When one car horns at 205 Hz, a person in the other car hears a frequency of 192Hz. What is the speed of the cars?

Solution: $f_L = \left(\frac{v - v_L}{v + v_s}\right) f_s \rightarrow$ Listener and source move away each other

and $v_L = v_s$

$$192 = \left(\frac{340 - v_s}{340 + v_s}\right) 204$$

$$\Rightarrow v_s = 10.3\text{m/s}$$

85. An Ambulance sires at 600Hz and is moving at 15 m/s towards a boy riding a bicycle at 6 m/s towards the ambulance. What frequency is heard by the boy?

Solution:

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$f_L = \left(\frac{v - v_L}{v - v_s}\right) f_s$ →Listener and source move towards each other

$= \left(\frac{340 + 6}{340 - 15}\right) 600$

$f_L = 639\text{Hz}$

Additional Examples

Simple Harmonic, Motion

86. A particle is executing SHM. Which of the following is maximum when it is at the mean position?
A. Period C. Velocity
B. Amplitude D. Acceleration
Explanation: At the mean (equilibrium) position, velocity is maximum and acceleration is minimum. **Answer: C**

87. The distance moved by a particle in SHM in one period is:
A. A B. 2A C. 4A D. zero
Answer: C

88. The period of a mass – spring system is 5 sec. If the spring is cut into four equal parts and the mass is suspended on one of the parts, the period will be:
A. 5 sec B. 2.5 sec C. 1.2 sec D. 10 sec

Solution: $T_1 = 2\pi\sqrt{m/k_1}$ ..... (*)

$T_2 = 2\pi\sqrt{m/k_2}$ .....(**)

Taking ratio

$$\frac{T_1}{T_2} = \frac{2\pi\sqrt{m/k_1}}{2\pi\sqrt{m/k_2}} = \sqrt{\frac{k_2}{k_1}}, \text{ but } k_2 = 4k_1$$

$$\Rightarrow \frac{T_1}{T_2} = \sqrt{4} \Rightarrow T_2 = \frac{T_1}{2} = \frac{5}{2} = 2.5\,\text{sec}$$

89. In SHM **Answer: B**
A. The velocity and acceleration are in the same phase.
B. The velocity and displacement are in the same phase.

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C. The velocity is ahead in phase of the displacement by π.
D. The acceleration is ahead of the displacement by a phase of π.

Answer: D

90. The velocity and acceleration of a particle executing SHM have a steady phase relationship. The acceleration leads velocity in phase by

A. π B. $\pi/2$ C. $-\pi/2$ D. $-\pi$

Solution: If, $x = A\sin\omega t$, $v = A\omega\cos\omega t$ and

$$a = -A\omega^2\sin\omega t = A\omega^2\cos\left(\pi/2 + \omega t\right)$$

Answer: B

91. A pendulum clock is taken to high altitude.
A. It will keep correct time.
B. Its length should be increased to keep correct time.
C. Its length should be decreased to keep correct time.
D. It can't keep correct time even if the length is changed.

Explanation: $T = 2\pi\sqrt{\ell/g}$ as altitude increases, g decreases so, ℓ should be decreased to keep T constant.

Answer: C

92. The equation of SHM of a particle is $a + 4\pi^2x = 0$ where a is acceleration at displacement x. The frequency of motion is

A. 1 Hz B. 4πHz C. $1/4$ D. 4Hz

Solution: $a = -A\omega^2\sin\omega t = -\omega^2x$ ..... (*)

since $x = A\sin wt$ and from at $4\pi^2x = 0$

$a = -4\pi^2x$ ..... (**)

$-\omega^2x = -4\pi^2x$

$\omega^2 = 4\pi^2$, but $\omega = 2\pi f$

$(2\pi f)^2 = 4\pi^2$

$4\pi^2f^2 = 4\pi^2 \Rightarrow f = 1\text{Hz}$

Answer: A

93. All the other factor being unchanged, when the displacement is halved, the acceleration will be:
A. Halved B. doubled C. tripled D. quadrupled

Explanation: $a = -A\omega^2\sin\omega t = -\omega^2x \Rightarrow a \sim x$

Answer: A

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94. The period of oscillation of a spring – mass system is 12 sec on earth. What will be its period on the moon where $g_e = \frac{1}{6} g_e$?

A. 12 sec B. 2 sec C. 72 sec D. $32\sqrt{3}$ sec

Solution: $T = 2\pi\sqrt{\frac{m}{k}}$, independent of g **Answer: A**

95. If the magnitude of the displacement is equal to that of the acceleration of a SHM, the period is then;

A. 1 sec B. πsec C. 2πsec D. 4πsec

Solution: $a = \omega^2 x$, since $x = a \Rightarrow \omega^2 = 1 = 4\pi^2 / T^2 \Rightarrow T = 2\pi\text{sec}$

Answer: C

96. The SHM of a particle is given by $y = 3 \sin \omega t + 4 \cos \omega t$. What is the amplitude?

A. 1 B. 7 C. 5 D. 12

Solution: $A = \sqrt{3^2 + 4^2} = 5$ **Answer: C**

97. The equation of two SHMs are $x = A \sin(\omega t - \theta)$ and $y = B \cos(\omega t - \theta)$. The phase difference between the two is

A. 180° B. 0° C. 45° D. 90°

Solution: $\cos(\theta - 90) = \sin\theta$ **Answer: D**

98. The displacement of a particle moving in SHM is given by $y = a \sin \omega t$. The acceleration after $t = T/4$ is

A. $a\omega^2$ B. $-a\omega^2$ C. $-a\omega$ D. $a\omega$

Solution: $\vec{a} = -a\omega^2 \sin\omega t = -a\omega^2 \sin\frac{2\pi}{T}\left(\frac{T}{4}\right) = -a\omega^2 \sin\frac{\pi}{2}$

$\vec{a} = -a\omega^2$

Answer: B

99. A particle in SHM has a period of 0.01 sec and an amplitude of 0.2m. What is the velocity at the center of oscillation?

A. 100m/s B. 100πm/s C. 20πm/s D. 40πm/s

Solution: At the center (equilibrium) position, v is maximum

$$v = A\omega = A\frac{2\pi}{T} = (0.2)\frac{(2\pi)}{0.01} = 40\pi\text{m/s}$$

Answer: D

100. The motion of a particle varies with time as $y = a(\sin\omega t + \cos\omega t)$ then, the motion is SHM with amplitude.

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a. $a\sqrt{2}$ B. a^2 C. 2a D. a

Solution: $A = \sqrt{a^2 + a^2} = a\sqrt{2}$

Answer: A

101. If the period of oscillation of mass M on a spring is 1 sec, the period of 4M will be

A. 2 sec B. 0.5sec C. 4 sec D. 0.25sec

Solution: $T_1 = 2\pi\sqrt{\frac{m}{k}}$ ..... (*), $T_2 = 2\pi\sqrt{\frac{4m}{k}}$ ..... (**)

taking ratio: $\frac{T_2}{T_1} = 2 \Rightarrow T_2 = 2T_1 = 2$ sec

Answer: A

102. The displacement of a particle executing SHM is $y = 5 \sin 20\pi t$. Its frequency is

A. 10Hz B. 20πHz C. 0.1Hz D. 20Hz

Solution: $y = A \sin \omega t = 5 \sin 20\pi t \Rightarrow \omega = 20\pi = 2\pi f \Rightarrow f = 10$Hz

Answer: A

103. The motion of a particle executing SHM is given by $x = 0.01 \sin 100\pi (t + 0.05)$. The period is

A. 0.2 sec B. 0.1 sec C. 0.02 sec D. 0.01sec

Solution: $x = A \sin(\omega t + \phi) = 0.01 \sin(100\pi t + 5\pi)$

Where, $\omega = 100\pi = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{100\pi} = 0.02$ sec

Answer: C

104. The pendulum suspended from the ceiling of a train has a period T when the train is at rest. When the train is accelerating the period of oscillation.

A. Increases C. Unchanged

B. Decreases D. Become infinite

Explanation: $T_1 = 2\pi\sqrt{\ell/g}$, train at rest

$T_2 = 2\pi\sqrt{\ell/\sqrt{g^2 + a^2}}$, train accelerating

Answer: B

Thus, $T_1 > T_2$

105. The potential energy of a body executing SHM is maximum at:

A. Equilibrium position

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B. Extreme position
C. Both at equilibrium and extreme positions.
D. Midway between equilibrium and extreme positions.

Explanation: In SHM $P_E \sim x^2$ **Answer: B**

106. If the mass of an oscillator is equal to its force constant, the frequency is:

A. π B. 2π C. $1/\pi$ D. $1/2\pi$

Solution: $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$, when $m = k \Rightarrow f = \frac{1}{2\pi}$ **Answer: D**

107. The motion of a particle is given by $x = A \sin \omega t + B \cos \omega t$. The motion of the particle is

A. Not simple harmonic
B. SHM with amplitude $A + B$
C. SHM with amplitude $\frac{A+B}{2}$
D. SHM with amplitude $\sqrt{A^2 + B^2}$

Explanation: Here A and B are perpendicular. **Answer: D**

108. The period of simple pendulum of infinite length is (Re = radius of Earth)

A. $T = 2\pi\sqrt{\frac{Re}{g}}$ C. $T = 2\pi\sqrt{\frac{Re}{2g}}$

B. $T = 2\pi\sqrt{\frac{2Re}{g}}$ D. $T = \infty$

Solution: $T = 2\pi\sqrt{\frac{Re}{g}} \simeq 84.3$ min **Answer: A**

109. What friction of the total energy is kinetic when the displacement is one – half of the amplitude?

A. $\frac{1}{4}$ B. $\frac{2}{4}$ C. $\frac{3}{4}$ D. $\frac{3.5}{4}$

Solution: at $x = A/2$, total energy $= \frac{1}{2}m\omega^2A^2$

$$\text{Kinetic energy} = \frac{1}{2}m\omega^2(A^2 - x^2)$$

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Taking ratio,

$$\frac{\frac{1}{2}m\omega^2(A^2-x^2)}{\frac{1}{2}m\omega^2A^2}=\frac{A^2-x^2}{A^2}=\frac{A^2-\left(\frac{A}{2}\right)^2}{A^2}=\frac{4A^2-A^2}{4A^2}$$

$$\frac{k_E}{T_E}=\frac{3}{4}, \text{ and } \frac{P_E}{T_E}=\frac{1}{4}$$

Answer: C

110. At what displacement will the kinetic and potential energies of a SHM equal?

A. $x=4$ B. $x=A/2$ C. $x=A/\sqrt{2}$ D. $x=A/\sqrt{3}$

Solution: $\frac{1}{2}kA^2=\frac{1}{2}kx^2+\frac{1}{2}mv^2$, but $P_E=K_E$

$$\frac{1}{2}kA^2=\frac{1}{2}kx^2+\frac{1}{2}kx^2$$

$$2x^2=A^2\Rightarrow x=A/\sqrt{2}$$

Answer: C

111. What is the phase difference between velocity and displacement in SHM?

A. 0 B. $\pi/2$ C. $\pi/4$ D. π

Solution: $x = A\sin\omega t \Rightarrow v = A\omega\cos\omega t = A\omega\sin\left(\omega t+\pi/2\right)$.

Thus, the velocity is $\pi/2$ ahead of the displacement in phase.

Answer: B

112. The displacement of a particle in SHM is $y = 10\sin\left(6t+\frac{\pi}{2}\right)$.

The initial velocity of the particle is

A. 15 m/s B. 10 m/s C. 30 m/s D. 5m/s

Solution: $y=10\sin\left(6+\pi/3\right), v=\frac{dy}{dt}=10\cos\left(6t+\pi/3\right)(6)$

$$v = 60\cos\left(6+\pi/3\right), \text{ at } t=0,$$

$$v = 60\cos\pi/3,\ \pi/3=60^\circ \text{ and } \cos 60^\circ=\frac{1}{2}$$

$$v = 30 \text{ m/s}$$

Answer: C

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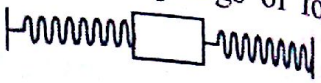
113. A block of mass 0.1 kg is held between two springs of force constant 8 N/m and 2 N/m. If the block is displaced to one side and released, the frequency of vibration is?



A. $\frac{5}{\pi}$Hz B. $\frac{4}{\pi}$Hz C. 5πHz D. 4πHz

Solution: $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{8+2}{0.1}} = \frac{5}{\pi}$Hz

Answer: A

114. Two identical springs each of spring constant k are connected in series and parallel. If a mass M is suspended from them, the ratio of their frequencies is

A. 2: 1 B. 1: 1 C. 1: 2 D. 4: 1

Solution: $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

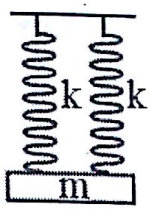
In series, $k_s = \frac{k}{2}, f_s = \frac{1}{2\pi}\sqrt{\frac{k_s}{m}}$

In parallel,

$k_P = 2k,\ f_P = \frac{1}{2\pi}\sqrt{\frac{k_P}{m}}$

Taking ratio,

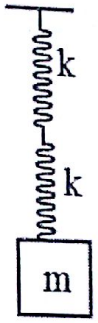
$\frac{f_s}{f_P} = \frac{1}{2} \Rightarrow f_s : f_P = 1:2$



$k_P = k_1 + k_2$

$= k + k$

$k_P = 2k$



$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2}$

$k_s = \frac{k_1 k_2}{k_1 + k_2} = \frac{k^2}{2k} = \frac{k}{2}$

Answer: C

115. A particle executes SHM of amplitude 5 cm and period 3 sec. What is the velocity of the particle at a distance of 4cm from the mean position? (take $\pi = 3$)

A. 8 cm/s B. 12cm/s C. 4cm/s D. 6 cm/s

Solution: $v = \omega\sqrt{A^2 - x^2} = \frac{2\pi}{T}\sqrt{A^2 - x^2} = \frac{(2)(3)}{3}\sqrt{25-16} = 6\text{m/s}$

Answer: D

116. The maximum acceleration in SHM is α and the maximum velocity is β. The amplitude is

A. $\frac{\beta^2}{\alpha}$ B. $\frac{\alpha}{\beta^2}$ C. $\alpha\beta$ D. $\frac{\beta}{\alpha}$

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Solution: $a = A\omega^2 = \alpha$ and $v = A\omega = \beta \Rightarrow \omega = \frac{B}{A}$ and $\omega^2 = \frac{B^2}{A^2}$

$$\Rightarrow \frac{\alpha}{A} = \frac{B^2}{A^2} \Rightarrow A = \frac{\beta^2}{\alpha}$$

Answer: A

117. The maximum speed of a particle executing SHM is 1 m/s and the maximum acceleration is 1.57 m/s^2. The period is?

A. 4 sec B. 0.25sec C. 1.5sec D. 1.57sec

Solution: $v = A\omega = 1$, $a = A\omega^2 = 1.57$

$$\Rightarrow A = \frac{1}{\omega} \Rightarrow A\omega^2 = 1.57 = \left(\frac{1}{\omega}\right)\left(\omega^2\right) = 1.57$$

$$\omega = 1.57 = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{1.57} = \frac{(2)(3.14)}{1.57} = 4\,\text{sec}$$

Answer: A

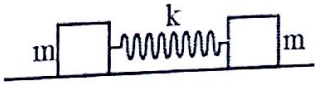
118. One spring has a force constant of 200 N/m. Another has force constant of 500N/m. If they are joined in series, the force constant will be about

A. 700N/m B. 300N/m C. 143N/m D. 100N/m

Solution: $\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} \Rightarrow \frac{1}{k_s} = \frac{1}{200} + \frac{1}{500} \Rightarrow k_s = 143\text{N/m}$

Answer: C

119. If the blocks shown are displaced in opposite directions and released. The period of oscillation is



A. $2\pi\sqrt{m/k}$ C. $2\pi\sqrt{\frac{m}{2k}}$

B. $2\pi\sqrt{m/5k}$ D. $2\pi\sqrt{m/9k}$

Explanation: The mass is reduced by half $\left(\frac{m}{2}\right)$ **Answer: C**

120. A pendulum clock on earth is taken to the moon. It will run

A. At same rate C. $\sqrt{6}$ times faster

B. 6 times faster D. $\sqrt{6}$ times slower

Explanation: $T_e = 2\pi\sqrt{\ell/g_e}$ and $T_m = 2\pi\sqrt{\ell/g_m}$, but $g_m = \frac{1}{6}g_e$

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$$\Rightarrow T_m = 2\pi\sqrt{6\ell/g_e}\text{, taking ratio,}$$

$$\frac{T_m}{T_e} = \frac{2\pi\sqrt{6\ell/g_e}}{2\pi\sqrt{\ell/g_e}} = \sqrt{6} \Rightarrow T_m = \sqrt{6}T_e$$

121. The amplitude and period of a SHM are 5cm and 6 sec respectively. At a distance of 2.5cm from the mean position, the phase will be **Answer: D**

A. $\pi/6$ B. $\pi/4$ C. $5\pi/12$ D. $\pi/3$

Solution: $x = A\sin\omega t \Rightarrow 2.5 = 5\sin\frac{2\pi}{6}t, \frac{2\pi t}{6} = \frac{\pi}{6} \Rightarrow t = \frac{1}{2}\text{sec}$

Phase difference at 6 sec is 2π and that at $\frac{1}{2}$ sec will be $\frac{2\pi}{12} = \frac{\pi}{6}$.

122. A particle in SHM has an amplitude A and maximum velocity v. What will be its velocity at displacement $A/2$. **Answer: A**

A. $v/2$ B. $\frac{\sqrt{3}}{2}v$ C. V D. $v/4$

Solution: $v = \omega\sqrt{A^2 - x^2} = \frac{A\omega}{A}\sqrt{A^2 - x^2} = v\frac{\sqrt{A^2 - x^2}}{A}$

$v = \frac{v\sqrt{A^2 - x^2}}{A}$, at $x = A/2 \Rightarrow v = \frac{v\sqrt{A^2 - \frac{A^2}{4}}}{A} = \frac{vA}{A}\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}v$

123. A particle executes SHM so that its period is 12 sec, what will it take to cover a distance equal to half its amplitude? **Answer: B**

A. 6 sec B. 4 sec C. 2 sec D. 1 sec

Solution: $x = A\sin\omega t$, when $x = A/2 \Rightarrow \frac{A}{2} = A\sin\frac{2\pi}{T}t$

$$\frac{1}{2} = \sin\frac{2\pi}{12}t \Rightarrow t = 1\text{sec}$$

Answer: D

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124. The maximum energy of a body executing SHM of amplitude 3cm but same period, the maximum energy will be

A. E B. $\frac{2}{3}E$ C. $\frac{3}{2}E$ D. $\frac{9}{4}E$

Solution: $E = \frac{1}{2}kA^2 \Rightarrow E \sim A^2$

$$\Rightarrow \frac{E_1}{E_2} = \frac{A_1^2}{A_2^2} \Rightarrow E_2 = \frac{E_1 A_2^2}{A_1^2} = \frac{E(3)^2}{(2)^2} = \frac{9}{4}E$$

Answer: D

125. A simple pendulum of length ℓ_1 has a period of 4 sec and another simple pendulum of length ℓ_2 has a period of 3 sec. What is the period of another pendulum of length $(\ell_1 - \ell_2)$?

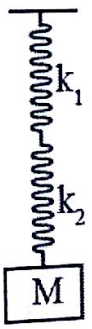
A. $\sqrt{3}$ sec B. 1 sec C. $\frac{\sqrt{3}}{2}$ sec D. $\sqrt{7}$ sec

Solution: $T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow T \sim \sqrt{\ell} \Rightarrow$ here, $\ell = \ell_1 - \ell_2 = T_1^2 - T_2^2$

$$\ell = 16 - 9 = 7 \Rightarrow T = \sqrt{7}\text{ sec}$$

Answer: D

126. Mass M is suspended by two springs of force constants k_1 and k_2 as shown. What is the total stretch of the springs.



A. $\frac{Mg}{k_1 + k_2}$ C. $\frac{Mgk_1}{k_1 + k_2}$

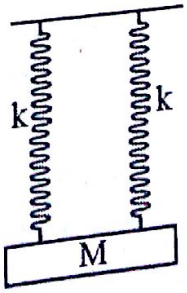
B. $\frac{Mg(k_1 + k_2)}{k_1 k_2}$ D. $\frac{k_1 + k_2}{Mgk_1 k_2}$

Solution: $x = x_1 + x_2 = \frac{Mg}{k_1} + \frac{Mg}{k_2} = Mg\left(\frac{1}{k_1} + \frac{1}{k_2}\right)$

$$x = Mg\left(\frac{k_1 + k_2}{k_1 k_2}\right)$$

Answer: B

127. What is the period of oscillation for the system shown



A. $2\pi\sqrt{\frac{m}{k}}$ C. zero

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B. $2\pi\sqrt{\frac{m}{2k}}$ D. $2\pi\sqrt{\frac{m}{4k}}$

Solution: When springs are in parallel, the effective force constant is, $k = k_1 + k_2 = k + k = 2k$

$$T = 2\pi\sqrt{\frac{m}{2k}}$$

Answer: B

128. The amplitude and period in a SHM are 0.5cm and 0.4 sec respectively. If the initial phase is $\pi/2$, then the equation of SHM will be

A. $y = 0.5 \sin 4\pi t$ C. $y = 0.5 \cos 5\pi t$

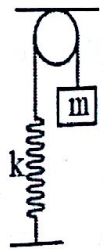
B. $y = 0.5 \sin 5\pi t$ D. $y = 0.5 \sin 2.5\pi t$

Solution: $y = A \sin(\omega t + \phi_o) = 0.5 \sin\left(\frac{2\pi}{0.4}t + \frac{\pi}{2}\right)$

$$y = 0.5 \sin\left(5\pi t + \frac{\pi}{2}\right) = 0.5 \cos 5\pi t$$

Answer: C

129. What is the period of oscillation of the system shown?



A. $2\pi\sqrt{\frac{m}{k}}$ C. $2\pi\sqrt{\frac{g}{k}}$

B. $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$ D. $2\pi\sqrt{\frac{m}{g}}$

Answer: A

130. The equation of SHM is $y = 5 \sin\left(0.314t - \frac{\pi}{3}\right)$. What is its period?

A. 20 sec B. 10 sec C. 0.314 sec D. $\pi/3$ sec

Solution: $y = A \sin(\omega t + \phi) = 5 \sin\left(0.314t - \frac{\pi}{3}\right)$, here $\omega = 0.314$

$$\omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega} = \frac{(2)(3.14)}{0.314} = 20\,\text{sec}$$

Answer: A

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131. The maximum energy of a body executing SHM is 500J and the potential energy of the body is 20J when displacement is 2cm. What is the amplitude of the SHM?

A. 10cm B. 20cm C. 40cm D. 100cm

Solution: $E = \frac{1}{2}mA^2\omega^2 = 500$ and $P_E = \frac{1}{2}mx^2\omega^2 = 20$

$$\frac{E}{P_E} = \frac{A^2}{A^2} \Rightarrow A^2 = 100 \Rightarrow A = 10\text{cm}$$

Answer: A

132. The potential energy of a particle in SHM is 2.6J when its displacement is half of the amplitude. What is the total energy of the particle?

A. 2.5J B. 18J C. 10J D. 12J

Solution: $P_E = \frac{1}{2}m\omega^2x^2, x = \frac{A}{2}, x^2 = \frac{A^2}{4}$

$$2.5 = \frac{1}{2}m\omega^2\frac{A^2}{4} \Rightarrow \frac{1}{2}m\omega^2A^2 = 10\text{J} \Rightarrow E = 10\text{J}$$

Answer: C

133. A particle is in SHM according to $x = A + B \sin \omega t$. What its amplitude?

A. $A + B$ B. A C. B D. $\sqrt{A^2 + B^2}$

Solution: $x = x_o + A \sin \omega t$, at $t = 0$, $x = x_o$ **Answer: C**

134. The mass and diameter of a planet are twice that of earth. What will be the period of oscillation of a pendulum on this planet if its period is 1 sec on earth?

A. $1/\sqrt{2}$ sec B. $\sqrt{2}$ sec C. 2 sec D. $1/2$ sec

Solution: $g_e = \frac{GM_e}{R_e^2}$ and $g_P = \frac{GM_P}{R_P^2} = \frac{G(2M_e)}{(2R_e)^2} = \frac{1}{2}\frac{GM_e}{R_e^2}$ and

$$\frac{T_e}{T_P} = \frac{2\pi\sqrt{\ell/g_e}}{2\pi\sqrt{\ell/g_P}} = \sqrt{\frac{g_P}{g_e}} = \sqrt{\frac{1}{2}} \Rightarrow T_P = \sqrt{2}\text{ sec}$$

Answer: B

135. The oscillation frequency of the system is f. If one spring is removed, the frequency will become

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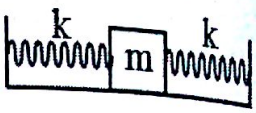
A. f B. 2f C. $f\sqrt{2}$ D. $f/\sqrt{2}$

Solution: Before removed $k_T = k_1 + k_2 = k + k = 2k$ after removed $k_T = k$



$$f_1 = \frac{1}{2\pi}\sqrt{\frac{k_T}{m}} = \frac{1}{2\pi}\sqrt{\frac{2k}{m}} \Rightarrow \frac{f_1}{f_2} = \sqrt{2} \Rightarrow f_2 = \frac{f_1}{\sqrt{2}} = \frac{f}{\sqrt{2}}$$

$$f_2 = \frac{1}{2\pi}\sqrt{\frac{k_T}{m}} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

Answer: D

136. The work done by the string of a simple pendulum during one complete oscillation is equal to

A. Total energy C. Potential energy
B. Kinetic energy D. Zero

Explanation: Work done = change in total energy
But the energy of simple pendulum is constant
$\Rightarrow W = 0$

Answer: D

137. A particle in SHM has acceleration given by $a = -bx$. What is the period of oscillation

A. $2\pi\sqrt{b}$ B. $2\pi/\sqrt{b}$ C. $2\pi b$ D. $2\sqrt{\pi}/b$

Solution: $a = -\omega^2 x = -bx \Rightarrow \omega^2 = b$

$$\left(\frac{2\pi}{T}\right)^2 = b \Rightarrow \frac{4\pi^2}{T^2} = b \Rightarrow T = \frac{2\pi}{\sqrt{b}}$$

Answer: B

138. In SHM, the acceleration has a phase lead over the velocity by

A. π B. $\pi/2$ C. $-\pi/2$ D. $-\pi$

Solution: If $x = A\sin(\omega t + \phi)$, then $v = A\omega\cos(\omega t + \phi)$
or $v = A\omega\sin(\omega t + \pi/2)$ and $a = -A\omega^2\sin(\omega t + \phi)$
or $a = A\omega^2\sin(\omega t + \phi + \pi)$, phase difference
$= \pi - \pi/2 = \pi/2$

Answer: B

139. The period of a SHM is 0.05 sec and the amplitude is 4 cm. The maximum velocity of the body is then

A. 1.6πm/s B. 2πm/s C. 3.1m/s D. 4πm/s

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Solution: $v = A\omega = (0.04)\left(\frac{2\pi}{0.05}\right) = 1.6\pi m/s$ **Answer: A**

140. A mass of 1 kg elongates a vertical spring by 0.8m. If the mass is further pulled down by 0.5m and released, its period of oscillation is about

A. $\frac{2}{7}$sec B. 2πsec C. $\frac{4}{7}\pi$sec D. $\frac{7}{2}$sec

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{m}{\frac{mg}{\ell}}} = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{0.9}{10}} \simeq \frac{4\pi}{7}$$

Answer: C

141. The kinetic energy of a particle in SHM is 16J at equilibrium position. If its mass and amplitude are 5.12 kg and 25cm respectively. What is its period of oscillation?

A. $\frac{\pi}{5}$sec B. 2πsec C. 20πsec D. 5πsec

Solution: $k_E = \frac{1}{2}m\omega^2A^2 \Rightarrow \omega = \frac{1}{A}\sqrt{\frac{2k_E}{m}} = \frac{2\pi}{T} \Rightarrow$

$$T = 2\pi A\sqrt{\frac{m}{2k_E}} = (2\pi)(0.25)\sqrt{\frac{5.12}{(2)(16)}} = \frac{\pi}{5}\text{sec}$$

Answer: A

142. A particle of mass 0.3 kg is subjected to a force $F = -kx$ where $k = 15$ N/m. What will be the initial acceleration if it is released from a point 20cm away from the origin?

A. $15\ m/s^2$ B. $10\ m/s^2$ C. $5\ m/s^2$ D. $3\ m/s^2$

Solution: $F = -kx \Rightarrow ma = -kx$

$$\Rightarrow a = \frac{kx}{m} = \left(\frac{15}{0.3}\right)(0.2) = 10 m/s^2$$

Answer: B

143. The maximum velocity of a particle executing SHM with amplitude 7mm is 4.4 m/s. What is the period of oscillation?

A. 0.01 sec B. 10 sec C. 0.1 sec D. 100sec

Solution: $v = \omega A = \frac{2\pi}{T}A \Rightarrow A = \frac{2\pi A}{v} = \frac{2\left(\frac{27}{7}\right)(7\times10^{-3})}{4.4}$

$T = 0.01$ sec **Answer: A**

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Waves

144. Two waves represented by $y_1 = a \sin \omega t$ and $y_2 = a \sin\left(\omega t + \pi/2\right)$ are superposed. What is the resultant amplitude?

A. a B. 4a C. $\sqrt{2}a$ D. Zero

Solution: $A = \sqrt{a^2 + a^2} = a\sqrt{2}$ **Answer: C**

145. An empty vessel is partially filled with water. The frequency of vibration of the vessel.

A. Decreases C. Depend on purity of water
B. Increases D. Unchanged

Explanation: Frequency inversely proportional to air column length. Hence, when vessel is partially filled with water, the length of air column is reduced so frequency increases $f_n = \frac{nv}{4L}$.

Answer: B

146. In an open pipe, first overtone is produced when the length of pipe is

A. $\ell/4$ B. $\ell/3$ C. $\ell/2$ D. ℓ

Solution: $f_n = \frac{nv}{2L}$, 1st overtone is $f_2 \Rightarrow f_2 = \frac{v}{L} \Rightarrow L = \lambda$

Answer: D

147. A 2kHz siren and a person are both at rest. What frequency does the person hear if the wind is blowing at 12 m/s from source to observer.

A. $2kH_z$ C. $< 24kH_z$
B. $> 2kH_z$ D. $> 24kH_z$

Explanation: There is no relative motion between source and observer. Hence, frequency unchanged. **Answer: A**

148. Which of the following properties makes difference between progressive and stationary waves?

A. Amplitude C. Energy Propagation
B. Frequency D. Wave phase

Explanation: There is no net flow of energy in stationary wave. **Answer: C**

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149. A man sets his watch by a whistle that is 2 km away. How much will his watch be in error if speed of sound is 330 m/s?

A. 3 sec fast C. 6 sec fast
B. 3 sec slow D. 6 sec flow

Solution: $s = vt \Rightarrow t = s/v = \frac{2000}{330} = 6\,\text{sec}$ slow

Answer: D

150. The equation of a wave a string is $y = 5 \sin 2\pi\left(\frac{t}{0.34} - \frac{x}{50}\right)$ cm.

The wavelength of the wave is

A. 15cm B. 10cm C. 25cm D. 50cm

Solution: $y = A\sin\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right)$, here, $\frac{2\pi}{50} = \frac{2\pi}{\lambda}$

$\Rightarrow \lambda = 50\text{cm}$ **Answer: D**

151. The first overtone of a stretched string is 320 Hz. What is the first harmonic?

A. 320Hz B. 640Hz C. 160Hz D. 480Hz

Solution: $f_n = \frac{nv}{2L}$, 1^{st} overtone is f_2

$f_2 = \frac{v}{L} = 2f_1 \Rightarrow f_1 = \frac{f_2}{2} = \frac{320}{2} = 160\text{Hz}$ **Answer: C**

152. It is possible to distinguish between transverse and longitudinal waves by studying the property of

A. Interference C. Reflection
B. Diffraction D. Polarization

Answer: D

153. When sound wave is refracted from air to water, which of the following will remain unchanged?

A. Wave number B. Wave length C. wave speed D. frequency

Answer: D

154. A string of 1m length fixed at both ends has a mass of 0.5g and is under a tension of 20N. It is plucked at 25cm from one end. What frequency is produced in the string

A. 400Hz B. 100Hz C. 200Hz D. 256Hz

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Solution:

$$f_n = \frac{nv}{2L} \Rightarrow f_2 = \frac{2}{2L}\sqrt{\left(\frac{T}{M}\right)}L = \frac{2}{2(1)}\sqrt{\left(\frac{20}{5\times10^{-4}}\right)}(1) = 200Hz$$

Answer: C

155. The equation of a progressive wave is $y = 0.02 \sin 2\pi\left(\frac{t}{0.01} - \frac{x}{0.30}\right)$. The velocity of the wave is

A. 300m/s B. 30m/s C. 400m/s D. 40m/s

Solution: $\frac{2\pi}{\lambda} = \frac{2\pi}{0.3} \Rightarrow \lambda = 0.3m$ and $\frac{2\pi}{T} = \frac{2\pi}{0.01} \Rightarrow T = 0.01sec$

$$v = \lambda f = \frac{\lambda}{T} = \frac{0.3}{0.01} = 30m/s$$

Answer: B

156. The ratio of intensity of sound A to B is 100: 1. Then intensity level of A is greater than that of B

A. 100 times B. 10 times C. 2 times D. $\sqrt{2}$ times

Solution: $L = \log\frac{I_o}{I} = \log 10^2 = 2$

Answer: C

157. A progressive wave on a string has a speed of 100 cm/s and frequency of 100Hz. What is the phase difference between two points 2.75cm apart in the string?

A. 0 B. $11\pi/2$ C. $\pi/4$ D. $3\pi/8$

Solution: $\lambda = \frac{v}{f} = \frac{100}{100} = 1cm$

$$\Delta\phi = \left(\frac{2\pi}{\lambda}\right)(x) = (2\pi)(2.75) = 5.5\pi = \frac{11}{2}\pi$$

Answer: B

158. The threshold of hearing for human ear is 10^{-12} ω/m^2 which is taken as a standard level. Express $1\omega/m^2$ in decibel

A. 10^{12}dB B. 12dB C. 240dB D. 120dB

Solution: $10hg\frac{I}{I_o} = 10\log\frac{1}{10^{-12}} = 10\log 10^{12} = 120dB$ **Answer: D**

159. The equation $y = 0.02 \sin(500\pi t)\cos(4.5x)$ represents

A. Progressive wave of frequency 250Hz.

B. Stationary wave of wavelength 1.4m.

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C. Progressive wave of amplitude 0.02m.
D. Progressive wave of speed about 350 m/s.

Solution: Stationary wave is given by $y = 2A \sin \omega t \, kx$

Answer: B

160. A person whistling at 272Hz runs at 18 km/h towards a reflecting surface. If the speed of sound in air is 345 m/s. The number of beats heard by him is

A. 4 B. 6 C. 8 D. zero

Solution: $V_s = 18\text{km/h} = 5\text{m/s}$.

Source moves towards listener and listener moves towards source

$$f_L = \frac{v + v_s}{v - v_s} f$$

$\text{Beat} = |f_L - f| = 280 - 272 = 8H_z$ **Answer: C**

160. The intensity level of sound is 4 dB. If the intensity is doubled, the intensity level would be.

A. 8 B. 16 C. 7 D. 14

Solution: When intensity doubles, intensity level increase by 3

$\Rightarrow 4 + 3 = 7$ dB **Answer: C**

161. For ordinary conversation the sound intensity is 10^{-6} w/m^2. Express this in decibel taking 10^{-12} w/m^2 as reference

A. 106 B. 60 C. 6 D. log 6

Solution: $10 \log \frac{I}{I_o} = 10 \log \frac{10^{-6}}{10^{-12}} = 10 \log 10^6 = 60$ dB

Answer: B

162. A traveling wave is given by $x = x_o \sin 2\pi \left(ft - \frac{x}{\lambda} \right)$ the maximum velocity will be four times if

A. $\lambda = \frac{\pi x_o}{4}$ B. $\lambda = 2\pi x_o$

C. $\lambda = \frac{\pi x_o}{2}$ D. $\lambda = 4\pi x_o$

Solution: $V = \frac{d}{dt}\left(x_o \sin 2\pi \left(ft - \frac{x}{\lambda} \right) \right) = 2\pi f x_o \cos 2\pi \left(ft - \frac{x}{\lambda} \right)$

$V = \lambda f$ and, $V_{max} = 2\pi f x_o = 4\lambda f \Rightarrow \lambda = \frac{\pi x_o}{2}$

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163. If the speed of a wave on a stretched string of length 1m is 60 m/s. What is the fundamental frequency of vibration

A. 10 Ht B. 30 Ht
C. 40 Hz D. 70 Hz

Answer: C

Solution: $f_1 = \frac{v}{2L} = \frac{60}{2} = 30H_z$

Answer: B

164. The intensity level of two sounds are 100dB and sod B. What is the ration of their intensities?

A. 10^1 B. 10^3 C. 10^5 D. 10^{10}

Solution: $L = 10 \log \frac{I}{Io}$

$L_1 = 10 \text{ Log } \frac{I_1}{Io}$ and $L_2 = 10 \log \frac{I_2}{I_o}$

$\frac{I_1}{Io} = 10^{10}$ $\frac{I_2}{Io} = 10^5$

Taking ration $\frac{\frac{I_1}{Io}}{\frac{I_2}{Io}} = \frac{10^{10}}{10^5} = 10^5$

Answer: C

165. The equations of two progressive wave are given by $x_1 = a \sin (\omega t - kx)$ and $x_2 = a \cos (\omega t - kx)$. What is the phase difference between the two waves

A. $\frac{\pi}{4}$ B. π C. $\frac{\pi}{2}$ D. $\frac{\pi}{6}$

Solution: $x^2 = a \sin (\omega t - kx + \pi/2) \Rightarrow \Delta\emptyset = \pi/2$

Answer: C

166. What is the change in intensity level when the intensity of sound increases 100 times its original intensity

A. 5 dB B. 10 dB C. 20 dB D. 30 dB

Solution: When $I = 100$ Io, then, $L = 10 \log \frac{100 \text{ Io}}{Io} = 20$ dB

Answer: C

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167. A source x of unknown frequency produces 8 beats / sec with a source of 250 Hz and 12 beats / sec with a source of 270 Hz. The frequency of source x is

A. 242 Hz B. 258 Hz
C. 282 Hz D. 262 Hz

Solution: $f_x - 250 = \pm 8 \Rightarrow f_x = 258$ or 242
$f_x - 270 = \pm 12 \Rightarrow f_x = 258$ or 282

Answer: B

168. A traveling wave is given by $y = 2\cos[\pi(0.5x - 200t)]$ cm which of the following is true?

A. $\lambda = 2$ cm, V = 400 cm/s B. $\lambda = 2$cm, A = 2cm
C. $\lambda = 4$ cm, f = 100 Hz D. A = 2cm, f = 200Hz

Solution: $k = 0.5\pi = \frac{2\pi}{\lambda} \Rightarrow \lambda = 4$cm, A = 2cm

$$f = \frac{\omega}{2\pi} = \frac{200\pi}{2\pi} = 100\text{Hz}$$

Answer: C

169. The equation of a travelling wave is $y = 20\sin\pi(0.02x - 2t)$ cm. The wavelength is

A. 50 cm B. 100 cm
C. 200 cm D. 5 cm

Solution: $\frac{2\pi}{\lambda} = 0.02\pi \Rightarrow \lambda = \frac{2\pi}{0.02\pi} = 100$cm. **Answer: B**

170. The equation of a progressive wave is $y = 10\sin\pi(0.01x - 2t)$ cm. Its frequency is

A. 10 Hz B. 2 Hz C. 1 Hz D. 0.01 Hz

Solution: $2\pi f = 2\pi \Rightarrow f = 1$ Hz **Answer: C**

171. A travelling wave is given by $y = A\sin(kx - \omega t)$. What is the maximum particle velocity

A. $A\omega$ B. ω/k C. ω/λ D. x/t

Solution: $V = \frac{dy}{dt} = A\omega\cos(kx - \omega t), V_{max} = A\omega$

Answer: A

172. The displacement of a particle is, $y = \frac{-6}{10}\sin(100t + 20x + \pi/4)$ m. The wave speed is

A. 200 m/s B. 20 m/s
C. 5π m/s D. 5 m/s

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Hint: $V = \lambda f = \frac{\omega}{k} = \frac{100}{50} = 5$

Answer: D

173. The displacement of a traveling wave is given by the equation $y = \frac{10}{\pi}\sin\left(2000\,\pi t - \frac{\pi x}{17}\right)$. The period and maximum velocity of the particle in the medium is

A. 10^{-3} sec, 330 m/s B. 10^{-3} sec, 200 m/s
C. 10^{-4} sec, 20 m/s D. 10^{-2} sec, 2000 m/s

Solution: comparing with $y = A \sin(\omega t - kx)$

$$\omega = 2000\pi,\ A = \frac{10}{\pi}$$

$$\omega = \frac{2\pi}{t} \Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi}{2000\pi} = 10^{-3}\text{sec}$$

$$V_{max} = A\omega = \left(\frac{10}{\pi}\right)(2000\pi) = 20000\ \text{cm/s} = 200\text{m/s}$$

Answer: B

174. The equation of a stationary wave is $y = 4 \sin\left(\frac{\pi x}{15}\right)\cos(96\,\pi t)$ the distance between a node and the next antinode is

A. 7.5 units B. 1.5 units
C. 22.5 units D. 30 units

Solution: comparing $y = 2yo \sin(kx)\cos(\omega t)$

$$y = 2\ yo \sin\left(\frac{2\pi}{\lambda}x\right)\cos\frac{(2\pi t)}{T}$$

$$\Rightarrow \frac{2\pi}{\lambda} = \frac{\pi}{15} \Rightarrow \lambda = 30\ \text{units}$$

The distance from a node to the next antinode is $\frac{\lambda}{4}$

$$\Rightarrow \frac{\lambda}{4} = \frac{30}{4} = 7.5\ \text{units}$$

Answer: A

175. Two forks A and B produce 3 beats/ sec. Forks B and C produce 5 beats / sec. What beats /sec will A and C produce

A. 8 B. 21 C. 6 D. 4

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Solution: $F_A - F_B = \pm 3$, $F_B - F_C = \pm 5$ adding, $f_A - f_C = \pm 8$

Answer: A

176. The frequency of the fundamental note of vibration of an open pipe is 400 Hz. When one end of the pipe is closed, the fundamental frequency is

A. 200 Hz B. 400 Hz
C. 600 Hz D. 800 Hz

Solution: $f_1 = \frac{v}{2L}$ open pipe $\Rightarrow L = \frac{V}{2f_1} = \frac{V}{2(400)} = \frac{V}{800}$

Closed pipe, $f_1 = \frac{v}{4L} = \frac{v}{4\left(\frac{v}{800}\right)} = \frac{800v}{4v} = 200\text{Hz}$

Answer: A

177. The equation of a stationary wave is given by

A. $y = 2yo \sin kx \cos \omega t$ B. $y = yo \sin k(x - ft)$
C. $y = yo \cos 2\pi\left(\frac{kx}{\lambda} - \frac{1}{T}\right)$ D. $y = yo \sin \frac{2fT}{\lambda}$

Answer: A

178. The equation of a stationary wave is $y = A \sin 100t \cos 0.01x$ The velocity of the wave is

A. 10^4 m/s B. 10^{-4} m/s
C. 1 m/s D. 0.1 m/s

Solution: $w = 2\pi f = 100 \Rightarrow f = \frac{so}{\pi}$, and

$\frac{2\pi}{\lambda} = 0.01 \Rightarrow \lambda = \frac{2\pi}{0.01}$ and $v = \lambda f = \left(\frac{50}{\pi}\right)\left(\frac{2\pi}{0.01}\right)$

$V = 104$ m/s

Answer: A

179. Standing waves are produced in a 10m long string. If the string vibrates in 5 segments with a speed of 20 m/s, what is the frequency produced

A. 2 Hz B. 4 Hz C. 5 Hz D. 10 Hz

Solution: $f_n = \frac{nv}{2L}$, $n = 5 \Rightarrow f_s = \frac{5v}{2L} = \frac{(5)(20)}{(2)(10)} = 5\text{Hz}$

Answer: C

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180. A steel wire 0.5m long has a mass of 0.01kg and is stretched under a tension of 800 N. The frequency of vibration of the string in its fundamental mode is

A. 2 Hz B. 4 Hz C. 100 Hz D. 200 Hz

Solution $f_1 = \frac{v}{2L} = \frac{1}{2L}\sqrt{\frac{T}{\mu}} = \frac{1}{2L}\sqrt{\frac{T}{m/L}} = \frac{1}{(2)(0.5)}\sqrt{\frac{(800)(0.5)}{0.01}}$

$f_1 = 200$ Hz **Answer: D**

Unit 10: Conceptual Question

1. If the frequency of a mass spring system is 5o Hz, how many times in 1 second will the mass pass through its rest position
2. Time period of simple pendulum at moon is less than that on earth. True or False? Explain
3. Time period of a simple pendulum in air is more than that in vaccum. True or False? Explain
4. If a spring of force constant K is cut into three equal parts. Then the force constant of each part is $\frac{k}{3}$. True or False
5. The time taken by a particle executing SHM of period T to move from the rest position to half the maximum displacent is $\pi/4$. True or False? Explain.
6. With increase of pressure, the speed of sound in air remains unchanged. True or False? Explain
7. In a stationary wave, there is no transfer of energy. True or False? Explain
8. There is no Doppler effect when source and listener are moving along the same direction with the same velocity. True or False? Explain

Unit 10: Problems

Simple Harmonic Motion

9. An object moving with SHM has an amplitude of 3cm and a frequency of 30 Hz. Calculate the
   a. period of oscillation
   b. acceleration at the center and at the maximum position
   c. velocity at the center and the maximum positions

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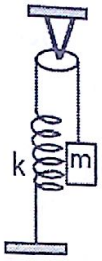
10. A body executing SHM has a maximum acceleration of 32π m/s^2 and a maximum speed of 6.4 m/s. Calculate
    a. The period
    b. The amplitude
11. Find the length of a simple pendulum which has a period of 1.2 π sec use g = 10 m/s^2
12. A simple pendulum is made from a bob of mass 0.04 kg and string length of 1.4m. If the pendulum is pulled to a height of 0.1m, calculate
    a. the total energy
    b. the amplitude
    c. the period of oscillation
13. If the 2 kg mass is released when the spring is upstretched, find the amplitude of the resulting oscillation
    (take, k = 300 N/m, m = 2kg)



Wave

14. A violin string is 32 cm long and has a mass per unit length of 2g/m. What tension is required to produce a fundamental frequency of 440 Hz?
15. The wave function of a travelling wave is given by $y = 8\sin\left(4\pi t - \frac{x}{3}\right)$. Find
    a. The wavelength
    b. The period of motion
16. An open organ pipe has adjacent harmonics of 440 Hz and 522 Hz. Calculate the first harmonic.
17. The intensity level of a sound is 90 dB. What is the intensity of this sound.
18. The intensity of sound is 2×10^{-6} w/m^2 at a distance of 12from an explosion. Find the distance from the source at which the intensity is 0.8×10^{-6} w/m^2
19. When a person wear a hearing aid, the sound intensity level increases by 20 dB. By what factor does the sound intensity increase?

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20. Two cars drive along the same road towards each other. One at 15 m/s and the other at 12m/s. Each car horns at 250 Hz. Calculate the frequency that the drivers of each car hears.

Unit 10: Explanation to Questions

1. In one cycle, it pass the equilibrium position two times. Hence, in 50 cycles, it pass the equilibrium position 100 times
2. False. Because, $T = 2\pi\sqrt{\ell/g}$, on the moon g is small.
3. True. Because, air damps vibration of simple pendulum
4. False. $K^1 = 3k$
5. False. $t = T/12$
6. True. As pressure increases, density increases proportionally.
7. True. Opposite waves superimpose
8. True. There is no relative motion between source and listener

Unit 10: Solution to Problems

9. a. $T = \frac{1}{f} = \frac{1}{30}\text{sec}$

   b. $\bar{a} = 0,\ \bar{a} = A\omega^2 = \left(\frac{3}{100}\right)(2\pi f)^2 = 54\ \pi\text{m/s}^2$

   c. $\bar{V} = A\omega = (0.03)(2\pi f) = 1.8\pi\ \text{m/s}, \bar{V} = 0\text{m/s}$

10. a. $\frac{a}{v} = \frac{A\omega^2}{A\omega} = \omega = \frac{2\pi}{T} = \frac{32\pi}{6.4} \Rightarrow T = 0.4\text{sec}$

    b. $V = A\omega = (A)\left(\frac{2\pi}{T}\right) \Rightarrow A = \frac{(0.4)(6.4)}{2\pi} = 0.04\text{m}$

11. $T = 2\pi\sqrt{\ell/g} \Rightarrow \ell = g\left(\frac{T^2}{4\pi^2}\right) = (10)\left(\frac{1.44\ \pi^2}{4\pi^2}\right) = 3.6\text{m}$

12. a. $E = mgh = (0.04)\ (10)\ (0.1) = 0.04\ \text{J}$

    b. $A = h = 0.1\text{m}$

    c. $T = 2\pi\sqrt{\ell/g} = 2\pi\sqrt{\frac{1.4}{10}} = 2.35\ \text{sec}$

13. $mgh = \frac{1}{2}KA^2,\ h = A \Rightarrow A = \frac{2mg}{k} = \frac{(2)(2)(10)}{300} = 0.13\text{m}$

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14. $f = \frac{1}{2L}\sqrt{T/\mu} \Rightarrow 440 = \frac{1}{(2)(0.32)}\sqrt{T/0.002} \Rightarrow 159N$

15. a. $\frac{1}{3} = \frac{2\pi}{\lambda} \Rightarrow \lambda = 6\pi m$

b. $4\pi = \frac{2\pi}{T} \Rightarrow T = 0.5sec$

16. $f_1 = 522 - 440 = 82\ Hz$

17. $90 = 10\log\frac{I}{Io} \Rightarrow I = 9\times10^{-19} w/m^2$

18. $\frac{I_1}{I_2} = \left(\frac{r_2}{r_1}\right)^2 = 19m$

19. $20 = 10\log\frac{I_1}{I_2} \Rightarrow \frac{I_1}{I_2} = 10^2 = 100 \Rightarrow I_1 = 100I_2$

20. $f_L = \left(\frac{V \pm V_L}{V \mp V_S}\right) f_s = \frac{(340)(256)}{340-3} = 258\ Hz$

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Unit 11

Wave Optics

From reading this book and viewing your image in a mirror to your use of a phone or computer and discovering the astronomical nature of outer space, all depends on the various ways light interact with the matter around you. What is more, by understanding diffraction of light, you can identify the resolving power of microscopes and telescopes. In fact, light has terminations applications in our every day life.

- Light is a wave and the study of light is called *optics*. Moreover, the study of light as a wave is called *wave optics*.

11.1 Wave Fronts and Hygen's Principle

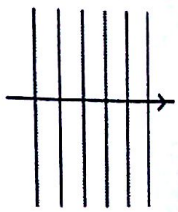
Wave Front: Is an imaginary line joining points of a travelling wave that are in phase.

Wave fronts can be spherical or plane.

Spherical Wave fronts: Form a set of concentric circles centered at the source.

Spherical wave front

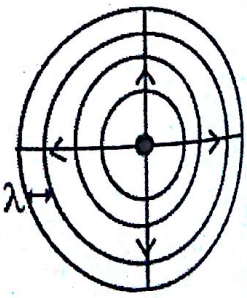
Plane wave fronts: Form parallel, flat and infinite extent lines.



plane wave front

Wave Fronts and Ray Diagrams

Wave front is a surface that describes how the energy carried by a wave travels through space. And a ray is a straight line indicating the direction of wave motion and is perpendicular to the wave fronts. Now, consider a simplified representation of wave fronts and ray diagrams in a water wave.



- Each circle represents a wave front
- The rays indicate the direction of wave motion,
- The distance between consecutive wave front is equal to the wavelength (λ) of the wave.

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- Rays are perpendicular to the wave fronts at any point.

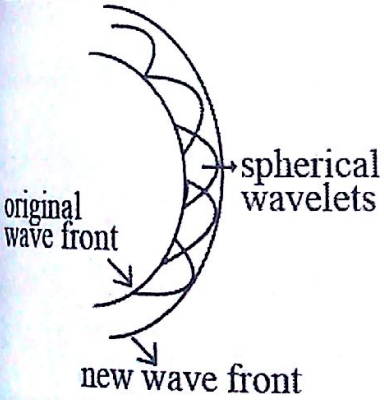
Huygens's Principle

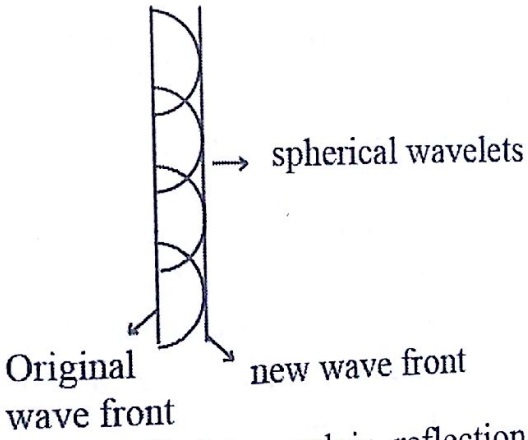
Newton believed that light consists of very small and fast moving particles called *corpuscles*. This particle theory of light remained the accepted theory for more than 100 years. And, the first person to advance a convincing wave theory for light was the Dutch physicists Christian Hygens, in 1678.

Huygen's principle state that:

- Every point on a wave front acts as a source of spherical secondary wavelets.
- The secondary wavelets spread in all directions with the same frequency, speed and wavelength as the original wave.
- A new wave front is formed as these wavelets combine.

Hygens's wave theory is based on a geometrical construction and if the original wave front is a plane wave then the wavelets combine to form another plane wave, and if the original wave front is spherical, then the new wave front will also be spherical.





Remember: Huygen's principle can be applied to explain reflection, refraction as well as diffraction of light wave.

11.2 Reflection and Refraction of Light Wave

Reflection: Is the bounding off a light wave from an obstacle such as mirror, water, …etc

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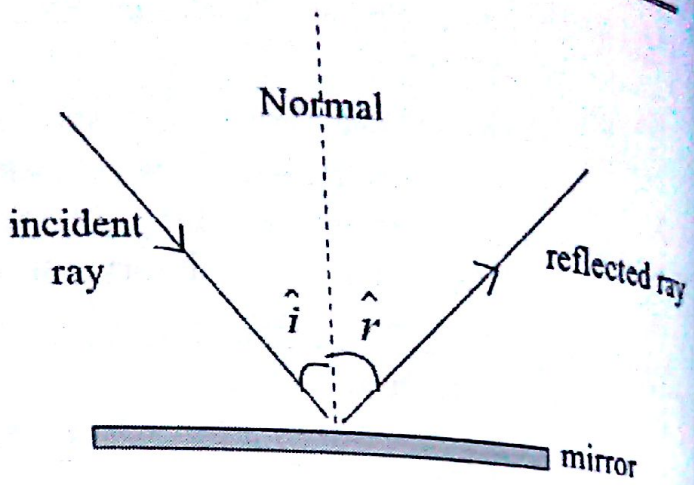
The following diagram demonstrates the reflection of light ray from a mirror.



- ✧ Incident ray, is a ray coming from the source and hits the surface.
- ✧ Normal line, is an imaginary line that is perpendicular to the point of incidence.

Law of Reflection; state that: "The angle of incidence is equal to the angle of reflection."

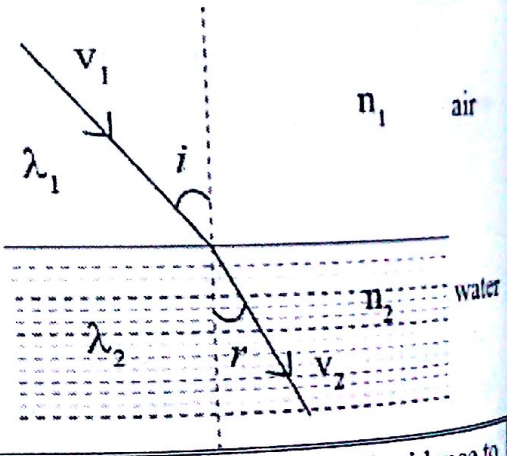
$$\hat{i} = \hat{r}$$

Remember, the angle of incidence and the angle of reflection are measured from the normal.

Refraction: Is the bending, that is a change in direction of a wave as it moves from one medium to the other.

The following diagram demonstrates the refraction of light as it moves from air into water.

- ✧ When light moves from optically rare medium such as air in to optically denser medium such as water, the speed decreases.



Snell's law states that: "The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant."

$$\frac{\sin i}{\sin r} = \text{constant}$$

In its general form, snell's law is written as,

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

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Where: v, stands for speed
λ, stands for wavelength, and
n, stands for index of refraction of each medium

$$\text{index of refraction} = \frac{\text{speed of light in vaccum}}{\text{speed of light in a medium}}$$

$$\Rightarrow \boxed{n = \frac{C}{v}},$$

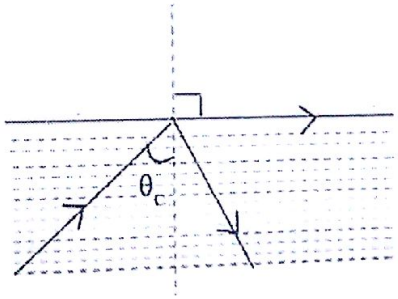
the index of refraction is always greater than or equal to one.

Remember: As light travels from one medium to another, its speed and wavelength change but its frequency does not change.

Total Internal Reflection

When light wave moves from a denser medium to a less dense medium and the angle of incidence is greater than the critical angle, then total internal reflection is said to occur.

Critical angle (θ_C): Is an angle of incidence for which the angle of refraction is 90°.



If $i > \theta_C$, total internal reflection takes place.

Conceptual Examples

1. What will happen to a light wave as it moves from a rare medium to a denser medium? Explain

   Explanation: The speed and wavelength of the wave decrease while, the frequency of the wave remains constant.

2. Does the angle of reflection depend on the wavelength of the light wave?

   Explanation: No, there is no such case so far.

Illustrative Examples

3. What is the speed of light in water (n = 1.33), assuming its speed in air is 3×10^8m/s?

   Solution: From snell's law, $\frac{v_1}{v_2} = \frac{n_2}{n_1} \Rightarrow \frac{3\times10^8}{v_2} = \frac{1.33}{1}$

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$$v_2 = \frac{3\times10^8}{1.33} = 2.25\times10^8 \text{ m/s}$$

→The index of refraction in air is one.

4. The index of refraction of alcohol is 1.4. Find the angle of refraction if light comes from air with an angle of incidence of 57°.

Solution: $\frac{\sin i}{\sin r} = \frac{n_2}{n_1} \Rightarrow \sin \hat{r} = \frac{\sin i}{n_2} = \frac{\sin 57°}{1.4} = 0.6$

$$\sin r = 0.6 \Rightarrow \sin^{-1}(0.6) = 37°$$

5. The speed of light in a certain medium $\frac{2}{5}C$ Find the index of refraction of the medium.

Solution: $\frac{v_1}{v_2} = \frac{n_2}{n_1} \Rightarrow n_2 = \frac{v_1}{\frac{2}{5}C} = \frac{5}{2}, v_1 = C = 2.5$

6. Light of wavelength 436nm enters a bowl filled with water (n = 1.33). What is the light wavelength in water.

Solution: $\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} \Rightarrow \lambda_2 = \frac{\lambda_1}{n_2} = \frac{436}{1.33} = 328\text{nm}$

7. A ray of light passes from Diamond to water ($n_{diamond} = 2.42$, $n_{water} = 1.33$) What is the critical angle.

Solution: $\frac{\sin\theta_C}{\sin 90°} = \frac{n_\omega}{n_D} = \frac{1.33}{2.42} \Rightarrow \sin\theta_C = \frac{1.33}{2.42} = 0.55$

$$\Rightarrow \theta_C = \sin^{-1}(0.55) = 33°$$

8. A wave has a speed of 50m/s and a wavelength of 25,000m. What is the frequncy of the wave: (UEE, 2005)

A) 2×10^{-3}Hz C) 5×10^{2}Hz

B) 6×10^{-1}Hz D) 1.8×10^{6}Hz

Solution: $v = \lambda f \Rightarrow f = \frac{v}{\lambda} = \frac{50}{25000} = 2\times10^{-3}\text{Hz}$

Answer: A

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9. A wave front is (UEE, 2003)
   A) a line that represents the direction of propagation of light
   B) the locus of all points of a travelling wave that are in phase
   C) the minimum distance between two points that are in phase
   D) the number of waves crossing a given plane in a unit time.

Answer: B

10. A wave traveling from a medium of index of refraction n_1 towards a medium of index of refraction n_2 undergoes partly transmission and partly refelction. If n_2 is greater than n_1, which one of the following statements is not correct?
   A) The speed of the wave in the medium with refractive index n_1 is smaller than with n_2.
   B) The incident and the reflected wave are in phase.
   C) The angle of reflection is the same as the angle of incidence
   D) The refraction angle is less than the incidence angle

Explanation: From snell's law, $\frac{v_1}{v_2} = \frac{n_2}{n_1}$, speed is inversely proportional to the refractive index.

Answer: A

11.3 Interference and Diffraction

The heart of this very unit is the concept of interference and diffraction.

Interference: Is the mixing up, combination or overlap of two or more waves of the same kind.

As usual, interference may be constructive or destructive.

- Constructive interference of light results in an increase in intensity or brighter regions called *maxima* and destructive interference gives rise to a drop in intensity, or dark regions called *minima*.

Conditions for Interference

For a sustained, that is stable interference pattern to occur, coherent sources of light must be used. In order for two waves to be coherent, they must

- be the same type of wave.

  For example, It is not possible to produce stable interference between light wave and sound wave.
- have the same frequency.

  Otherwise 'beats' will occur. Which is unnecessary case.
- maintain a constant phase relationship.

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The above statement doesn't mean the waves must be in phase. But, it means, their phase change must be constant, o, π or 2π. This ensures that at any instant, the interference is always the same, constructive or destructive.

- A satisfactory interference to occur, the light must be
  Monochromatic: that is, it should have a single color and hence a single frequency.
  Coherent; that is, there must be constant phase relationship between the mixing waves.

Laser emit light that is both monochromatic and coherent but, in candescent and fluorescent light sources are incoherent and hence doesn't produce stable interference.

Diffraction: Is the spreading out of a wave as it passes around obstacles or through a narrow gap.

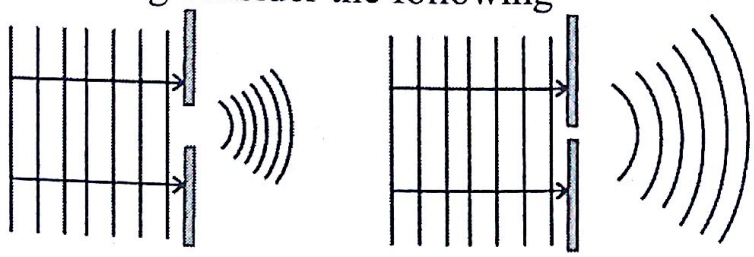
- It appears that light does not diffract! This is common sense.

A common example is that, while you are in the class room you can hear the conversation made by peoples in the outside without seeing them. That means, the sound wave diffracted and reached you but not the light wave since you do not see the people who are talking. But, the fact is, light is a wave and hence it can diffract.

Condition for the diffraction of light

- To notice the diffraction of light, the gap through which it passes should be as small as the wavelength of light. In simple terms, the closer the size of the gap to the wavelength, the more pronounced the diffraction. Got it? Good.

For better understanding consider the following



wide gap, hence less diffraction

narrow gap, hence more diffraction

Conceptual Examples

11. Explain what a laser is?

Explanation: Laser is a device that produces a beam of coherent and monochromatic light.

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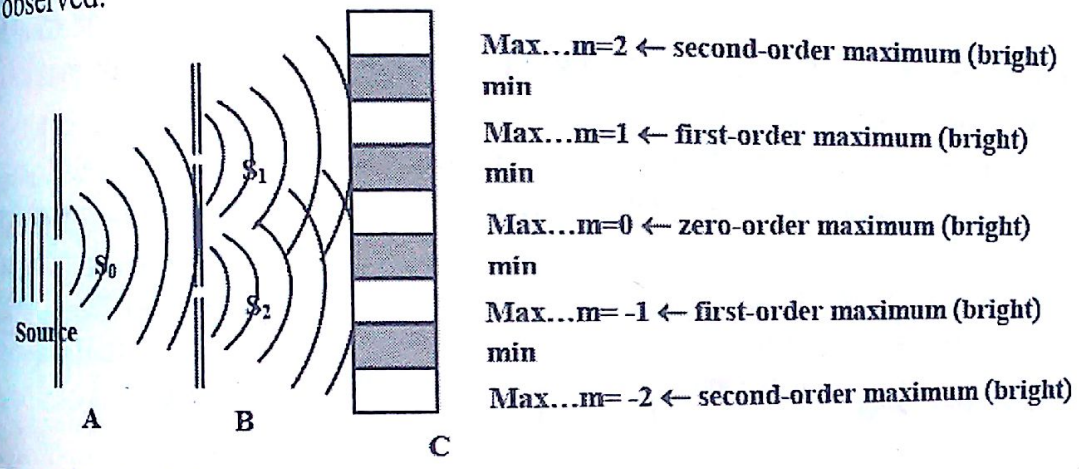
12. Explain the difrraction of light wave interms of wavelength and width of the gap.

Explanation: The longer the wavelength, the greater the diffraction. And, the narrower the gap, the more pronounced the diffraction.

11.4 Young's Double Slit Experiment

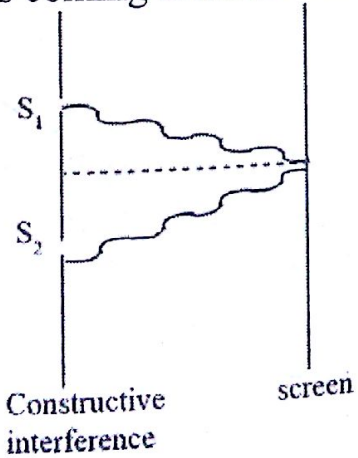
In 1801, the English physician and physicist, Thomas Young experimentaly proved that light undergoes intereference. The experiment, in simplest from, consists of a beam of monochromatic light that passes through a single slit and then illuminated on two slits. After the light diffracts through the two slits it interers and shines on a distant screen where an intererence pattern of bright and dark "fringes" is observed.



Young's double slit experiment

Looking carefully at the interference pattern, the alternating maxima (bright) and minima (dark) spots are formed due to a difference in path and phase between the interfering waves.

- A Constructive interference or bright band is formed on the screen when the light waves coming from the two slits are in phase.



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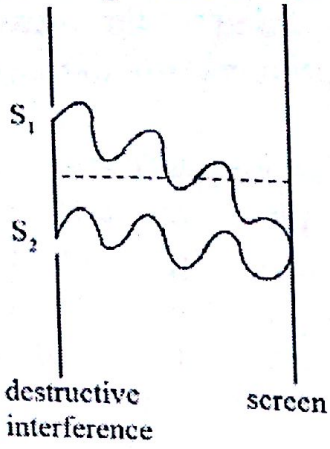
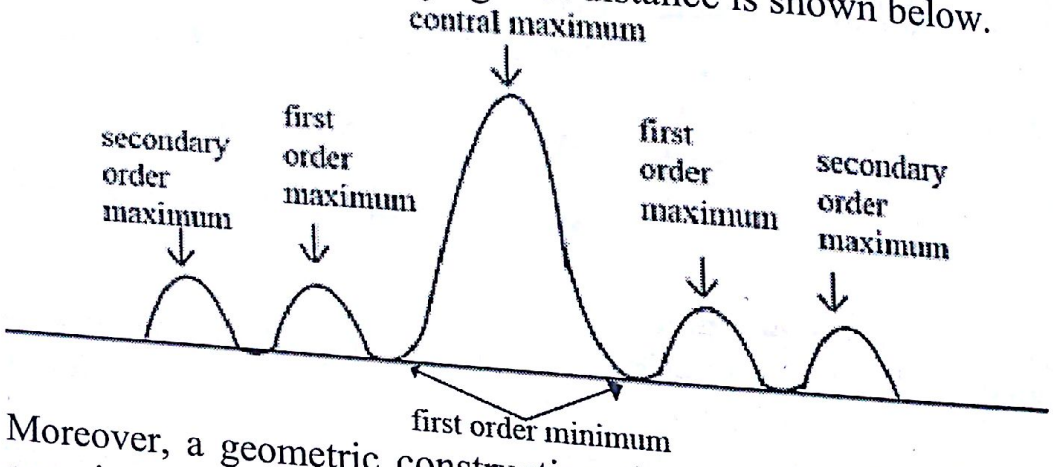
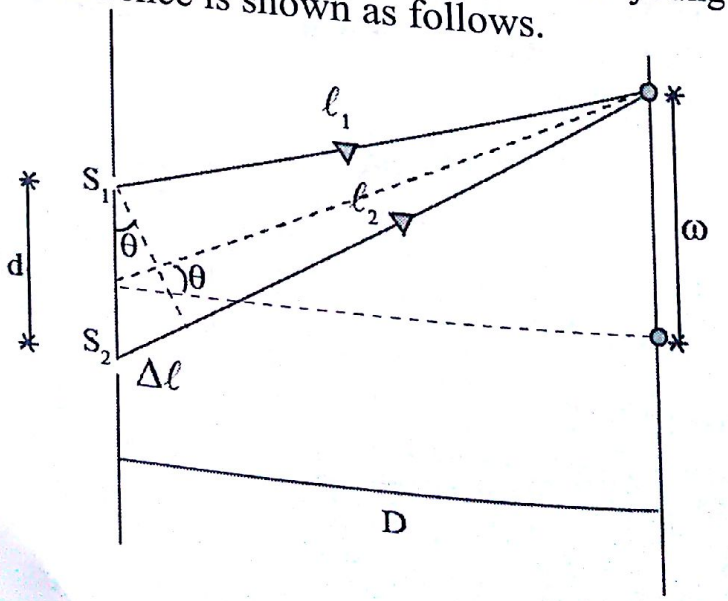
- A Destructive interference or dark band is formed on the screen when the light waves coming from the two slits are out of phase.



The brightest fringe so formed is located in the middle and is called *central maximum* or the *zero order* maximum and the next bright fringes on either side of the central maximum are called the *first order* maxima, followed by the second order maxima and so on.
A simple sketch of intensity against distance is shown below.



Moreover, a geometric construction that describes young's double slit experiment path difference is shown as follows.



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We can see that,

$\Delta\ell = \ell_2 - \ell_1 = d\sin\theta$ ← path difference between the two rays (waves)

The condition for constructive and destructive interferences is then found to be as follows.

i) Constructive Interference

$m\lambda$ ← path difference

$2\pi m$ ← phase difference

ii) Destructive Interference

$\left(m + \frac{1}{2}\right)\lambda$ ← path difference

$(2m - 1)\pi$ ← phase difference

Where, m = 0, 1, 2,...and is called *order number* for example, m = 0 implies central or zero order maximum and the path difference is zero since the waves from the slits reach the screen simultaneously and m = 1 means first order maxima, etc and here the path difference between the mixing wave is λ.

The path and phase differences of maxima and minimas can be illustrated as follows.

Furthermore, from the previous sketch, for small angle θ, sin θ = tanθ. Hence, the relationship among the distance from the slits to the screen (D), the distance between successive bright or dark fringes (ω), the distance between slits (d) and the wavelength (λ)is given by,

$$\frac{\lambda}{d} = \frac{\omega}{D}$$

Specifically,

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$\omega = \frac{\lambda D}{d} m$, for constructive, and

$\omega = \frac{\lambda D}{d}\left(m + \frac{1}{2}\right)$, for distructive interference

Keep in mind the following points

Slit, implies narrow opening or small gap.

Path difference, means, the difference in distance of the waves coming from slit 1 and 2 to the screen.

Phase difference, is the angle difference between the interfering waves.

Conceptual Examples

13. Light waves are emitted coherently from two sources and detected at a spot equidistant from the two sources. Do these waves interfere consturctively or destructively? Explain

Explanation: The path difference is zero because, $\ell_1 = \ell_2$ so we have, $\ell_1 - \ell_2 = m\lambda = 0$, $\therefore$ interfere constructively.

14. Suppose the distance between the slits is halfed. How would this change affect the interference pattern? Explain

Explanation: From, $\frac{\lambda}{d} = \frac{\omega}{D}$, $\omega = \frac{\lambda D}{d}$, if, $d = \frac{1}{2}d$

then, $\frac{\lambda}{\frac{1}{2}d} = \frac{\omega^1}{D} \Rightarrow \omega^1 = \frac{2\lambda D}{d} = 2(\omega)$

Thus, the fringe width doubled.

15. Light of wavelength λ_1 illuminates a double slit and inteference pattern is observed on a screen. When the wave length is changed to λ_2, the fringes get closer together. What can you say about λ_2.

Explanation: From $\frac{\lambda}{d} = \frac{\omega}{D} \Rightarrow \omega = \frac{\lambda D}{d}$

as can be seen, $\omega \sim \lambda$. hence, the fringe width (ω) got smaller since the wavelength decreased, $\lambda_2 < \lambda_1$.

16. If double slit experiment is done in water, what will happen to the fringe width? Explain.

Explanation: In water, wavelength of light decreases. Hence, the fringes become closely spaced when the experiment is done in water.

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Illustrative Examples

17. Two separate coherent sources produce waves whose wavelengths are 0.10m. The two waves spread out and intersect at a certain point. Identify whether constructive or desctructive interference occur at this point when.
   a) One wave travels 4.25m and the other travels 4.05m.
   b) One wave travels 5.20m and the other travels 5.05m.
   c) One wave travels 5.20m and the other travels 4.95m?

Solution: The answer basis its root on the concept of path difference. That is,

If, $\ell_1 - \ell_2 = 0,\ \lambda,\ 2\lambda, 3\lambda ... m\lambda \rightarrow$ constructive

And if,

$$\ell_1 - \ell_2 = \frac{\lambda}{2},\ 3\left(\frac{\lambda}{2}\right),\ 5\left(\frac{\lambda}{2}\right) ... \left(m + \frac{1}{2}\right)\lambda \rightarrow \text{destructive}$$

a) $\ell_1 - \ell_2 = 4.25 - 4.05 = 0.20\text{m}$. Since, $\lambda = 0.10\text{m}$

$$0.20\text{m} = 2(0.10) = 2\lambda \rightarrow \text{constructive}$$

b) $\ell_1 - \ell_2 = 5.20 - 5.05 = 0.15 = \frac{3}{2}(0.1) = \frac{3}{2}\lambda \rightarrow$ destructive

c) $\ell_1 - \ell_2 = 5.20 - 4.95 = 0.25 = \frac{5}{2}(0.1) = \frac{5}{2}\lambda \rightarrow$ destructive

18. A laser produces an interference pattern on a screen 10m away form a pair of slits. The slits are 0.25mm apart and the fringe width is 26mm. Caculate the wavelength.

Solution: $\frac{\lambda}{d} = \frac{\omega}{D} \Rightarrow \lambda = \frac{\omega d}{D} = \frac{\left(26\times10^{-3}\right)\left(0.25\times10^{-3}\right)}{10} = 650\text{nm}$

19. A viewing screen is 1.2m from two slits whose separation is 0.08mm. If the first dark is 0.68cm from the center of the screen, caluclate the wavelength of the light.

Solution: For destructive interference, $\omega = \frac{\lambda D}{d}\left(m + \frac{1}{2}\right)$

$$\text{now, } m = 1 \Rightarrow \omega = \frac{3}{2}\frac{\lambda D}{d}$$

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$$\Rightarrow \lambda = \frac{2\omega d}{3D} = \frac{(2)\left(0.68\times10^{-2}\right)\left(8\times10^{-5}\right)}{10}$$

$$\lambda = 302\text{nm}$$

20. Light with frequency of 6×10^5GHz passes through two narrow slits of 0.2mm separation and forms interference patten on a screen 2m form the slits. Calculate the fringe width. (UEE, 2003).
a) 3mm B) 5mm C) 4mm D) 3.5mm

Solution: $v = \lambda f \Rightarrow \lambda = \frac{v}{f} = \frac{3\times10^8}{6\times10^5\times10^2} = 5\times10^{-7}\text{m}$

and from $\frac{\lambda}{d} = \frac{\omega}{D}$, $\omega = \frac{\left(5\times10^{-7}\right)(2)}{\left(0.2\times10^{-3}\right)} = 5\text{mm}$

Answer: B

21. In a young's double slit experiment, the separation between the slits is doubled and the distance between the slits and the screen is halfed. What would be the new fringe width?

Solution: $\frac{\lambda}{d_1} = \frac{\omega_1}{D_1} \Rightarrow \omega_1 = \frac{\lambda D_1}{d_1}$

When, $d_2^1 = 2d_1$ and $D_2^1 = \frac{D_1}{2}$

$$\Rightarrow \omega_2^1 = \frac{D_2}{d_2} = \frac{\lambda\left(D_1/2\right)}{2d_1} = \frac{\lambda D_1}{4d_1} = \frac{1}{4}\left(\frac{\lambda D_1}{d_1}\right) \Rightarrow \omega_2 = \frac{1}{4}\omega_1$$

22. A double slit interference pattern is observed on a screen 1m from two slits spaced 0.3mm. If the distance between the central maximum and the ninth order maximum is 1.6cm. Calculate the wavelength of the wave.

Solution: In double slit interference, the fringe widths are equally spaced. Thus, $(9^{th} - 0^{th})$ order = 1.6cm

$\Rightarrow$ 9 fring width = 1.6cm = 9ω

$\Rightarrow \omega = \frac{1.6\text{cm}}{9} = 1.8\times10^{-3}\text{m}$, next using

$$\frac{\lambda}{d} = \frac{\omega}{D}, \lambda = \frac{\omega d}{D} = \frac{\left(1.8\times10^{-3}\right)\left(0.3\times10^{-3}\right)}{1} = 540\text{nm}.$$

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23. Red light (λ = 664nm) is used in young's double slit experiment with slits spacing of 1.2×10^{-4}m and screen being 2.75m from the slits. Calculate the distance from the central maximum to the third order maximum.

Solution: $\frac{\lambda}{d} = \frac{\omega}{D} \Rightarrow \omega = \frac{D\lambda}{d} = \frac{(2.75)(664\times10^{-9})}{(1.2\times10^{-4})}$

$\Rightarrow \omega = 1.52\times10^{-2}$m.

Now, the distance beetwen the central maximum and the third order maximum is $3\omega = 3(1.52 \times 10^{-2}) = 4.56 \times 10^{-2}$m.

24. Light from a laser beam (λ = 633nm) illuminates two slits spaced 0.4mm apart. If a viewing screen is 2m from the slits. Calculate the distance between

a) two second order bright fringes
b) two second order dark fringes

Solution: a) $\frac{\lambda}{d} = \frac{\omega}{D} \Rightarrow \omega = \frac{\lambda D}{d} = \frac{(633\times10^{-9})(2)}{4\times10^{-4}} = 3.15$mm

the distance between the central maximum and the second order maxima is, $2\omega = 2(3.15) = 6.3$mm. Hence the distance between the two second order maxima is $2(6.3) = 12.6$mm.

b) For dark fringe,

$$\omega = \left(m + \frac{1}{2}\right)\left(\frac{\lambda D}{d}\right), \quad m = 2$$

$$\omega = \left(2 + \frac{1}{2}\right)(3.15) = 7.9\text{mm}$$

Hence, the distance between the two secondary minima is $2(7.9) = 15.8$mm.

25. Monochromatic light illuminates a double slit system with slit separation of 0.3mm and slit to screen distance 1m. If the second order maxima occurs at 4mm from the central maximum

Calculate: a) The wavelength
b) The position of the third order maxima

Solution: a) For constructive interference,

$$\omega = m\frac{\lambda D}{d}, \text{ now, } m = 2$$

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$$\lambda = \frac{\omega d}{mD} = \frac{\left(4\times10^{-3}\right)\left(0.3\times10^{-3}\right)}{(2)(1)} = 600\text{nm}$$

b) $2\omega = 4\text{mm} \Rightarrow \omega = 2\text{mm}$

the third order maxima is at $3\omega = 3(2\text{mm}) = 6\text{mm}$

26. What happens if the monochromatic light used in Young's slit experiment is replaced by white light? (UEE, 2004)

A) No fringes are observed.
B) All bright fringes become white.
C) All bright fringes have colors between violet and red.
D) Only the central fringe is white, all the frigs are colored.

Answer: B

27. In Young's double slit experiment, both the separation between the slits and the distance between the slits and the screen are halved then what is the fringe width?

A) halved B) Unchanged C) zero D) doubled

Solution: $\frac{\lambda}{d_1} = \frac{\omega_1}{D_1} \Rightarrow \omega_1 = \frac{\lambda D_1}{d_1}$

when $d_2 = \frac{1}{2}d_1$ and $D_2 = \frac{1}{2}D_1$

$$\omega_2 = \frac{\lambda D_2}{d_2} = \frac{\lambda\left(\frac{1}{2}D_1\right)}{\frac{1}{2}d_1} = \frac{\lambda D_1}{d_1} = \omega_1$$

Answer: B

28. A viewing screen is separated from a double slit by 4.8cm. The distance between slits is 0.03mm. The first dark fringe is 4.5cm from the centerline on the screen. What is the wavelength of the light? (UEE, 2006)

A) 1250nm B) 1875nm C) 2500nm D) 5625nm

Solution: For dark fringe, $\omega = \frac{\lambda D}{d}\left(m + \frac{1}{2}\right)$ for $m = 1$,

$$\omega = \frac{3}{2}\frac{\lambda D}{d} \Rightarrow \lambda = \frac{2\omega d}{3D} = \frac{2\left(4.5\times10^{-2}\right)\left(0.03\times10^{-3}\right)}{3\left(4.8\times10^{-2}\right)}$$

$\lambda = 1875\text{nm}$

Answer: B

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29. Two separate coherent sources produce waves whose wavelengths are 0.10m. Which statement is correct about points that are 3.2m away from one source?

A) constructive interference occurs if the points are 3m from the other source

B) constructive interference occurs if the points are 3.05m from the other source

C) constructive interference occurs if the points are 2.95m from the other source

D) constructive interference occurs if the points are 3.2m from the other source

Solution: If $\ell_1 - \ell_2 = 0, \lambda, 2\lambda \ldots m\lambda \rightarrow$ constructive

And If $\ell_1 - \ell_2 = \frac{\lambda}{2}, \frac{3}{2}\lambda, \frac{5}{2}\lambda \ldots \left(m + \frac{3}{2}\right)\lambda \rightarrow$ distractive

now, observe that, $3.2 - 3 = 0.2 = 2(0.1) = 2\lambda, \lambda = 0.1m$

Answer: A

30. In Young's double slit experiment interference pattern which one of the following activity move the successive bright fringes farther apart?

A) Decreasing the wavelength of the light

B) Decreasing the distance between the screen and the slits

C) Increasing the slit spacing

D) Decreasing the slit spacing

Explanation: From $\frac{\lambda}{d} = \frac{\omega}{D}$, $\omega = \frac{\lambda D}{d}$

Hence, ω increases when d is decreased.

11.5 Single Slit Diffraction

A single slit can also produce a diffraction pattern, but this time, the central maximum is twice as wide as the first order maxima and much brighter. That is, for a single slit.

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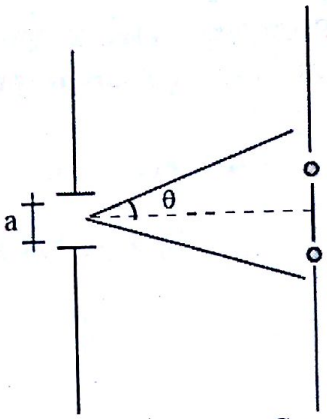

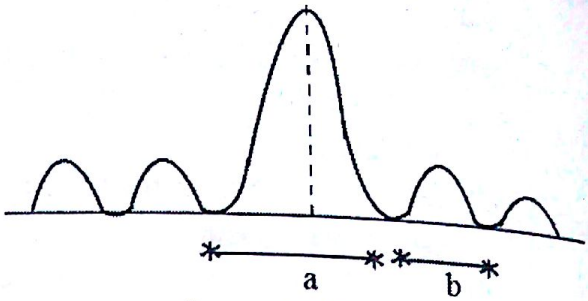

for a single slit, a = 2b

And, the condition that determines the location of dark fringes in a single slit is given by,

$$\boxed{a \sin\theta = m\lambda}$$

Where, a is slit width and for small angle, θ, it is true that $\sin\theta = \tan\theta$ and hence.

$$\boxed{\frac{\lambda}{a} = \frac{\omega}{D}}$$

Again, ω is fringe width.

To observe a pronounced diffraction pattern, the width of the slit must have the same order of magnitude as the wavelength of the light used.

☛ **Remember,** in single slit diffraction

- The interference pattern consists of a large bright central maximum.
- The width of the fringe depends on the width of the slit. The narrower the slit, the wider the fringe and the sharper the image.
- For a given slit width (a), the longer the wavelength, the wider the diffraction pattern.
- For a given wavelength (λ) the narrower the slit, the wider the diffraction.
- If the slit width (a) is less than or equal to the wavelength (λ), the diffraction pattern is well observed and the fringes are far apart.
- If the slit width (a) is much greater than the wavelength (λ), the fringe move closer together and the diffraction pattern become blurred and difficult to observe.

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- The angle between the center and the first fringe (m = 1) is,

$a \sin\theta = m\lambda \Rightarrow \sin\theta = \lambda/a$

Conceptual Examples

31. Explain the effect on the central bright fringe if the width of a single slit is reduced.

Explanation: The smaller the slit width, the wider the central fringe.

32. Explain diffraction of light interms of wavelength and slit width.

Explanation: The longer the wavelength, the greater the diffraction and the narrower the slit, the more the diffraction.

33. Explain the effect on the interference pattern when using
    a) light with low frequency
    b) light with wide slit

Explanation: We know, $v = \lambda f \Rightarrow \lambda = \frac{v}{f}$ the lower the frequency, the longer the wavelength.

a) $\frac{\lambda}{a} = \frac{\omega}{D}$, the lower the frequency the longer the wavelength

hence for low frequency, $\omega = \frac{\lambda D}{a}$, the fringe width increase.

b) For wide slit, $\omega = \frac{\lambda D}{a}$, the fringe width decreases

34. How can you compare the interference pattern produced by single slit and double slit.

Explanation: In a double slit, all the fringes produced do have the same width. However, in a single slit the width of the central maximum is twice of that of the first order maxima.

Illustrative Examples

35. A monochromatic light passes through a single slit of width 1.2×10^{-5}m. If the first dark fringe is produced at an angle of $\theta = 3.25°$. What is the wavelength of the light.

Solution: $a \sin\theta = m\lambda, \quad m = 1$

$\lambda = a \sin\theta = (1.2 \times 10^{-5})(\sin 3.25°) = 680\text{nm}$

36. A screen is placed 60cm from a single slit, which is illuminated with light of wavelength 700nm. If the distance between the first and the third order minima is 5mm, calculate the width of the slit.

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Solution: The distance between the first and third minima is

$$3^{rd} - 1^{st} = 2\omega = 5mm \Rightarrow \omega = \frac{5}{2} = 2.5mm$$

And using, $\frac{\lambda}{a} = \frac{\omega}{D} \Rightarrow a = \frac{\lambda D}{\omega} = \frac{\left(700\times10^{-9}\right)(0.6)}{\left(2.5\times10^{-3}\right)} = 168\mu m.$

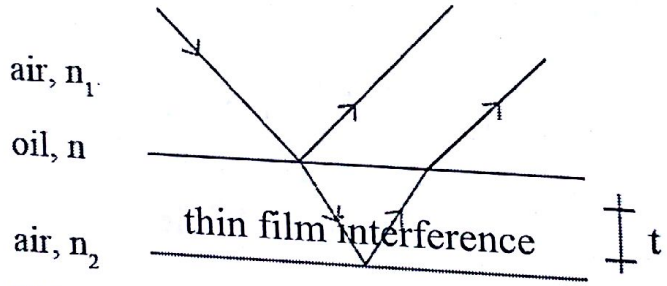
11.6 Thin Film Interference

Thin film interference has important application in the optics industry. Thin film coating less than 1μm thick are used for the anti-reflection coating on the lenses in cameras, microscopes and other optical equipment.

Thin Film Interference: Is a colourful pattern caused by the interference of light waves reflecting off the top surface of a thin film with those that have reflected off the bottom surface.

The beautiful, rainbow or multicolors of a soap bubble and an oil slick are due to thin film interference.

Consider a thin film with thickness (t) and index of refraction (n).



The condition for constructive and destructive interference in a thin film.

a) For $n_1 < n$ and $n > n_2$.
such as air, $n_1 = 1$, oil $n = 1.33$ and air $n_2 = 1$

$$2nt = \left(m + \frac{1}{2}\right)\lambda \leftarrow \text{constructive interference}$$

$$2nt = m\lambda \leftarrow \text{destructive interference}$$

b) For $n_1 < n < n_2$.
such as air, $n_1 = 1$, oil $n = 1.33$ and glass $n_2 = 1.5$

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$2nt = m\lambda \leftarrow$ constructive interference

$2nt = \left(m + \frac{1}{2}\right)\lambda \leftarrow$ destructive interference

Where, $m = 0, 1, 2, \ldots$

n, is refractive index of the film.

t, is the thickness of the thin film.

Conceptual Examples

37. The absence of a particular color in a region of thin film interference is due to_____

Explanation: destructive interference

38. Thin films like the soap bubble have a multicolored appearance that often change with time. Why is this happening?

Explanation: The multicolor is due to interference and the color change is because of the thickness change.

39. Explain what caused the colors in a soap bubble.

Explanation: The multicolors of soap bubbles and oil film are due to the interference of reflected rays from the film surfaces.

Illustrative Examples

40. Suppose a soap bubble with index of refraction 1.33 is illuminated by light of 602nm wavelngth. Calculate the minimum thickness that enables it produce constructive interference.

Solution: For constructive interference,

$$2nt = \left(m + \frac{1}{2}\right)\lambda$$

For minimum thickness, $m = 0$.

$$2nt = \frac{\lambda}{2} \Rightarrow t = \frac{\lambda}{4n} = \frac{602}{4(1.33)} = 11.3\text{nm}$$

41. A beam consisting of red light (λ = 662nm) and blue light (λ = 465nm) is directed on to a soap bubble of n = 1.33. Find the smallest non-zero thickness for which it appears red.

Solution: The film to look red, the blue light must cancel due to destructive interference.

For destructive interference,

$$2nt = m\lambda \quad , m = 1 \Rightarrow t = \frac{m\lambda}{2n} = \frac{465}{2(1.33)} = 175\text{nm}$$

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42. White light across the visible wavelength range of 400nm to 700nm is incident on to a film of n = 1.33 and t = 320nm. At what wavelength λ is the reflected light become bright?

Solution: It look bright when interfere constructively.

$$2nt = \left(m + \frac{1}{2}\right)\lambda, \Rightarrow \lambda = \frac{2nt}{m + \frac{1}{2}},$$

$$m = 0 \Rightarrow \lambda = \frac{(2)(1.33)(320)}{0 + \frac{1}{2}} = 1700\text{nm}$$

$$m = 1 \Rightarrow \lambda = \frac{(2)(1.33)(320)}{1 + \frac{1}{2}} = 567\text{nm}$$

$$m = 2 \Rightarrow \lambda = \frac{(2)(1.33)(320)}{2 + \frac{1}{2}} = 340\text{nm}$$

Hence, the wavelength is when m = 1 which is λ = 567nm.

43. A glass is coated on one side with a thin film of magnesium fluoride (MgF_2) to reduce reflection. If the index of refraction of MgF_2 is 1.38 and that of the glass is 1.5. What is the least coating thickness that eliminates the reflection of light of wavelength 600nm.

Solution: Eliminating means canceling, hence destructive

But now, $n_1 < n < n_2$, since $1 < 1.38 < 1.5$.

Thus, use, $2nt = \left(m + \frac{1}{2}\right)\lambda$

$$t = \frac{\left(m + \frac{1}{2}\right)\lambda}{2n},$$

$$\text{For } m = 0,\ t = \frac{\left(0 + \frac{1}{2}\right)(600)}{2(1.38)} = 108.7\text{nm}$$

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44. Which phenomenon is responsible for the colors seen on a thin film of oil on the water surface? (UEE, 2004)
A) Diffraction C) Superposition
B) Interference D) Refraction **Answer: B**

45. When a soap bubble is in air, it exhibits different colors under sunlight. The absence of a particular color is due to.
A) double refraction
B) straight line propagation of light
C) destructive interference of light of that color
D) constructive interference of light of that color **Answer: C**

46. Suppose that the index of refraction of a soap-bubble film is 1.25. What is the minimum thickness of the soap-bubble that results in constructive interference if the film is illuminated with light of 600nm wavelength in free space (UEE, 2006).
A) 120nm B) 150nm C. 300nm D. 450nm
Solution: For minimum thickness, m = 0

$$2nt = \left(m + \frac{1}{2}\right)\lambda \Rightarrow t = \frac{\left(m + \frac{1}{2}\right)\lambda}{2n} = \frac{\left(0 + \frac{1}{2}\right)600}{2n} = 120nm$$

Answer: A

11.7 Diffraction Grating

Light can be caused to diffract and interfere with more than two slits.

Diffraction Grating: Is a set of many equally spaced slits in which the slit separation is very small.

As more and more parallel slits are used, the principal maxima decrease in width, at the same time, the number of subsidiary maxima increases and their intensity decreases.

- This means that, the principal maxima are very narrow, and there are so many subsidiary maxima that they are so faint to be seen.

As an example, consider the following

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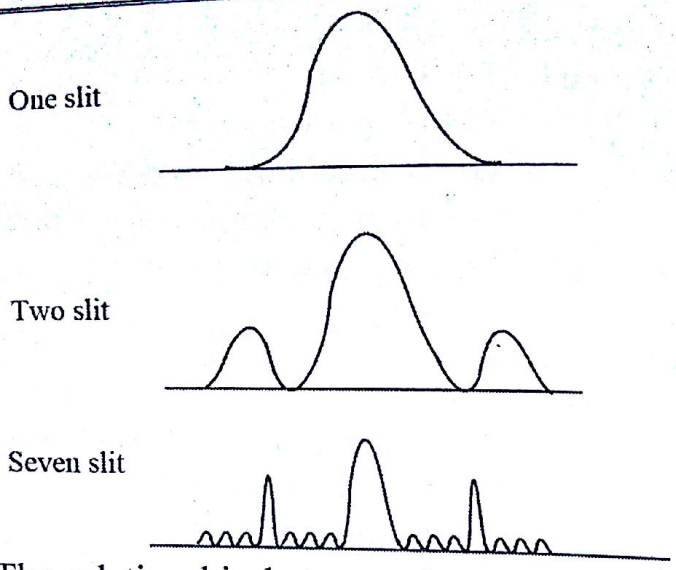
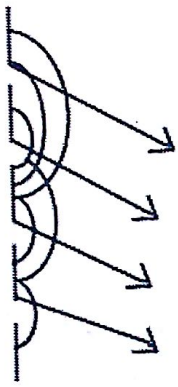


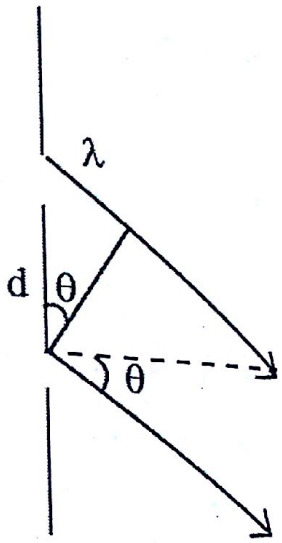
As the number of slits increases, the intensity and sharpness of the principal maxima increase while the intensity of the subsidiary maxima decreases.

The relationship between the angle (θ) at which the maxima occur, the slit separation (d) and the wavelength of light can be obtained as follows.



Taking portion of the grating

$$\sin\theta = \frac{\lambda}{d}$$



Hence, the first maxima is produced at an angle of, $\sin\theta = \frac{\lambda}{d}$ from the central maximum.

In general, the m^{th} maxima will occur at an angle θ_m from the central maximum and is given by,

$$\boxed{\sin\theta_m = \frac{m\lambda}{d}}$$, Where m is order number.

The spacing of the slits in a grating is sometimes expressed in terms of the number of slits or lines per millimeter, centimeter or meter (N). For a grating with N slits per meter, the slit spacing is N^{-1}.

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That is, $\boxed{d = \frac{1}{N}}$

To find out the highest order of the principal maxima we can use the fact that the maximum value of $\sin\theta = 1$. This means that, we can write.

Since, $\sin\theta \leq 1$

and, $\sin\theta_m = \frac{m\lambda}{d}$

It follows that

$$\frac{m\lambda}{d} \leq 1$$

Hence the maximum order produced is then

$\boxed{m \leq \frac{d}{\lambda}}$ ← highest order spectrum

Remember! Since m must be a whole number, since there is no such thing as half or quarter order maxima, we calculate the value of d/λ and round it down to the next whole number.

Conceptual Examples

47. How can you compare the interference pattern produced by a diffraction grating and double slit.

Explanation: The bright fringes produced by a diffraction grating are much narrower than those produced by a double slit.

Illustrative Examples

48. Suppose light with wavelength of 780nm incident on a diffraction graing has 4000 slits per centimer. Find the angles at which the third and fourth maximas produced.

Solution: $\sin\theta_m = \frac{m\lambda}{d}$, $d = \frac{1}{N} = \frac{1}{400}\text{cm} = 2500\text{nm}$

For the 3rd maxima, $m = 3$

$$\sin\theta_3 = \frac{3\left(780\times10^{-9}\right)}{2500\times10^{-9}}$$

$$\sin\theta_3 = 0.936 \Rightarrow \theta_3 = 69.4^\circ$$

For the 4th maxima, $m = 4$

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$$\sin\theta_4 = \frac{4\left(780\times10^{-9}\right)}{2500\times10^{-9}}$$

$$\sin\theta_4 = 1.248$$

$$\theta_4 = \sin^{-1}\left(1.248\right) \leftarrow \text{ No solution}$$

$\sin\theta$ can never be greater than one. Hence the forth order maxima cannot be produced.

49. A grating with 250 lines /mm is used with a visible light of wavelength range of 400nm to 700nm. How many orders can one see in
    a) The entire visible spectrum
    b) The short wavelength region

Solution: $d = \frac{1}{N} = \frac{1}{250}\text{mm} = 4000\text{nm}.$

a) $m \le \frac{d}{\lambda} \Rightarrow m \le \frac{4000\text{nm}}{700\text{nm}} = 5.7 \Rightarrow m = 5$

b) $m \le \frac{d}{\lambda} \Rightarrow m \le \frac{4000\text{nm}}{400\text{nm}} = 10 \Rightarrow m = 10$

50. Find the slit spacing necessaryfor 450nm light to have a second order maxima at 30°.

Solution: $\sin\theta_m = \frac{m\lambda}{d} \Rightarrow d = \frac{m\lambda}{\sin\theta}, \; m = 2$

$$d = \frac{(2)\left(450\times10^{-9}\right)}{\sin 30^\circ} = 1.8\times10^{-6}\text{m} = 1.8\mu\text{m}$$

51. When 546nm light passes through diffraction grating, a second order principal maxima is observed at an angle of $\theta = 30°$. How many lines per centimeter does this grating has?

Solution: $\sin\theta_m = \frac{m\lambda}{d}, \; N = \frac{1}{d}$

$$d = \frac{m\lambda}{\sin\theta} = \frac{(2)\left(546\times10^{-9}\right)}{\sin 30^\circ}$$

$$d = 2.18\times10^{-6}\text{m} = 2.18\times10^{-4}\text{cm}$$

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Thus, $N = \frac{1}{d} = \frac{1}{2.18\times10^{-4}\text{cm}} = 4580$ slits/cm

52. A ligth of wavelength 430nm is directed normally at a diffraction grating. The first order maxima is at 30° from the zero order maxima. Calculate
   a) The number of slits per millimeter
   b) The angle of diffraction of the second order maxima
   c) Can the third order maxima produced

Solution: a) $\sin\theta = \frac{m\lambda}{d} \Rightarrow d = \frac{m\lambda}{\sin\theta} = \frac{(1)(430\times10^{-9})}{\sin 30°}$

$d = 860\text{nm} \Rightarrow N = \frac{1}{d} = \frac{1}{8.60\times10^{-9}} = 1.16\times10^{6}\text{slit/m}$

$N = 1.16\times10^{3}\text{slit/mm}.$

b) $\sin\theta_2 = \frac{(2)(430\times10^{-9})}{860\times10^{-9}} = 1 \Rightarrow \theta_2 = 90°$

c) No. $\sin\theta_3 > 1$.

Additional Examples

Reflection and Refraction.

53. A ray of light travels through a transparent slab with a speed of 2×10^8m/s. This implies that the refractive index of the slab is
   A) 1.5 B) 0.667 C) 2 D) 6

Solution: $\frac{v_1}{v_2} = \frac{n_2}{n_1} \rightarrow$ snell's law

$\frac{3\times10^8}{2\times10^8} = \frac{n_2}{n_1} \Rightarrow n_2 = 1.5$ **Answer: A**

54. When a wave passes from one medium to another, there is a change in
   A) Frequency and velocity
   B) Frequency and wavelength
   C) Wavelength and velocity **Answer: C**
   D) Frequency, wavelength and velocity

55. Light of wavelength 720nm in air has a wavelength of ____ in a glass of refractive index 1.5.
   A) 720nm B) 480nm C) 620nm D) 1080nm

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Solution: From snell's law, $\frac{\lambda_1}{\lambda_2}=\frac{n_2}{n_1} \Rightarrow \lambda_2=\frac{\lambda_1}{n_2}$

$$\lambda_2=\frac{720}{1.5}=480\text{nm}$$

Answer: B

56. The sun is visible to us a little before the actual sunrise and a little after the actual sunset. This is because of atmospheric.
A) Reflection C) Scattering
B) Refraction D) Diffraction

Answer: B

57. Bright colour from spider's web when exposed to sun light is due to
A) reflection C) Interference
B) refraction D) Diffraction

Answer: C

Single and Double slits

58. In young's double slit Experiment, the separation between the slits is halved and the distance between the slits and the screen is doubled. The fringe width will be
A) doubled C) quadrupled
B) halved D) remain same

Explanation: $\frac{\lambda_1}{S_1}=\frac{\omega_1}{D_1} \Rightarrow \omega_1=\frac{\lambda_1 D_1}{S_1}$ .....(*)

$$\frac{\lambda_2}{S_2}=\frac{\omega_2}{D_2} \text{ .....(**)}$$

but $\lambda_1=\lambda_2,\ S_2=\frac{1}{2}S_1,\ D_2=2D_1$

$\Rightarrow$ equation (**) becomes

$$\frac{\lambda_1}{\frac{1}{2}(S_1)}=\frac{\omega_2}{2D_1} \Rightarrow \omega_2=\frac{4\lambda_1 D_1}{S_1}, \text{ but } \frac{\lambda_1 D_1}{S_1}=\omega_1$$

$$\Rightarrow \omega_2=4\omega_1$$

Answer: C

59. Young's double slit experiment establishes that light consists of
A) particles C) neither particle nor wave
B) waves D) both particle and waves

Answer: B

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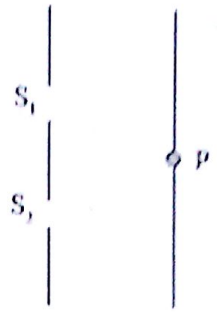
60. Coherent light is incident on two slits as shown. If a dark fringe occurs at p, which of the following gives possible phase differences for the light waves arriving at p.

A) $\pi, 2\pi, 3\pi\ldots$ C) $\frac{1}{2}\pi, \frac{3}{2}\pi, \frac{5}{2}\pi\ldots$

B) $2\pi, 4\pi, 6\pi\ldots$ D) $\pi, 3\pi, 5\pi$

Explanation: For a dark (destructive) interference,

phase difference $= \left(m + \frac{1}{2}\right)2\pi, m = 0, 1, 2\ldots$



Answer: D

61. To demonstrate the phenomenon of interference, require two sources which emit radiation of

A) the same frequency and definite phase difference
B) different frequency
C) different wavelength
D) same frequency

Explanation: The sources should be coherent. **Answer: A**

62. Two monochromatic light waves are coherent if

A) They are in phase
B) They have a constant phase difference
C) They travel equal distances
D) They have equal amplitude

Answer: B

63. Soap bubble appears coloured due to the phenomenon of

A) Interference C) Dispersion
B) Diffraction D) Reflection

Answer: A

64. In double slit experiment, for light of which colour, the fringe width will be minimum?

A) Violet B) Red C) Green D) Yellow

Explanation: $\frac{\lambda}{S} = \frac{\omega}{D} \Rightarrow \omega = \frac{\lambda D}{S}$, ω is minimum for minimum λ, keeping other constant.

Answer: A

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65. Fringe width observed in the Young's double slit experiment is ω. If the frequency of the source is doubled, the fringe width will

A) remain ω C) become $\frac{\omega}{2}$

B) become 2ω D) become $\frac{3}{2}\omega$

Explanation: $v = \lambda f \Rightarrow \lambda = \frac{v}{f}$, when f is doubled, λ halfed

and $\frac{\lambda}{S} = \frac{\omega}{D} \Rightarrow \omega = \frac{\lambda D}{S}$, when λ is halfed, ω below half

Answer: C

66. In an interference experiment, the spacing between successive maxima or minima is

A) $\frac{\lambda S}{D}$ B) $\frac{\lambda D}{S}$ C) $\frac{SD}{\lambda}$ D) $\frac{\lambda S}{4D}$

Explanation: $\frac{\lambda}{S} = \frac{\omega}{D} \Rightarrow \omega = \frac{\lambda D}{S}$

Answer: B

67. In young's double slit experiment, yellow light is replaced by red light of the same intensity. The fringe width shall

A) increase C) unchanged

B) decrease D) become zero

Explanation: Red light has greater wavelength than yellow light.

And since $\omega = \frac{\lambda D}{S}$.

Answer: A

68. Monochromatic green light of wavelength 5×10^{-7}m illuminates a pair of slits 1mm apart. The separation of nearest bright fringes on the screen 2m away is

A) 0.25mm B) 0.5mm C) 0.1mm D) 1.0mm

Solution: $\omega = \frac{\lambda D}{S} = \frac{\left(5 \times 10^{-7}\right)(2)}{10^{-3}} = 10^{-3}\,\text{m} = 1\text{mm}$ **Answer: D**

69. If the source of light used in a Young's Double slit experiment is changed from red to blue, then

A) the fringe will become brighter

B) consecutive fringes come closer

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C) the number of maxima increases
D) the central bright fringe disappear.

Explanation: The wavelength of blue light is less than that of the red light, hence the fringe width decreases and the number of maxima increases.

Answer: C

70. If torch is used in place of monochromatic light in Young's experiment, what will happen?
A) Fringe will occur as the monochromatic source
B) Fringes will appear for a moment and then disappear
C) No fringe will appear
D) Only bright fringe will appear

Explanation: For interference, coherent sources are required.

Answer: C

71. What is the path difference between two interfering waves if the corresponding phase difference is 2π?

A) λ B) 3λ C) $\lambda/2$ D) $\lambda/4$

Explanation: $\left.\begin{array}{l}\text{path difference} = (m)\lambda \\ \text{phase difference} = (m)2\pi\end{array}\right\}$ constructive

When m = 1, phase difference = 2π and path difference = λ

Answer: A

72. Two light sources are said to be coherent if they are obtained from
A) Two sources emitting light of the same wavelength
B) A single point source
C) A wide source
D) Two ordinary bulbs emitting light of different wavelength

Answer: B

73. Two coherent sources of light produce destructive interference when phase difference between tehm is

A) 2π B) π C) $\pi/2$ D) 0

Explanation: Condition for destructive interference is that the phase difference be an odd multiple of π. i.e $\left(m+\frac{1}{2}\right)2\pi$.

Where, m = 0, 1. 2,...

Answer: B

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74. In Young's double slit experiment, the band width of fringe is independent of
A) distance between the coherent sources
B) distance of the screen from the source
C) wavelength of light emitted by the source
D) order of the fringe.
Explanation: All the interference fringe are of equal width.
Answer: D

75. The fringe width in Young's double slit experiment can be increased if we decrease the;
A) separation of slits
B) width of slits
C) distance between slits and screen
D) wavelength of light
Explanation: $\omega = \frac{\lambda D}{S}$, ω increases as S decreases. **Answer: A**

76. A light of wavelength 600nm falls on a single slit 1mm wide and is observed on a screen 2m away. What is the distance between two first order dark fringes on either side of the central bright fringe?
A) 1.2cm B) 1.2mm C) 2.4cm D) 2.4mm
Solution: $\omega = \frac{2\lambda D}{S}$, $\omega = \frac{2 \times 2 \times 600 \times 10^{-9}}{10^{-3}} = 24 \times 10^{-4}\,m = 2.4mm$
Answer: D

77. The penetration of light into the region of geometrical shadow is called
A) polarisation C) diffraction
B) interference D) refraction **Answer: C**

78. To observe diffraction, the size of an obstacle
A) should be the same order as wavelength
B) should be much larger than the wavelength
C) has no relation to wavelength
D) should be exactly $\lambda/2$ **Answer: A**

79. Both, light and sound show wave characteristic but diffraction is much hard to observe in light. That is because
A) speed of light is far large
B) wavelength of light is smaller

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C) light does not require a medium
D) waves of light are transverse

Answer: B

80. The wavelength of light can be experimentally found using
A) ripple tank
B) diffraction grating
C) plane mirror
D) glass prism

Answer: B

81. The frequency of light can be experimentally found using
A) ripple tank
B) diffraction grating
C) plane mirror
D) glass prism

Answer: B

82. In a Young's double slit experiment, the fringe width is found to be 0.4mm. If the system is immersed in water of refractive index $4/3$, the new fringe width will be

A) 0.3mm B) 0.4mm C) 0.53mm D) 0.45mm

Solution: $\frac{\lambda_1}{\lambda_2}=\frac{n_2}{n_1}\Rightarrow\lambda_2=\left(\frac{n_1}{n_2}\right)(\lambda_1)=\left(\frac{1}{4/3}\right)\lambda_1$

$$\lambda_2=\frac{3}{4}\lambda_1$$

And from, $\frac{\lambda}{S}=\frac{\omega_2}{D}\Rightarrow\omega_2=\frac{\lambda D}{S}=\frac{3}{4}\left(\frac{\lambda D}{S}\right)=\frac{3}{4}(\omega_1)$

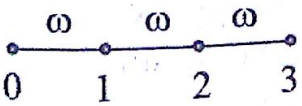
$$\omega_2=\left(\frac{3}{4}\right)(0.4)=0.3\text{mm}$$

Answer: A

83. If young's double slit experiment carried out with light of wavelength 500nm, the distance between the slits is 0.2mm and the screen is at 200cm from the slits. The central maximum is at x = 0. The third maximum will be at x equal to.

A) 1.67cm B) 1.5cm C) 0.5cm D) 5cm

Solution: $\frac{\lambda}{S}=\frac{\omega}{D}\Rightarrow\omega=\frac{\lambda D}{S}=\frac{\left(500\times10^{-9}\right)(2)}{\left(0.2\times10^{-3}\right)}=5\times10^{-3}\text{mm}$



the 3rd order is at 3ω.

$\Rightarrow x=3\omega=3\times5\times10^{-3}=15\times10^{-3}\text{ mm}=1.5\text{cm}$

Answer: B

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84. In Young's double slit experiment, the phase difference between the light waves reaching third bright fringe from the central fringe will be

A) zero B) 2π C) 4π D) 6π

Solution: Path difference of 3λ corresponds to phase difference of 6π. That is $\phi = 2m\pi$, when $m = 3$, $\phi = 6\pi$. **Answer: D**

85. The path difference between two interfering waves at a point on a screen is 70.5 times the wavelength. The point is

A) dark B) bright C) green D) white

Solution: Path difference $= 70.5\lambda = 141\frac{\lambda}{2}$. Since the path difference is odd multiple of $\frac{\lambda}{2}$, the point is dark. **Answer: A**

86. A light of wavelength 510nm is incident on to a double slit. If the overall separation of 10 bright fringes on a screen 3m away is 3cm. What is the slits separation?

A) 0.051mm
B) 0.51mm
C) 5.1mm
D) 5.1cm

Solution: $\omega = \frac{3}{10} = 0.3\text{cm}$

$$\frac{\lambda}{S} = \frac{\omega}{D} \Rightarrow S = \frac{\lambda D}{\omega} = \frac{\left(510\times10^{-9}\right)(3)}{\left(0.3\times10^{-2}\right)}$$

$S = 0.51\text{mm}$ **Answer: B**

87. In a Young's experiment, the distance of the second dark bank is observed at a distance of 0.3cm from the central bright band. The distance of the 4^{th} bright band from the centre is

A) 0.2cm B) 0.4cm C) 0.6cm D) 0.8cm

Solution: The 3^{rd} maxima is at $3\omega = 0.3\text{cm}$

$\Rightarrow \omega = 0.1\text{cm}$

The 4^{th} maxima is at 4ω from the center

$\Rightarrow y = 4\omega = (0.1)4 = 0.4\text{cm}$ **Answer: B**

88. In the interference pattern, energy is

A) created at the points of maxima
B) destroyed at the points of minima
C) conserved but is redistributed
D) unknown

Answer: C

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89. To demonstrate the phenomenon of interference, we require
A) two sources which emit light of the same frequency
B) two sources which emit light of nearly the same frequency
C) two sources which emit light of the same frequency and have definite phase relationship
D) two sources which emit light of different wavelength.

Answer: C

Thin film and Diffraction grating

90. On a diffraction grating having 4.5×10^5 lines per meter the second order is at 30° from the center. What is the wavelength of the light?
A) 200nm B) 566nm C) 430nm D) 589nm

Solution: $\sin\theta = \frac{n\lambda}{d}$, $d = \frac{1}{N} = \frac{1}{4.5\times10^5}$ m

$$\lambda = \frac{d\sin\theta}{n} = \frac{\sin 30}{2\left(4.5\times10^5\right)} = 566\text{nm}$$

Answer: B

91. In a single slit experiment, the width of the slit is equal to twice the wavelength of light used. What is the angle of diffraction of the first minimum?
A) 30° B) 45° C) 60° D) 90°

Solution: $\sin\theta = \frac{n\lambda}{d} = \frac{(1)\lambda}{2\lambda} = \frac{1}{2} \Rightarrow \theta = \sin^{-1}\left(\frac{1}{2}\right)$

$\theta = 30°$

Answer: A

92. In an interference pattern, the 16^{th} order maxima is observed for $\lambda_1 = 600$nm. What order will be visible for a light of wavlenegth $\lambda_2 = 480$nm?
A) 16 B) 18 C) 20 D) 24

Solution: $n_1 = \frac{d}{\lambda_1} \ldots\ldots (*)$, and

$n_2 = \frac{d}{\lambda_2} \ldots\ldots (**)$, taking ratio,

$$\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} \Rightarrow n_2 = \frac{n_1\lambda_1}{\lambda_2} = \frac{(16)(600)}{(480)} = 20$$

Answer: C

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93. What is the least thickness of a soap film which will appear black when viwed with soidum light of wavelength light 589.3nm. take refractive index of soap solution to be 1.38?

A) 213.5nm
B) 428nm1
C) 617nm
D) 512nm

Solution: $m\lambda = 2nt$, $m = 1$

$$t = \frac{m\lambda}{2n} = \frac{(1)(589.3)}{(2)(1.38)} = 213.5\text{nm}$$

Answer: A

Unit 11: Conceptual Questions

1. Expalin the difference in the interference pattern of light produced by a single slit and a double slit.
2. What will happen to the interference pattern if the width of a slit is decreased.
3. When two light waves interfere destructively, what happens to their energy? Explain
4. Describe the changes that would be observed in the two slit interfrence pattern if the entire experiment were to be submerged in water.
5. How would you expect the interference pattern of a two slit experiment to change if white light is used instead of monochromatic light?

Unit 11: Problems

Single slit and Double slit Experiments.

6. A screen is placed 40cm from a single slit, which is illuminated with light of 680nm wavelength. If the distance distance between the first and third order minima is 3mm, what is the width of the slit?
7. A double slit expeirment is set up with a light of wavelength 475nm. If the slit separation is 0.4mm and the slit to screen distnce is 2.2m. Calculate
   a) the fringe width
   b) the angle between the central and the second maxima
8. A laser produces an interference pattern on a screen 10m from a pair of slits. If the slits separation is 0.25mm and the fringe width is 26mm. Determine
   a) the wavelength
   b) the frequency of the laser light

Thin Film Interference

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9. A light wave of 690nm wavelength is incident on to a soap bubble (n = 1.33). What is the smallest non-zero film thickness for which the reflected light undergoes constructive interference?
10. A mix of red light (λ = 660nm) and green light (550nm) is directed on a soap film (n = 1.33) suspended is air. What is the minimum non-zero thickness of the film, so that destructive interference causes it to look red in the reflected light.

Diffraction Grating

11. A grating has 767 lines per centimeter. Find the angle of the first principal maxima above the central fringe when illuminated with 655nm light.
12. A grating with 1500 slits per centimeter is illuminated with light of wavelength 500nm. What is the highest order that can be observed in a screen?

Unit 11: Explanation to Questions

1. The fringes in a double slit have the same width. However, in the case of a single slit, the central maximum is twice as wide as the first order maxima.
2. The central bright band will become broader. That is, the diffraction will be more pronounced.
3. At the point of destructive interference, there is no energy. In fact, the energy is not lost. Rather, it is redistributed.
4. Since the wavelength of light decreases in water than in air, the fringes so formed will be closely spaced.
5. White light is not monochromatic, that is single color and frequency. As a result, there will not be interference pattern of alternate bright and dark bands.

Unit 11: Solution to problems

6. $\frac{\lambda}{a}=\frac{\omega}{D}\Rightarrow \omega = 3^{rd}-1^{st}=2\omega=3mm\Rightarrow\omega=1.5mm$

$$a=\frac{\lambda D}{\omega}=\frac{\left(680\times10^{-9}\right)(0.4)}{1.5\times10^{-3}}=181mm.$$

7. a) $\frac{\lambda}{d}=\frac{\omega}{D}\Rightarrow\omega=\frac{\lambda D}{d}=\frac{\left(475\times10^{-9}\right)(2.2)}{0.4\times10^{-3}}=2.6mm.$

b) $$\sin\theta=\frac{m\lambda}{d}=\sin\theta_2=\frac{(2)\left(475\times10^{-9}\right)}{0.4\times10^{-3}}=0.14^{\circ}$$

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8. a) $\frac{\lambda}{d}=\frac{\omega}{D}\Rightarrow\lambda=\frac{\omega d}{D}=\frac{\left(26\times10^{-3}\right)\left(0.25\times10^{-3}\right)}{10}=650\text{nm}$

b) $f=\frac{v}{\lambda}=\frac{3\times10^{8}}{650\times10^{-19}}=4.6\times10^{14}\text{Hz}$

9. $2nt=\left(m+\frac{1}{2}\right)\lambda,\ m=0\Rightarrow t=\frac{\left(0+\frac{1}{2}\right)690}{2\times1.33}=130\text{nm}$

10. Looks red implies, the green is canceled due to destructive interference, hence, $2nt=m\lambda_{\text{green}}\Rightarrow t=\frac{\lambda}{2n}$, $m=1$.

$t=\frac{550\text{nm}}{2(1.33)}=270\text{nm}$

11. $\sin\theta=\frac{m\lambda}{d}=\sin\theta_1=\frac{(1)\left(655\times10^{-9}\right)}{\left(1.3\times10^{-5}\right)}\Rightarrow\theta=2.9°.\ d=\frac{1}{N}$

12. $d=\frac{1}{N}=\frac{1}{1500}\text{cm}=6.7\times10^{-6}\text{m}$

$m\le\frac{d}{\lambda}=\frac{6.7\times10^{-6}}{500\times10^{-9}}=13.4\Rightarrow m=13$

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Unit 12

Electrostatics

Paint spray, which is often used to paint car body and bike frames, chimney filter, which traps dust particles and soot from cement or textile factory, and photocopiers and printers which fix paper and ink all work on the bases of the concept of electrostatics.

Electrostatics: Is the study of charges at rest.

12.1 Electric Charge and Coulomb's Law

All objects are made up from tiny building blocks called *atoms*. An atom in turn consists of elementary particles such as electron, proton and neutron. Protons are positively charged, electrons are negatively charged but neutrons have no net charge. And as to mass, the mass of an electron is 9.1×10^{-31}kg, the mass of proton is 1.67×10^{-27}kg and the mass of neutron is 1.675×10^{-27}kg. This indicates that, proton is 1836 times massive than the electron and neutron is 1838 times massive than the electron.

A proton and an electron have equal and opposite charges which is given by the symbol e.

The SI unit of charge is the coulomb (c), and the value of e has been determined experimentally and found to be.

$$e = 1.6 \times 10^{-19}c$$

Hence, For an electron, $e = -1.6 \times 10^{-19}c$

For a proton, $e = 1.6 \times 10^{-19}c$

- The charge on an electron or a proton is the smallest amount of charge that has been discovered so far and is called the *fundamental* or *basic charge*.

In nature, atoms are electrically neutral since they have equal number of electrons and protons, and the sum of positive and negative charges goes to zero indicating that the atom carries no net charge under normal condition. However, when an atom loses or gains an electron, its electrical balance is disturbed and hence the atom is said to be *charged*.

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☛ An atom is said to be positively charged when it lose electron and negatively charged when it gain electron.

Hence, charges of large magnitude are build up on an object by adding or removing electrons. Thus, the charge on a body is obtained by.

$Q = Ne$ Where N = number of electrons lost or gained

$e = 1.6 \times 10^{-19}c$

Conservation of Charge

Electric charge cannot be created or destroyed, only transferred. The number of positive charges on an atom is the same as the number of negative charge on the atom. The atom is neutral. The charges from an atom can be removed from the atom and transferred to another atom by the processes of charging but the overall number of positive and negative charges does not change. This is the law of conservation of charge. $Q_i = Q_f$

Fundamental law of Electrostatics states that:

"Like charges repel each other".

That is, two positive charges repel each other, and two negative charges repel each other. See?

"Unlike charges attract each other".

That is, positive charge attract negative charge and vice versa.

☛ **Important points to remember:**

- It was Benjamin Franklin who first assigned positive and negative signs to the charges.
- The word electric is derived from the Greek word elektron meaning amber, which is a material charged by rubbing.

Conceptual Example

1. If a body is charged positively, what will happen to its mass? Explain

   Explanation: Its mass decreases. This is because a positive charge is due to loss of electrons, each of which has a small mass.

2. When an object that was neutral becomes charged, does the total charge of the universe change? Explain.

   Explanation: No. When an object becomes charged, it is because of a transfer of charge between it and another object.

Illustrative Examples

3. What is the net charge on a conductor having 20 protons and 12 electrons?

   Solution: $Q = (N_p - N_e)e = (20 - 12)(1.6 \times 10^{-19}) = 1.28 \times 10^{-18}$

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How many electrons are there is one coulomb of negative charge.

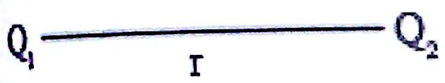
Solution: $Q = Ne \Rightarrow N = \frac{Q}{e} = \frac{-1}{-1.6\times10^{-19}} = 6.25\times10^{18}$ electrons.

12.2 Coulomb's Law

Qualitatively, we know that like charges repel and unlike charges attract each other. Quantitatively, the amount of electrostatic force with which charges attract or repel each other is experimentally discovered by the French physicist Charles Augustine de Coulomb in 1785.

Coulomb's Law is stated as: "The electrostatic force between two charges is directly proportional to the product of the magnitude of the charges, and inversely proportional to the square of the distance between them."

That is, for the charge arrangement shown,



Then, according to coulomb's law,

$$F \sim \frac{|Q_1 \| Q_2|}{r^2}$$

$$\Rightarrow \boxed{F = \frac{1}{4\pi\epsilon_0}\frac{|Q_1 \| Q_2|}{r^2}} \leftarrow \text{coulomb's law}$$

Where $\epsilon_o = 8.85\times10^{-12} C^2/Nm$ and is called *permittivity of free space* and,

$$\frac{1}{4\pi\epsilon_o} = \frac{1}{(4)(3.14)(8.85\times10^{-12})} \simeq 9\times10^9 = k$$

Hence $\boxed{F = \frac{k|Q_1 \| Q_2|}{r^2}}$, $k = 9\times10^9 \frac{Nm^2}{c^2}$ and it is called the *electrostatic constant*.

- **Keep in mind that:**
- The attractive or repulsive force that two point charges exert on one another is equal. That is, from Newton's third law, the force

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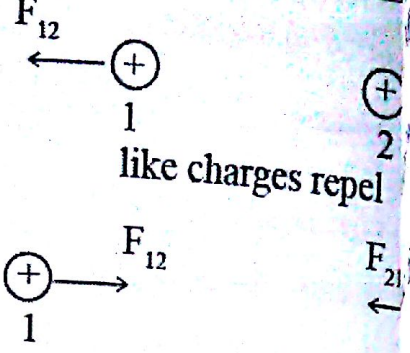
exerted by Q_1 on Q_2 (F_{12}) is equal but opposite to the force exerted by Q_2 on Q_1 (F_{21}). $\boxed{F_{12} = -F_{21}}$



- The resultant electrostatic force due to many point charges is obtained by a vector sum.

Conceptual Example

5. Object A has a charge of 2μc and object B has a charge of 6
Which statement is true?

a) $\vec{F}_{AB} = -3\vec{F}_{BA}$ c) $3\vec{F}_{AB} = -\vec{F}_{AB}$

b) $\vec{F}_{AB} = -\vec{F}_{BA}$ d) $\vec{F}_{AB} = \vec{F}_{BA}$

Explanation: From Newton's third law, $\vec{F}_{AB} = -\vec{F}_{BA}$

6. Why do we experience gravitational force in everyday life bu the electrostatic force?

Explanation: Because, gravitational force is between masses electrostatic force is between charges. And under no condition, that is naturally, we and things around us do pos mass and not net charge, and hence the above case.

Illustrative Examples

7. What is the electrostatic force between two charges of 4μc −6μc placed 2cm apart in vacuum?

Solution: $F = \dfrac{kQ_1Q_2}{r^2} = \dfrac{\left(9\times10^{9}\right)\left(4\times10^{-6}\right)\left(-6\times10^{-6}\right)}{\left(2\times10^{-2}\right)^2} = -540$

The minus sign indicates that the charges attract each other.

8. Compare the electrostatic force and the gravitational fo between an electron and a proton in vacuum.

Solution: Electrostatic force, $F_E = \dfrac{Ke^2}{r^2}$ .....(*)

Gravitational force, $F_G = \dfrac{GM_eM_p}{r^2}$ ..... (**

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Talking ratio of equation (*) to equation (**), we have;

$$\frac{F_E}{F_G} = \frac{Ke^2}{GM_eM_p} = \frac{(9\times10^9)(1.6\times10^{-19})^2}{(6.67\times10^{-11})(9.1\times10^{-31})(1.67\times10^{-27})}$$

$$\frac{F_E}{F_G} \simeq 2\times10^{39}$$

- Electrostatic force is about 2×10^{39} times the gravitational force.

9. A charge of $q = 3\mu c$ is placed half way between $Q_A = 8\mu c$ and $Q_B = 6\mu c$, which are 20cm apart. Calculate the force on q due to Q_A and Q_B.

Solution: Q_A ——— r_A (F_B) — q — r_B (F_A) ——— Q_B; $r_A = r_B = \frac{20}{2} = 10cm^2 = 0.1m$

The force on q due to Q_A and Q_B are assigned to be F_A and F_B and they are repulsive.

$$F_{net} = F_A - F_B = \frac{KQ_Aq}{r_A^2} - \frac{KQ_Bq}{r_B^2} = \frac{Kq}{r^2}(Q_A - Q_B)$$

$$F_{net} = \frac{(9\times10^9)(3\times10^{-6})}{(0.1)^2}\left[8\times10^{-6} - 6\times10^{-6}\right] = 5.4N$$

10. Three point charges are placed on the x-axis at x = 0, x = 10cm and x = 30cm. Calculate the force on Q_3 due to the other two.

Solution: $Q_1 = 4\mu c$, $x = 0$ ——— $Q_2 = -2\mu c$, $x = 10cm$ ——— $Q_3 = 5\mu c$, $x = 30cm$

The force due to Q_1 is , $F_1 = \frac{KQ_1Q_3}{r_1^2}$, which is repulsive and,

The force due to Q_2 is, $F_2 = \frac{KQ_2Q_3}{r_2^2}$, which is attractive.

Thus, the force on Q_3 is,

$$F_{net} = F_1 - F_2 = \frac{KQ_1Q_3}{r_1^2} - \frac{KQ_2Q_3}{r_2^2}, \quad r_1 = 30cm = 0.3m$$

$$r_2 = 30 - 10 = 20cm$$

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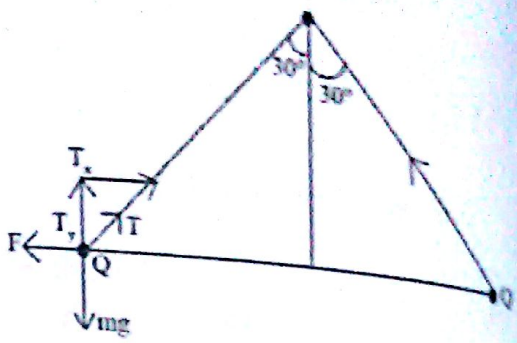
$$(9\times10^{9})(5\times10^{-6})\left[\frac{4\times10^{-6}}{(0.3)^2}-\frac{2\times10^{-6}}{(0.2)^2}\right]=-0.25\text{N}$$

$= 0.25$N, to the left.

11. Two charged identical balls of mass 30g each when suspended on strings make an angle of 30° from the vertical. If the charges on each are the same, calculate the charge on each ball.



Solution: Three forces acting on each ball are the weight (W) downward, the electrostatic repulsive force (F) horizontally, and the tension in the string (T).

Since the system is at rest we can apply the equilibrium condition.

$$\sum F_x = 0 \Rightarrow F - T_x = 0$$

$$F - T\sin30° = 0 \Rightarrow F = 0.5\,T \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(*)$$

$$\sum F_y = 0 \Rightarrow T_y - mg = 0 \ , \cos 30° = 0.86$$

$$T\cos30° - (0.03)\,(10) = 0 \Rightarrow T = 0.35\text{N} \quad \ldots\ldots\ldots(**)$$

From equations (*) and (**),

$F = 0.5T = (0.5)\,(0.35) = 0.715$N

From coulomb's law, $F = \dfrac{KQ\bullet Q}{r^2}$.

$$\Rightarrow 0.715 = \frac{(9\times10^{9})\ (Q^2)}{(0.4)^2}$$

Solving the above, one obtains, $Q = 3.6\mu$c.

12. Two point charges Q_1 and Q_2 separated by a distance of 90cm exerts a force of 0.05N on each other. If $Q_1 + Q_2 = 4.5\mu$c, find the magnitude of Q_1 and Q_2

Solution: $F = \dfrac{KQ_1Q_2}{r^2}$, but $Q_1 + Q_2 = 4.5 \Rightarrow Q_1 = 4.5 - Q_2$

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$$\Rightarrow 0.5 = \frac{(9\times10^{9})(4.5\times10^{-6} - Q_2)(Q_2)}{(0.9)^2}$$

$$4.5\times10^{-12} = 4.5\times10^{-6}Q_2 - Q_2^{\,2}$$

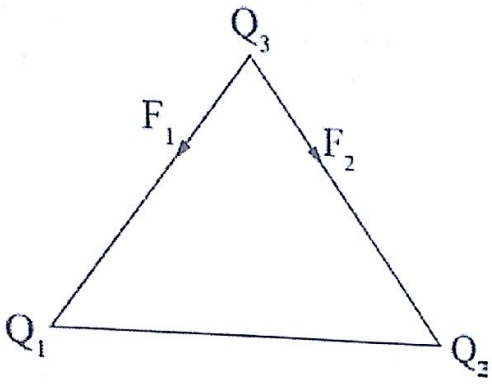
$$\Rightarrow Q_2^{\,2} - 4.5\times10^{-6}Q_2 + 4.5\times10^{-12} = 0$$

Using the general quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$Q_2 = 3\mu c$ or $1.5\mu c$

Hence, $Q_1 = 1.5\mu c$ or $3\mu c$

13. Charges of $Q_1 = 2\mu c$, $Q_2 = 3\mu c$ and $Q_3 = -8\mu c$ are placed at the vertices of an equilateral triangle of side 10cm. Calculate the magnitude of the force acting on Q_3 due to the other two charges.



Solution: Let the force on Q_3 due to Q_1 be F_1.

$$\Rightarrow F_1 = \frac{KQ_1Q_3}{r^2} = \frac{(9\times10^{9})(2\times10^{-6})(8\times10^{-6})}{(0.1)^2} = 14.4N$$

and the force on Q_3 due to Q_2 is F_2

$$\Rightarrow F_2 = \frac{KQ_2Q_3}{r^2} = \frac{(9\times10^{9})(3\times10^{-6})(8\times10^{-6})}{(0.1)^2} = 21.6N$$

now, using the cosine law,

$$F = \sqrt{F_1^{\,2} + F_2^{\,2} + 2F_1F_2\cos\theta}$$

$$= \sqrt{(14.4)^2(21.6)^2 + 2(14.6)(21.6)\cos60^\circ} = 31.4N$$

14. An equal number of electrons is placed on two metal spheres 4mm apart. How many electrons are there on each sphere if the repulsive force is 400N?

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Solution: $F = \frac{KQ^2}{r^2}, Q_1 = Q_2 = Q$

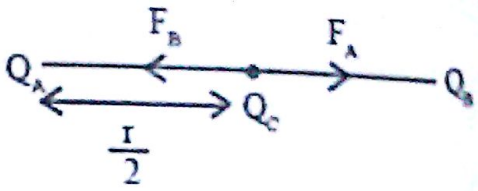
$$\Rightarrow 400 = \frac{(9\times10^9)(Q^2)}{(0.004)^2} \Rightarrow Q = 843\times10^{-9}c$$

but, $Q = ne \Rightarrow n = \frac{Q}{e} = \frac{843\times10^{-9}}{1.6\times10^{-19}} = 5.27\times10^{12}$ electrons

15. Two equally charged metal spheres A and B repel each other with a force of $2\times10^{-5}N$. If uncharged sphere C is touched to A and then placed on the midpoint between A and B, what is the net force on C?

Solution: $F_{AB} = \frac{KQ^2}{r^2} = 2\times10^5 N$

the charge on sphere A and C after touching is,



$$Q_A = Q_C = \frac{O+Q}{2} = \frac{Q}{2}$$

and, the charges arrangement is then,

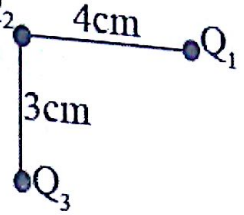
The force on C due to A is, $F_A = \frac{KQ_AQ_C}{r_A^2} = \frac{K\left(\frac{Q}{2}\right)\left(\frac{Q}{2}\right)}{\left(\frac{r}{2}\right)^2}$

$$F_A = \frac{KQ^2}{r^2} \quad \ldots\ldots\ldots\

$$F = F_B - F_A = \frac{2KQ^2}{r^2} - \frac{KQ^2}{r^2} = \frac{KQ^2}{r^2}$$

$$\therefore F = 2\times10^5 N$$

16. Determine the net force on Q_2, given that $Q_1 = -3\mu c$, $Q_2 = 6\mu c$ and $Q_3 = 1.5\mu c$.



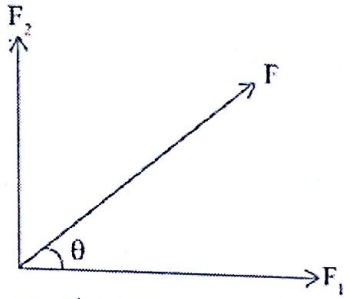
Solution: Let the force on Q_2 due to Q_1 is F_1

$$F_1 = \frac{KQ_1Q_2}{r_1^2} = \frac{(9\times10^9)(3\times10^{-6})(6\times10^{-6})}{(4\times10^{-2})^2} = 1\times10^2 N$$

and, the force on Q_2 due to Q_3 is F_2

$$F_2 = \frac{KQ_3Q_1}{r_2^2} = \frac{(9\times10^9)(1.5\times10^{-6})(6\times10^{-6})}{(3\times10^{-2})^2}$$

$$= 9\times10^1 N$$



The net force F is then obtained by Pythagoras theorem, as

$$F = \sqrt{F_1^2 + F_2^2} = \sqrt{(1\times10^2)^2 + (9\times10^1)^2} = 130N$$

And, its direction is, $\tan\theta = \frac{F_2}{F_1} = \frac{90}{100} \Rightarrow \theta = 42°$ above x-axis.

17. Two negative point charges are 2m apart and repel each other with a force of 2N. When the distance between the charges is doubled, the force between them is. (UEE: 2002).

A) One fourth as great
B) One half as great
C) twice as great
D) four times as great

Explanation: $F_1 = \frac{KQ_1Q_2}{r_1^2}$

$$F_2 = \frac{KQ_1Q_2}{r_2^2}, \text{ when } r_2 = 2r_1$$

$$F_2 = \frac{KQ_1Q_2}{(2r_1)^2} = \frac{KQ_1Q_2}{4r_1^2} = \frac{1}{4}\left(\frac{KQ_1Q_2}{r_1^2}\right) = \frac{1}{4}F_1$$

Answer: A

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18. What is the magnitude of the electric force between two protons separated by 2×10^{-15}m. (UEE: 2004).

A) 67.5N B) 115.0N C) 57.5N D) 83.4N

Solution:

$$F = \frac{KQ_1Q_2}{r^2} = \frac{(9\times10^{9})(1.6\times10^{-19})(1.6\times10^{-19})}{(2\times10^{-15})^2} = 57.5N$$

Answer: C

12.3 Electric Field

Electric Field: Is a region around a charged object where another charged object will experience a force.

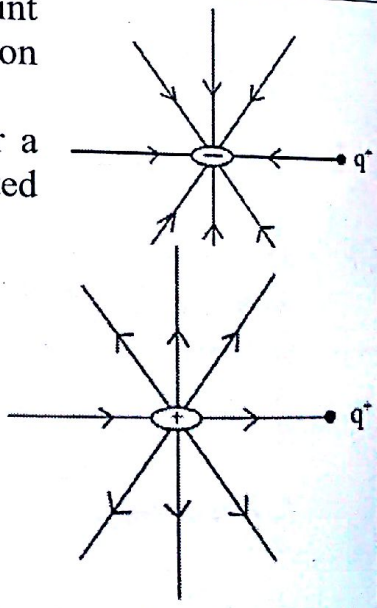
Imaginary lines that are used to visualize an electric field were introduced by Michael Faraday and are called *field lines* or *lines of force.*

Electric Field Lines: Are imaginary lines drawn in such a way that their direction at any point is the same as the direction of the force on a test charge placed at that point.

- The direction of the electric field at a point is the same as the direction of the force on a test charge placed at that point.

For example, if a test charge is placed near a negative point charge, it will be attracted inward. Hence,

- The electric field near a negative point charge is directed radially into the point charge. With the same reasoning, if a test charge is placed near a positive point charge, it is repelled outward. Thus,
- The electric field near a positive point charge is directed radially away from the charge.



away from the charge

Characteristics of Field Lines

- Field lines never cross each other.
- The space between field lines represents the strength of the field Hence, the closer the lines are together the stronger is the field.

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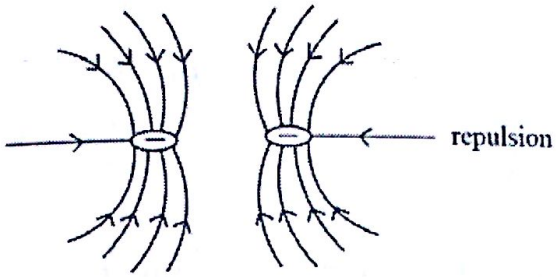
- It is possible for two electric fields to cancel each other and create a neutral point.

Test charge (q^+): Is a very small positive charge that do not exert a force on a nearby charge.

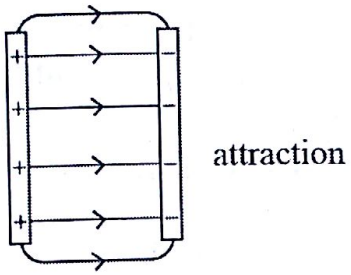
Conceptual Examples

19. Draw the electric field lines between
    a) two negative point charges
    b) two oppositely charged parallel plates

Explanation: a)



b)



12.3 Electric Field Strength

At a point within an electric field, there is a field intensity or a field strength.

Electric Field Strength: Is the force per unit positive charge acting on a test charge placed in the field.

Mathematically, this may be expressed as,

$$E = \frac{F}{Q^+}$$

Electric field strength is a vector quantity and its unit is Newton/Coulomb, N/c.

Assuming that a point charge Q and a test charge q^+ are a distance r apart as shown then we have,

$$F = \frac{KQq^+}{r^2} \text{ and } E = \frac{F}{q^+}$$

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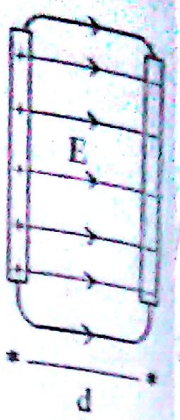
$$E = \frac{KQ}{r^2}$$ ← electric field due to a point charge (variable field).

The electric field strength between oppositely charged parallel plates of separation d and plates potential difference v is obtained as follows.



$$E = \frac{V}{d}$$ ← field between parallel plate (uniform field).

Important points to remember

- A positive charge experiences a force in the direction of the field.
- A negative charge experiences a force in the opposite direction to the field.
- The magnitude of the force acting on charge Q is $F = E|Q|$.
- The net electric field at a point due to many point charges is obtained by a vector sum of each field at that point, that is based on the *superposition principle*.

Illustrative Examples

20. A charge of 2μc placed at point p in an electric field experiences a force of 8×10^{-4} N. What is the electric field intensity at that point?

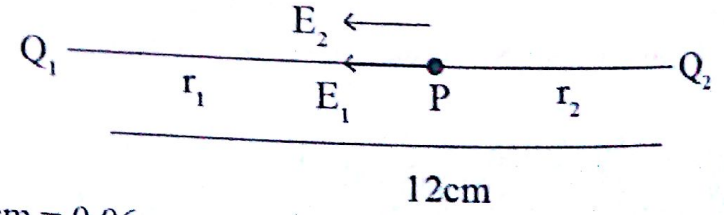
Solution: $E = \frac{F}{Q} = \frac{8\times10^{-4}}{2\times10^{-6}} = 400 N/c$

21. Find the electric field strength at a distance of 30cm from a point charge of 2μc?

Solution: $E = \frac{KQ}{r^2} = \left(9\times10^{9}\right)\frac{\left(2\times10^{-6}\right)}{\left(0.3\right)^2} = 2\times10^{5} N/c$

22. What is the electric field at the mid-point of the two charges $Q_1 = -2\mu c$ and $Q_2 = 2\mu c$.

Solution:



$r_1 = r_2 = 6cm = 0.06m$

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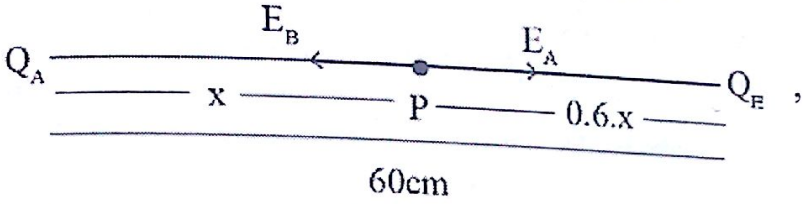
$$E_P = E_1 + E_2 = \frac{KQ_1}{r_1^2} + \frac{KQ_2}{r_2^2} = \frac{2KQ}{r^2}$$

$$E_P = \frac{2\left(9\times10^9\right)\left(2\times10^{-6}\right)}{\left(0.06\right)^2} = 1\times10^7 \text{ N/c}$$

23. Two point charges $Q_A = 2\mu c$ and $Q_B = 8\mu c$ are placed 60cm apart. At what distance from Q_A will the field be zero.

Solution:



Let the field be zero at a distance x from Q_A

$$E_P = E_A + E_B = 0$$

$$\Rightarrow E_A = E_B$$

$$\frac{KQ_A}{x^2} = \frac{KQ_B}{\left(0.6-x\right)^2} = \frac{2}{x^2} = \frac{8}{\left(0.6-x\right)^2}$$

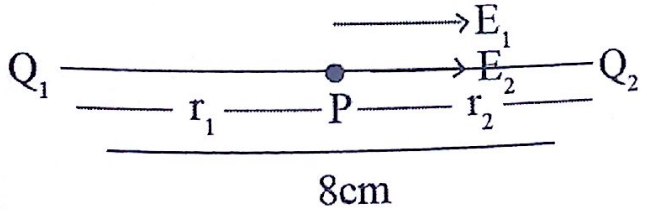
$$\Rightarrow \frac{1}{x^2} = \frac{4}{0.36-1.2x+x^2} \Rightarrow 3x^2 + 1.2x - 0.36 = 0$$

Using the general quadratic formula, $x = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$

$$x = 0.2\text{m} = 20\text{cm}.$$

24. Two point charges of equal magnitude are 8cm apart. At the midpoint of the line connecting them, the net electric field is 45N/c. Find the magnitude of the charges.

Solution:



$$E_P = E_1 + E_2$$

$$E_P = \frac{KQ_1}{r_1^2} + \frac{KQ_1}{r_2^2}, \; r_1 = r_2 = 4\text{cm} = 0.04\text{m}$$

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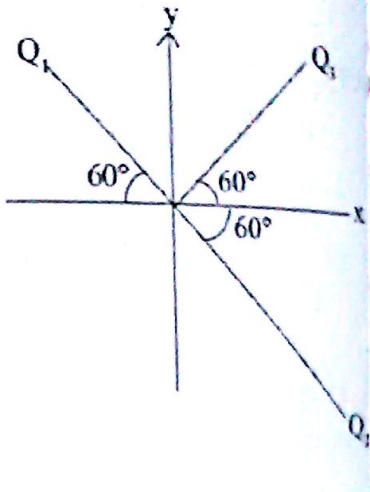
$$E_P = \frac{2KQ}{r^2} = \frac{2\left(9\times10^9\right)(Q)}{(0.04)^2}$$

$$\Rightarrow 45 = 1.125\times10^{13}Q \Rightarrow Q = 4\times10^{-19}c$$

$$Q_1 = Q_2 = 4\times10^{-9}c = 4nc$$

25. Three point charges $Q_1 = 3Q$, $Q_2 = -3Q$ and $Q_3 = -6Q$ are each at a distance of d from the origin. What is the net electric field at the origin.



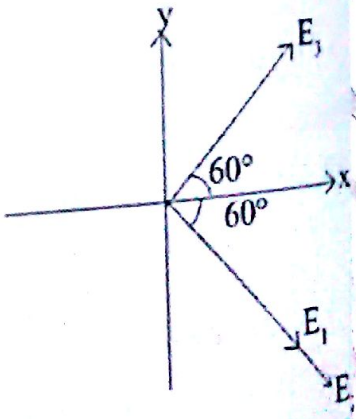
Solution: $\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$

$$E_1 = \frac{KQ_1}{r_1^2} = \frac{3KQ}{d^2}, \text{ to the origin}$$

$$E_2 = \frac{KQ_2}{r_2^2} = \frac{3KQ}{d^2}, \text{ towards } Q_2.$$

$$E_3 = \frac{KQ_3}{r_3^2} = \frac{6KQ}{d^2}, \text{ towards } Q_3.$$

So, we have



Now, resolving E_1, E_2 and E_3 into x and y components.

<u>x – components</u>

$E_{1x} = E_1\cos60°$, $E_{2x} = E_1\cos60°$ and $E_{3x} = E_3\cos60°$

$E_x = E_{1x} + E_{2x} + E_{3x}$

$$= \frac{3KQ}{d^2}\left(\frac{1}{2}\right) + \frac{3KQ}{d^2}\left(\frac{1}{2}\right) + \frac{6KQ}{d^2}\left(\frac{1}{2}\right)$$

$$E_x = \frac{6KQ}{d^2}$$

<u>y-components</u>

$E_{1y} = E_1\sin60°$, $E_{2y} = E_2\sin60°$ and $E_{3y} = E_3\sin60°$

$E_y = E_{1y} + E_{2y} + E_{3y}$

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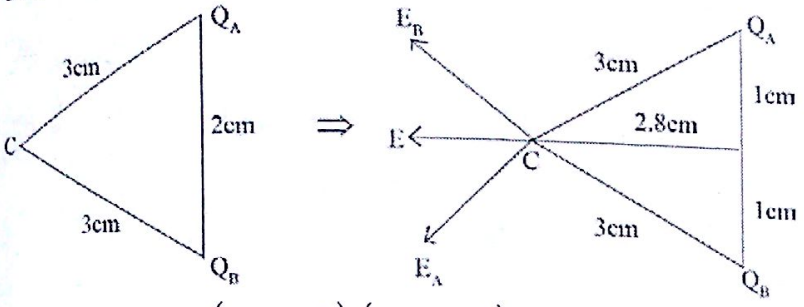
$$= \frac{3KQ}{d^2}(\sin 60°) + \frac{3KQ}{d^2}(\sin 60°) - \frac{6KQ}{d^2}(\sin 60°)$$

$$E_y = \frac{6KQ}{d^2}\ \sin 60° - \frac{6KQ}{d^2}\ \sin 60° = 0$$

$$\therefore \vec{E}_{net} = \vec{E}_x + \vec{E}_y = \frac{6KQ}{d^2}$$

26. If $Q_A = 5\mu c$, $Q_B = 1\mu c$, what is the net field at point C

Solution:



$$E_A = \frac{KQ_A}{r_A^{\,2}} = \frac{(9\times10^9)(5\times10^{-6})}{(0.03)^2} = 5\times10^5 N/c$$

$$E_B = \frac{KQ_B}{r_B} = \frac{(9\times10^9)(1\times10^{-6})}{(0.03)^2} = 1\times10^7 N/c$$

<u>x – components</u>

$$E_x = E_A\cos\theta + E_B\cos\theta,\ \cos\theta,\ = \frac{2.8}{3}$$

$$E_x = 5.6\times10^7 N/c$$

<u>y – components</u>

$$E_y = E_A\sin\theta + E_B\sin\theta,\ \sin\theta = \frac{1}{3}$$

$$E_y = 1.4\times10^7 N/C$$

$$E = \sqrt{E_x^{\,2} + E_y^{\,2}} = \left[(5.6\times10^7)^2 + (1.4\times10^7)^2\right]^{\frac{1}{2}} = 5.8\times10^7 N/c$$

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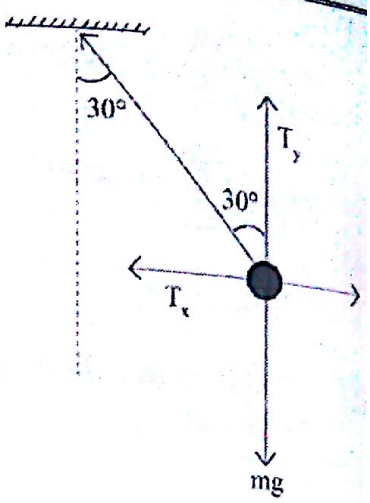
27. An electreic field of 800N/C causes a charged ball of 50g mass to make an angle of 30° from the vertical. What is the charge on the ball?



Solution: The system is in equilibrium

$$\sum F_x = 0 \Rightarrow F - T_x = 0$$

$$F - T\sin 30 = 0$$

$$F = 0.5T \quad \ldots\ldots\ldots\ldots\ldots\ldots (*)$$

$$\sum F_y = 0 \Rightarrow T_y - mg = 0$$

$$T\cos 30 - (0.05)(10) = 0,\ \cos 30 = 0.86 = 0.6\text{N}$$

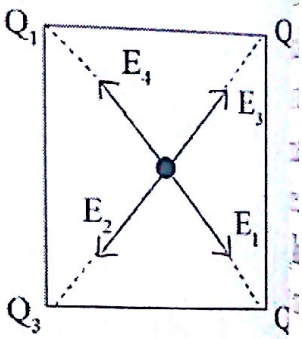
Substituting this into equation (*), we have,

$$F = 0.5T = (0.5)(0.6) = 0.3\text{N}$$

Also, from the fact that,

$$E = \frac{F}{Q} \Rightarrow Q = \frac{F}{E} = \frac{0.3}{800} = 3.8\times10^{-4}\text{C}$$

28. Four point charges of each 4μC are placed at the four corners of a square of side 20cm. Find the electric field intensity at the centre of the square.



Explanation: $E = E_1 + E_2 + E_3 + E_4$

As can be seen, $E_1 = -E_4$ and $E_2 = -E_3$

Thus, $E = 0$.

29. An oil drop of 3.2×10^{-13}N weight is observed to be balanced an electric field of 5×10^{5} N/C. What is the charge of the drop?

Solution: $F = W \Rightarrow EQ = mg$

$$\Rightarrow Q = \frac{mg}{E} = \frac{3.2\times10^{-13}}{5\times10^{5}} = 6.4\times10^{-19}\text{C}$$

30. In the Millikan experiment, an oil drop carries four electron charges of mass 1.8×10^{-12}g. If the drop is stopped between plates of separation 1.8cm, what voltage must there be between the plates?

Solution: $F = mg \Rightarrow EQ = mg$, but, $E = \frac{V}{d}$

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$$\Rightarrow \frac{v}{d}Q = mg \Rightarrow v = \frac{mgd}{Q} = \frac{(1.8\times10^{-12})(10)(1.8\times10^{-2})}{4\times1.6\times10^{-19}} = 506v$$

12.4 Motion of Charged Particles in a Uniform Electric Field

When a particle of charge Q and mass m is placed in an electric field E, the electric force exerted on the charge is given by,

$\boxed{F = EQ}$, and this force causes the particle to accelerate.

And, from Newton's second law

$\boxed{F = ma}$, from the two equations, we can have

$ma = EQ$

$\Rightarrow \boxed{a = \frac{EQ}{m}}$, and from equations of motion

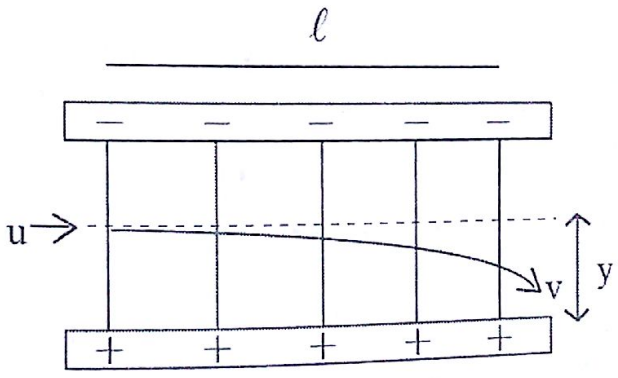
For a uniformly accelerated motion, we can calculate the time (t), speed (v) and distance (s) of the particle.

Important points to remember

- If the electric field is uniform, the acceleration of the particle becomes constant.
- If the particle has a positive charge, its acceleration is in the direction of the electric field.
- If the particle has a negative charge, its acceleration is in the opposite direction to the field.

Illustrative Examples

31. An electron enters the region of a uniform electric field of 200N/C with a speed of 3×10^6m/s. If the length of the plates is 0.1m. Calculate.
    a) the acceleration
    b) the time to travel through the field
    c) the verticle displacement
    d) the speed as it emerges



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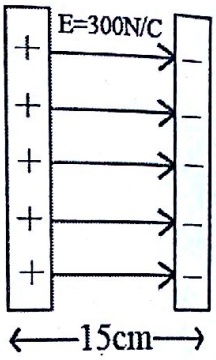
Solution: a) $a = \frac{EQ}{m} = \frac{(200)(-1.6\times10^{-19})}{9.1\times10^{-31}} = -3.51\times10^{13}\,m/s^2$

b) $t = \frac{\ell}{u} = \frac{0.1}{3\times10^{6}} = 3.3\times10^{-8}\,sec$

c) $y = \frac{1}{2}at^2 = \frac{1}{2}(3.51\times10^{13})(3.3\times10^{-8})^2 = 1.95cm$

d) $v = \sqrt{u^2+(at)^2} = 3.22\times10^{6}\,m/s$

32. A proton is shot with speed 2×10^5m/s towards the negative plate from the positive plate. What is its speed just before hitting the negative plate.



Solution: $a = \frac{EQ}{m} = \frac{(300)(1.6\times10^{-19})}{(1.67\times10^{-27})}$

$a = 2.9 \times 10^{11}\,m/s^2$

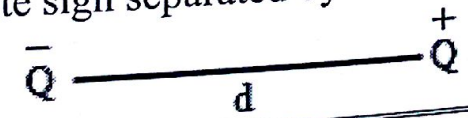
Now, using $v^2 = u^2 + 2as$

$v = \sqrt{u^2 + 2as}$

$= \left((2\times10^5)^2 + 2(2.9\times10^{11})(0.15)\right)^{\frac{1}{2}}$

$v = 3.55 \times 10^5\,m/s$

Electric Dipole: Is a system that consists of two point charges of equal magnitude and opposite sign separated by a small distance.



And, the product of the charge Q and the separation d is called *dipole moment*. And is given by, $\boxed{P = Qd}$

☛ **Remember:** Dipole moment is a vector quality and its direction is from the negative charge towards the positive charge.

Conceptual Examples

33. Consider an electric dipole in a uniform electric field what is the net?

a) Force on the dipole?

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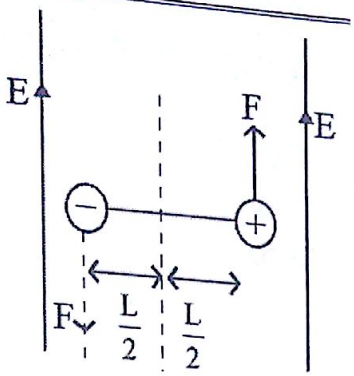
b) Torque developed?



Explanation: a) The negative charge exerts force opposite to the field and the positive charge exerts force in the direction of the field. The two charges experience forces of equal magnitude, F = QE, but opposite directions. Therefore the net force is zero.

b) $\tau = \tau_+ + \tau_- = F\left(\frac{L}{2}\right) + F\left(\frac{L}{2}\right)$

$$\tau = QE\left(\frac{L}{2}\right) + QE\left(\frac{L}{2}\right) = QEL$$

12.5 Electric Flux and Gauss's Law

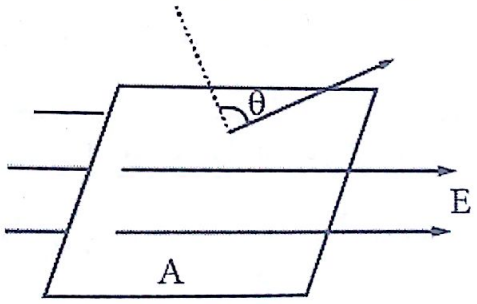
Electric Flux

The word "flux" comes from the Latin word "fluere" which means to flow. Electric flux is then the measure of the number of field lines flowing through a surface.

Electric Flux: Is the number of field lines crossing a given surface.

That is, flux = (electric field) (perpendicular area component)

$$\phi = E \cdot A = EA\cos\theta$$



Flux (ϕ, phi), is a scalar quantity and its unit is Nm^2/C.

In the equation, $\phi = EA\cos\theta$, θ is the angle between a normal line to the area and the electric field.

☛ **Note that:** Flux is maximum when the electric field is perpendicular to the area, that is when $\theta = 0°$.

Gauss's Law

This law relates the electric flux through a closed surface and the total charge inside that surface and was developed by the German Mathematician Karl Friedrich Gauss.

Gauss's law states that: "The net electric flux through any closed surface is equal to the net charge inside the surface divided by the permittivity of free space".

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$\sigma_+ = 3.4\frac{\mu C}{m^2}$ and $\sigma_- = 2.3\frac{\mu C}{m^2}$. Find the electric field

a) at point L
b) at point B
c) at point R

Solution: The electric field due to each plate is,

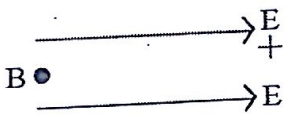
$$E_+ = \frac{1}{2}\frac{\sigma_+}{\epsilon_o} = \frac{3.4\times10^{-6}}{(2)(8.85\times10^{-12})} = 1.92\times10^5 N/C$$

and, $$E_- = \frac{1}{2}\frac{\sigma_-}{\epsilon_o} = \frac{2.3\times10^{-6}}{(2)(8.85\times10^{-12})} = 1.3\times10^5 N/C$$

The resultant field at the three points is

a) at point L,

$$E_L = E_+ - E_- = (1.92-1.3)10^5 = 0.62\times10^5 N/C \text{ to the left}$$

b) at point B 

$$E_B = E_+ + E_- = (1.92\times10^5)+(1.3\times10^5) = 3.22\times10^5 N/C$$

c) at point R,

$$E_R = E_+ - E_- = (1.92-1.3)10^5 = 0.62\times10^5 N/C \text{ the the right}$$

43. What is the SI unit of electric filed (UEE, 2001)
A) Am^{-1} B) Vm^{-1} C) NC^{-1} D) Cn

Explanation: $\vec{E} = \frac{\vec{F}}{Q} \Rightarrow$ unit of $\vec{E} = \frac{N}{C} = NC^{-1}$ **Answer:** (

44. Consider a uniformly charged insulating spherical shell of in radius R_1 and outer radius R_2. What is the electric field at $r >$ and $r < R$ respecctively. (UEE, 2001)

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Mathematically, Gauss's law is given by,

$$\boxed{\phi_{net} = \frac{Q_{net}}{\epsilon_o}} \leftarrow \text{Gauss's law}$$

And, relating flux and charge, Gauss's law becomes

$$\boxed{E.A = \frac{Q}{\epsilon_o}}$$

Where the symbol ($\oint$) represents integral taken over any closed surfac that encloses the charge.

Specifically, the electric field may be computed as,

$$\boxed{E = \frac{Q}{A\,\epsilon_o}}$$

And, for a point charge, the electric field at a distance r from the charg is assumed to act in all directions covering a spherical area, $A = 4\pi r^2$ Hence,

$$\boxed{E = \frac{Q}{4\pi r^2\,\epsilon_o} = \frac{1}{4\pi\,\epsilon_o}\frac{Q}{r^2}}$$

Moreover, the electric field strength can be expressed, using the concep of charge density.

Charge density: Is charge per unit area.

That is, $\boxed{\sigma = \frac{Q}{A}} \leftarrow \sigma$ (rho) charge density

Therefore, the electric field can be written as,

$$E = \frac{Q}{A}\left(\frac{1}{\epsilon_o}\right) \Rightarrow \boxed{E = \frac{\sigma}{\epsilon_o}}$$

☛ **Distribution of Charges on a Charged Conductor**

- The net electric charge of a conductor resides entirely on its surface: This is due to the repulsion of like charges.
- The electric field inside the conductor is zero: If there were to be any charge inside, this would cause a net force and acceleration.

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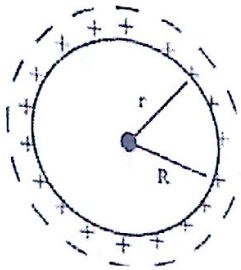
- The electric field at the surface of the conductor is perpendicular to that surface. If the field were to act at an angle then there would be a horizontal component of the force and this would again cause the charges to move.

The electric field due to a charged spherical conductor.

Consider a charged spherical conductor of radius R. Then, the electric field at a distance r where r > R is given by $\boxed{E = K\frac{Q}{r^2}}$



Important points to remember

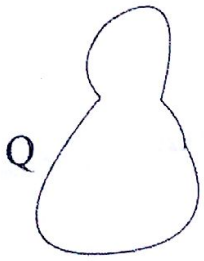
- The imaginary surface, such as sphere, cylinder…etc. on which the charges resides on is called *Gaussian Surface*.
- If there is no charge on the gaussian surface, flux is zero.
- If there is equal and opposite charges on the surface, the flux is zero.
- If the filed lines do not cross the given area, then the flux is zero.
- Zero flux does not mean zero electric field.
- The electric field is zero inside a charged conductor.
- The electric field strength is largest at sharp coners.
- The flux is positive for a positve charge and tells that the field lines leave the enclosed surface.
- The flux is negative for a negative charge and tells that the field lines enter the enclosed surface.
- Permitivity of free space, is a measure of the ease of an electric field to pass through vacuum.

Conceptual Examples

34. If more electric field lines are leaving a Gaussian surface than entering, what can you conclude about the net charge by that surface? Explain

Explanation: The net enclosed charge must be positive.

35. The electric field by a point charge Q outside a Gaussian surface is passed through the surface what is the net flux through the surface.



Explanation: The charge is not enclosed by the surface. Hence the net flux is zero.

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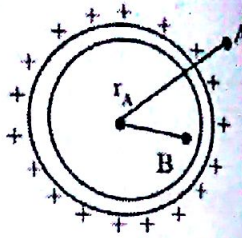
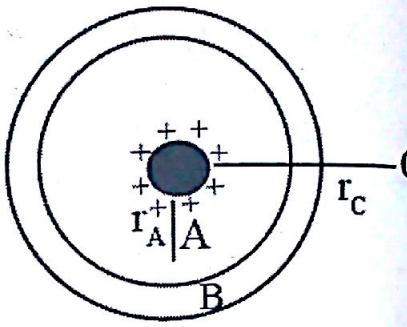
36. Find the electric field at points A and B with regard to charged spherical shell shown.



Explanation: At point B, there is no charge hence the electric field at point B is zero. $E_B = 0$. But, at point A, the electric field is,

$$E_A = \frac{KQ}{r_A^2}.$$

37. Small sphere carrying charge Q is located at the center of a spherical cavity in a large uncharged metal sphere. Determine the electric field.



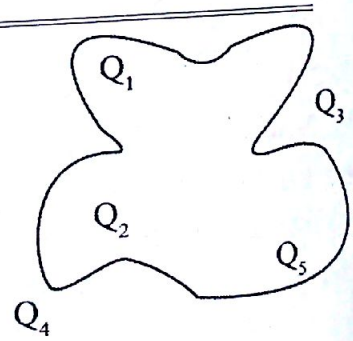
a) at point A
b) at point B
c) at point C

Explanation: a) $E_A = K\frac{Q}{r_A^2}$ b) $E_B = 0$, no charge at B

c) $E_C = K\frac{Q}{r_C^2}$

Illustrative Examples

38. What is the net electric fiux through the Gaussian surface if $Q_1 = Q_3 = 1.5\mu C$ and $Q_2 = Q_4 = -3.2\mu C$ and $Q_5 = -2\mu C$.



Solution: $\phi_{net} = \frac{Q_{net}}{\epsilon_o}$

$$= \frac{Q_1 + Q_2 + Q_5}{\epsilon_o}$$

$$\phi_{net} = \frac{(1.5 - 3.2 - 2)10^{-6}}{8.85 \times 10^{-12}} = 4.2 \times 10^5 Nm/C^2$$

39. A point charge causes an electric flux of $-750 Nm^2/C$ to pass through a sperical Gaussian surface of radius 10cm what is the value of the

a) point charge b) electric field

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Solution: a) $\phi = \frac{Q}{\epsilon_o} \Rightarrow Q = \phi \epsilon_o = (-750)(8.85\times10^{-12}) = -6.6\text{nC}$

b) $E = \frac{Q}{A\epsilon_o} = \frac{Q}{4\pi\epsilon_o r^2} = \frac{\phi}{4\pi r^2} = \frac{750}{4\pi(0.1)^2} = -6000\text{N/C}$

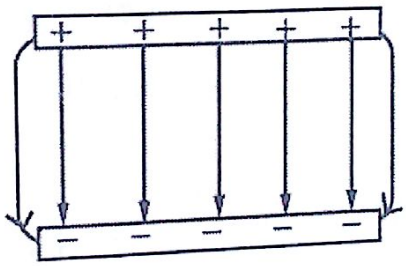
40. A surface of area 4m^2 is placed in a uniform electric field of 620N/C. Calculate the flux when the electric field is
   a) Perpendicular to the surface
   b) Parallel to the surface
   c) 60° to the normal

Solution: a) $\phi = EA = (620)(4) = 2.48\times10^3\text{Nm}^2/\text{C}$

b) $\phi = 0$

c) $\phi = EA\cos\theta = (620)(4)(\cos60°) = 1.24\times10^3\text{Nm}^2/\text{C}$

41. Suppose two parallel oppositely charged metal plates are separated by a distance of 8cm and charged with a battery of 120v. Calculate
   a) The electric field between the plates
   b) The charge density of each plate



Solution: a) $E = \frac{v}{d} = \frac{120}{0.08} = 1500\text{v/m}$

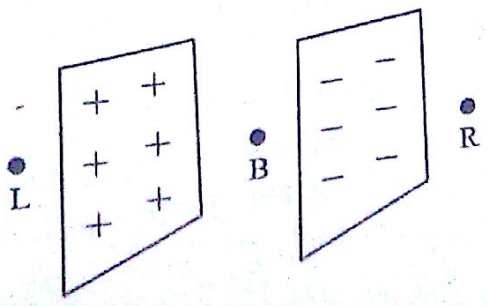
b) • The charge density of the positive plate

$$E = \frac{1}{2}\frac{\sigma}{\epsilon_o} \Rightarrow \sigma_+ = 2E\epsilon_o = 2.6\times10^{-8}\ \text{c/m}^2$$

• The charge density of the negative plate

$$E = \frac{1}{2}\frac{\sigma}{\epsilon_o} \Rightarrow \sigma_- = 2E\epsilon_o = -2.6\times10^{-8}\ \text{c/m}^2$$

42. Two large parallel non conducting sheet each with a surface charge density of



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$\sigma_+ = 3.4\frac{\mu C}{m^2}$ and $\sigma_- = 2.3\frac{\mu C}{m^2}$. Find the electric field

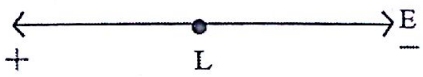
a) at point L
b) at point B
c) at point R

Solution: The electric field due to each plate is,

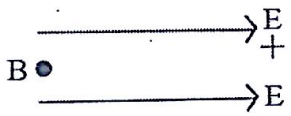
$$E_+ = \frac{1}{2}\frac{\sigma_+}{\epsilon_o} = \frac{3.4\times10^{-6}}{(2)(8.85\times10^{-12})} = 1.92\times10^5 N/C$$

and, $$E_- = \frac{1}{2}\frac{\sigma_-}{\epsilon_o} = \frac{2.3\times10^{-6}}{(2)(8.85\times10^{-12})} = 1.3\times10^5 N/C$$

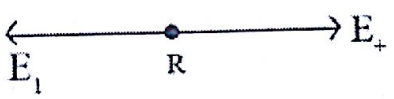
The resultant field at the three points is

a) at point L, 

$$E_L = E_+ - E_- = (1.92-1.3)10^5 = 0.62\times10^5 N/C \text{ to the left}$$

b) at point B 

$$E_B = E_+ + E_- = (1.92\times10^5) + (1.3\times10^5) = 3.22\times10^5 N/C$$

c) at point R, 

$$E_R = E_+ - E_- = (1.92-1.3)10^5 = 0.62\times10^5 N/C \text{ the the right}$$

43. What is the SI unit of electric filed (UEE, 2001)
A) Am^{-1} B) Vm^{-1} C) NC^{-1} D) Cn

Explanation: $\vec{E} = \frac{\vec{F}}{Q} \Rightarrow$ unit of $\vec{E} = \frac{N}{C} = NC^{-1}$ **Answer:**

44. Consider a uniformly charged insulating spherical shell of in radius R_1 and outer radius R_2. What is the electric field at r > and r < R respecctively. (UEE, 2001)

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A) $\frac{q}{4\pi \in_o r^2}, 0$

B) $\frac{q}{4\pi \in_o R_2^{\,2}}, 0$

C) $\frac{q}{4\pi \in_o R^2}, 0$

D) $\frac{q}{4\pi \in_o \left(R_2^{\,2} - R_1^{\,2}\right)}, 0$

Solution: at $r < R_1$, no charge, thence $E = 0$

$$\text{at } r > R,\ E = \frac{q}{4\pi \in_o r^2}$$

Answer: A

45. Consider a uniformly charged conducting spherical shall of inner radius R_1 and outer radius R_2 carrying a total of negative charge q. A point positive charge of q is placed at the center of the shell. What are the electric field at $r < R_1$, $R_1 < r < R_2$ and $r > R_2$ respectively. (UEE, 2001)

A) $\frac{q}{4\pi \in_o R_2^{\,2}},\ 0,\ \frac{q}{4\pi \in_o \left(R_2^{\,2} - R_1^{\,2}\right)}$

B) $\frac{q}{4\pi \in_o R_2^{\,2}},\ 0,\ \frac{q}{4\pi \in_o r^2}$

C) $\frac{q}{4\pi \in_o r^2},\ 0,\ 0$

D) $\frac{q}{4\pi \in_o \left(R_2^{\,2} - R_1^{\,2}\right)},\ 0,\ 0$

Solution: The electric field at $r < R_1$ is $E = \frac{q}{4\pi \in_o r^2}$

But at $R_1 < r < R_2$ and $r > R_2$, the field is zero.

Answer: C

46. Which of the following is NOT property of a charged conductor (UEE, 2002)

A) The electric filed is zero everywhere inside the conductor

B) Any net charge on the conductor resides entirely on its surface

C) On an irregularly shaped conductor, the surface charge density is greatest where the radius of curvature is smallest

D) The electric field outside the conductor is tangent to its surface.

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Explanation: The electric field outside the conductor is perpendicular to the surface of the conductor. **Answer: D**

47. Consider a long cylinderical charge distribution of radius R with uniform change density ρ. What is the magnitude of the electric field at distance r from the axis, where r < R (UEE, 2002).

A) $\frac{\rho r}{3\epsilon_o}$ B) $\frac{\rho r}{2\epsilon_o}$ C) $\frac{\rho r}{7\epsilon_o}$ D) $\frac{\rho r}{5\epsilon_o}$

Solution: $E = \frac{Q_{in}}{A\epsilon_o}$, $Q_{in} = \rho\pi r^2\ell$, $A = 2\pi r\ell$

$$E = \frac{\rho r}{2\epsilon_o}$$

Answer: B

48. One of the following statements is NOT characteristics of electric field (UEE 2003)

A) an electric field is the region around a charged body where another charged body experience a force

B) lines of electric field have a direction which move away from negative towards positive

C) the direction of electric field is in the direction of the force acting on a small positive charge

D) lines of electric flux do not cross over each other.

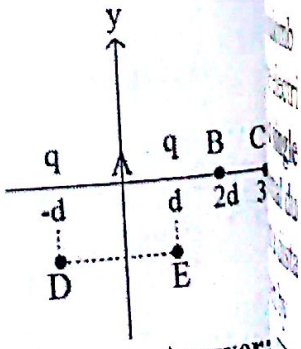
Explanation: Electric field lines run from positive to negative point charge.

Answer:

49. Two positive charges of magnitude q are each a distance d from the origin A of a coordinate system as shown below. The magnitude of the electric field is minimum at point (UEE, 2006)

A) A B) B C) C D) D

Explanation: at A, E = 0



Answer:

50. Which one of the following statements is correct (UEE, 2007)

A) The electric field strength due to two point charges of eq magnitude and the same sign is zero only at one point on line joining the two charges.

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B) If a point charge of mass m is released from rest in a uniform electric field, it will move along a line of force with uniform speed

C) The variation of the electric field due to an electric dipole is inversely proportional to the square of the distance from centre of mass

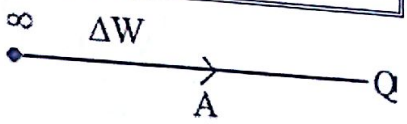
D) An electric dipole moment p in a uniform electric field E is in stable equilibrium if P and E are in opposite directions.

Answer: D

12.6 Electric Potential, Potential Difference and Electric Potential energy.

Electric Potential: Is the work done per unit positive charge to move a positive test charge from infinity to its current position within an electric field.

That is, the electric potential (V) at point A from a charge Q is equal to the work done per unit charge in moving a positive test charge from infinity (∞) to point A.



$$V = \frac{\Delta W}{Q}$$

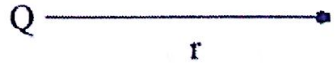
- Electric potential is a scalar quantity and its unit is volt (v). Thus,

$$1V = \frac{1J}{C}$$

Hence, an electric potential of 1000v means that there has been 1000J of work done per coulomb of charge to move a test charge to its current position within the electric field.

Potential due to a single point charge

- Electric potential due to a point charge (Q) at a point a distance (r) from the charge is given by



$$V = \frac{1}{4\pi \in_o}\left(\frac{Q}{r}\right) = K\frac{Q}{r}$$ ← absolute potential

- **Remember**, the potential due to a positive charge is positive, and the potential due to a negative charge is negative.

Illustrative Examples

51. What is the electric potential 30cm from a 6μC point charge?

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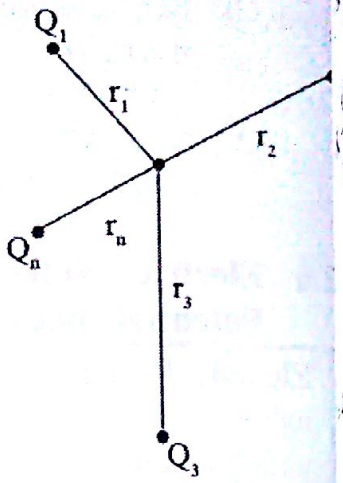
Solution: $V = K\frac{Q}{r} = (9\times10^{9})\left(\frac{6\times10^{-6}}{0.3}\right) = 180{,}000V$

$V = 180KV$

Potential due to many point charges



- The potential at a point in space due to many point charge $Q_1, Q_2,\ldots Q_n$ that are at $r_1, r_2,\ldots r_n$ distances from the point is equal to the algebraic sum of the potentials due to each charge.

$$V = V_1 + V_2 + \ldots V_n$$

$$V = K\frac{Q_1}{r_1} + K\frac{Q_2}{r_2} + \ldots + K\frac{Q_n}{r_n}$$

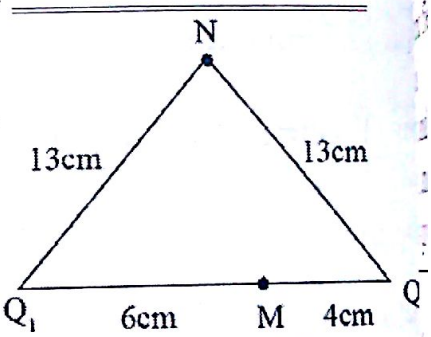
$$\boxed{V = K\left[\frac{Q_1}{r_1} + K\frac{Q_2}{r_2} + \ldots\frac{Q_n}{r_n}\right]}$$

Illustrative Examples

52. Given the electric dipole, $Q_1 = 12nc$ and $Q_2 = -12nC$. Calculate the electric potential at point

A) N

B) M



Solution: A) $V_N = V_1 + V_2\, n$

$$V_N = K\frac{Q_1}{r_1} + K\frac{Q_2}{r_2} = 9\times10^{9}\left[\frac{12\times10^{-9}}{0.13} + \left(\frac{-12\times10^{-9}}{0.13}\right)\right] = 0$$

B) $V_M = V_1 + V_2$

$$V_M = K\frac{Q_1}{r_1} + K\frac{Q_2}{r_2} = 9\times10^{9}\left[\frac{12\times10^{-9}}{0.06} + \frac{-12\times10^{-9}}{0.04}\right]$$

$V_M = -900V$

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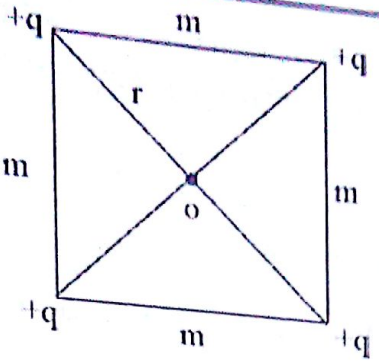
53. Four charges, each +q, are placed at the four corners of a square of side m. Calculate the potential at the center of the square.



Solution: $V_O = V_1 + V_2 + V_3 + V_4$

$$V_o = K\frac{Q_1}{r_1} + K\frac{Q_2}{r_2} + K\frac{Q_3}{r_3} + K\frac{Q_4}{r_4}$$

$$V_o = K\frac{Q}{r} + K\frac{Q}{r} + K\frac{Q}{r} + K\frac{Q}{r}$$

$$= 4K\frac{Q}{r}$$

$$V_o = 4\frac{KQ}{\frac{m}{\sqrt{2}}} = 4\sqrt{2}\frac{KQ}{m}$$

$m^2 + m^2 = (2r)^2$

$2m^2 = 4r^2$

$m^2 = 2r^2$

$$r = \frac{m}{\sqrt{2}}$$

Potential Difference

The potential difference is nothing but the difference of potentials of two points in an electric field.

- The potential difference between point A and B, that are at distances of r_A and r_B from a point charge Q is given by

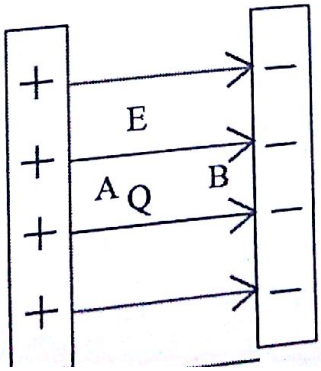
$$V_{AB} = V_B - V_A = \frac{KQ}{r_B} - \frac{KQ}{r_A} = kQ\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

For example, if the potential at point A is 40V and the potential at point B is 100V, the potential difference between them is, $V_{AB} = V_B - V_A =$ 60V. This means that 60J of work will be done by the field on each coulomb of positive charge moved from A to B.

Hence, another way of defining potential difference is,

Potential difference: Is the work per unit positive charge required to move a test charge from a point of higher potential to a point of lower potential.

- The potential difference between points A and B in a parallel plates of electric field E



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is equal to the work required to move a unit positive charge Q from A to B and is given by,

$$V_{AB} = V_B - V_A = \frac{W}{Q} \text{ or } \boxed{V = \frac{W}{Q}}$$

The charge Q placed in the region between the parallel plates will experience a force given by

$$F = QE$$

The work done by this force in moving the charge from one plate to the other is then given by,

$$W = FS = QED \quad \ldots\ldots\ldots\

Suppose now, the test charge q is moved from infinity to a point r from the charge Q. The work done becomes

$$W = KQq\left(\frac{1}{r} - \frac{1}{\infty}\right)$$

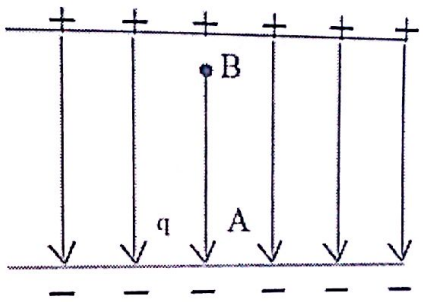
$$\boxed{W = \frac{KQq}{r}}$$

Which is then stored as the potential energy of the positive charge. Therefore,

$$\boxed{U = \frac{KQq}{r}} \leftarrow \text{potential energy.}$$

- The potential energy of the system is equal to the work done against the electric field in moving the charge q from infinity to that point.

Now, let's consider a positive charge q resting at point A in a uniform electric field E between two oppositely charged plates. An electric force qE acts downward on the charge. The work done against the electric field in moving the charge from A to B is equal to the product of the force qE and the distance d. Hence, the potential energy at B relative to A is $\boxed{U = qEd}$



When the charge is released, the electric field will perform this amount of work and the charge q will have a kinetic energy.

$$U = KE$$

$$\boxed{\frac{1}{2}mv^2 = qEd}$$

- **Remember:** Whenever a positive charge is moved against an electric field, the potential energy increases, and, whenever a negative charge moves against an electric field, the potential energy decreases.

Conceptual Examples

55. If an electron is released from rest in a uniform electric field, what will happen to the potential energy?

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Explanation: The potential energy decreases and its kinetic energy increases.

56. If the electric potential at some point is zero, can you conclude that no charge exists at that point?

Explanation: No. For example, the potential at the mid-point between two charges of equal magnitude but opposite sign is zero.

57. If the electric field is zero in some region, what can you say about the electric potential in that region.

Explanation: From $E = \frac{\Delta V}{\Delta X}$, $E = 0 \Rightarrow \Delta V = 0$, thus $E = 0$ implies V is constant.

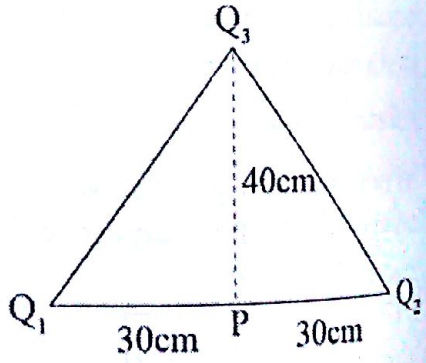
Illustrative Examples

58. What is the change in potential energy of a charge of 2nc as it moved from 20cm to 8cm from a 4μc point charge.

Solution: $\Delta U = KQq\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$

$$\Delta U = \left(9\times10^{9}\right)\left(2\times10^{-9}\right)\left(4\times10^{-6}\right)\left[\frac{1}{0.08} - \frac{1}{0.2}\right] = 54\times10^{15}\text{J}$$

59. Three charges $Q_1 = 2\mu c$, $Q_2 = -3\mu c$ and $Q_3 = 4\mu c$ are placed at the vertices of a triangle. Calculate



a) The electric potential at point P

b) The potential energy of a 5μc charge as it moves from infinity to point P.

Solution: a) $V_p = V_1 + V_2$

$$V_P = 9\times10^{9}\left[\frac{2\times10^{-6}}{0.3} + \frac{-3\times10^{-6}}{0.03} + \frac{4\times10^{-6}}{0.4}\right]$$

$$V_P = 6\times10^{4}\text{V}$$

b) $U = Vq = \left(6\times10^{4}\right)\left(5\times10^{-6}\right) = 0.3\text{J}$

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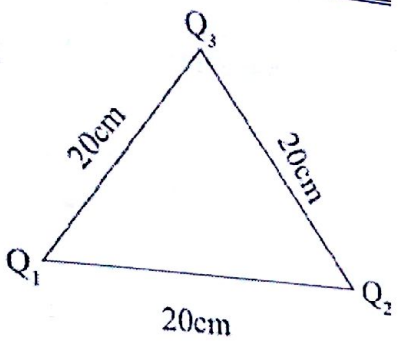
60. Three charges $Q_1 = 2\mu c$, $Q_2 = -3\mu c$ and $Q_3 = -4\mu c$ are placed in vertices of equilateral triangle of 20cm. Find the total potential energy of the system.



Solution: The total potential energy is

$$U = U_{12} + U_{13} + U_{23}$$

$$U = K\frac{Q_1Q_2}{r_1} + K\frac{Q_1Q_3}{r_2} + K\frac{Q_2Q_3}{r_3}$$

$$U = \frac{9\times10^9}{0.2}\left[-6\times10^{-6} - 8\times10^{-6} + 12\times10^{-6}\right] = -0.9\times10^3\,J$$

61. A proton is moved from the negative to the positive plate. If the plates are 2.5cm apart and the electric field is 1800N/C. Calculate
   A) The change in potential energy
   B) The speed of the proton as it reached the negative plate if it is released from rest from the positive plate.

Solution: A) $\Delta U = QEd = (1.6\times10^{-19})\,(1800)\,(0.0025)$

$\Delta U = 7.2\times10^{-18}J$

B) Transformation of energy. $K_E = U$

$$\frac{1}{2}mv^2 = QEd = 7.2\times10^{-18} \Rightarrow V = 9.3\times10^5\,m/s$$

62. An ion accelerated through a potential difference of 115V experiences an increase in kinetic energy of 7.37×10^{-17}J. What is the charge on the ion (UEE: 2004)

A) $6.41\times10^{-19}C$ C) $4.51\times10^{-15}C$

B) $2.32\times10^{-15}C$ D) $2.0\times10^{-21}C$

Solution: $$Qv = \frac{1}{2}mv^2 = 7.37\times10^{-17}J$$

$$Q = \frac{7.37\times10^{-17}}{115} = 6.41\times10^{-19}C$$

Answer: A

63. When a p.d of 150v is applied to the plates of a parallel plate. The plates carry a surface charge density of $30nc/cm^2$. What is the spacing between the plates (UEE: 2004)

A) 2.24μm B) 5.32μm C) 12.24μm D) 4.42μm

B)

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Solution: $V = Ed = \frac{Q}{A\epsilon_o} d = \frac{\sigma d}{\epsilon_o}$

$$\Rightarrow d = \frac{V\epsilon_o}{\sigma} = \frac{(150)(8.85\times10^{12})}{(30\times10^{-19})} = 4.42\mu m$$

Answer: D

64. Points A and B are each a distance d from each of two equal and opposite charges as shown. What will be the work required to move a negative charge q from A to B? (UEE: 2005)

A) zero

B) $\frac{Qq}{4\pi\epsilon_o d^2}$

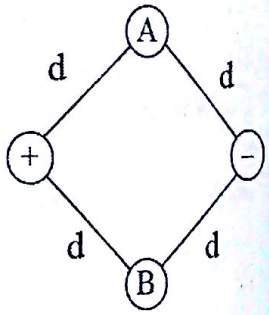
C) $\frac{Qq}{4\pi\epsilon_o d}$

D) $\frac{Qq}{4\pi\epsilon_o (4d^2)}$

Solution: $V_A = \frac{KQ}{d} + \frac{K(-Q)}{d} = 0$

$$V_B = \frac{KQ}{d} + \frac{K(-Q)}{d} = 0$$

$V_{AB} = V_B - V_A = 0$. Thus, $W = QV = 0$.



Answer: A

65. In a certain region of space, a uniform electric field is directed in the positive x direction. If a particle with negative charge is carried from x = 20cm to x = 60cm, what can we conclude about the electric potential and the electric potential energy of the charge field system? (UEE: 2007)

A) The electric potential at x = 20cm is larger than that at x = 60cm.
B) The electric potential at x = 20cm is the same as that at x = 60cm.
C) The electric potential energy decreases
D) The electric potential energy increases.

Answer: D

66. A proton is released from rest at point P, where the potential is 0v. Afterward, the proton moves.

A) Towards B with an increasing speed

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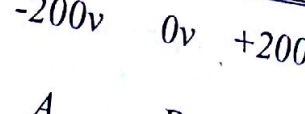
B) Towards A with an increasing speed
C) Towards A with a decreasing speed
D) Towards B with a steady speed



Explanation: Its speed increases while moving from P to A

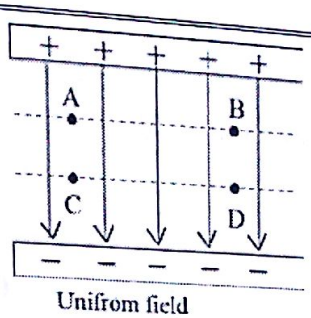
Answer: B

Equipotential Lines and Surfaces

Equi means equal or same. Thus,

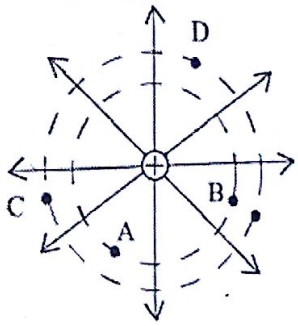
Equipotential line: Is a line joining points within an electric field that have the same potentials.

Uniform Field: In a uniform field the equipotential lines are equidistant parallel lines. For example the line joining point A and B is equipotential line. The same is true for the line joining points C nad D.



Unifrom field

Radial Field: In a radial field, such as the field around a point charge, the lines of equipotential are concentric circles centred around the single charge. For example the circle joining point A and B is equipotential line. The same is true for the circle joining points C and D.



variable field

Equipotential Surface: Is a surface containing equipotential lines.

Important points to remember

- Equipotential lines or surfaces are perpendicular to the electric field lines at every point.
- No work is done in moving a charge along an equipotential line or surface. This is because, work done is computed as, $W = QV$, where $V = V_{AB} = V_B - V_A$ but points A and B do have the same potentials thus, $V_B - V_A = 0$. This implies work done is zero.
- In a uniform field, if the electric field strength is increased, then the equipotential lines move closer together.
- In a variable field, the equipotential lines move further apart as the distance from the point charge increases.

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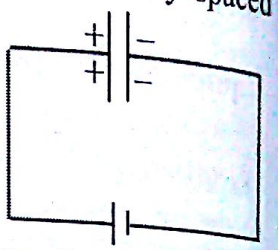
12.7 Capacitors

Capacitors are devices usually formed by two oppositely charged conductors and are used to store charge and electric energy.

Charging a capacitor is often done by connecting the closely spaced conducting plates with a battery. The battery then pumps charges from one plate and deposits them on to the other plate. The conductors finally carry equal and opposite charges.



Parallel plate capacitor

The battery charges the capacitor until the potential difference between the plates is equal to the terminal voltage of the battery.

The charge stored on each plate of a capacitor is given by

$$\boxed{Q = CV} \leftarrow \text{Capacitor equation}$$

Where Q: is the charge stored on plates of a capacitor
V: is the potential difference across the plates of a capacitor
C: is the capacitance of the capacitor

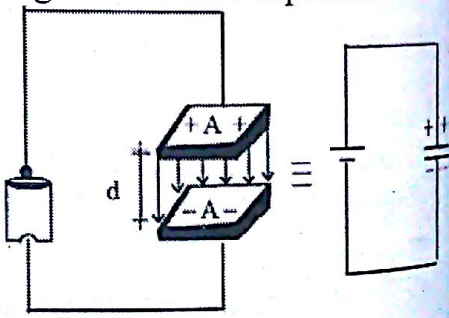
Capacitance: Is the measure of a capacitor's ability to store charge.

The SI unit of capacitance is the Farad (F) in honor of Micheal Faraday.

Conceptual Examples

67. What can you say about the net charge stored on a capaictor.

Explanation: A capacitor's plates carry equal and opposite charges and hence, the net charge on the capacitor as a whole is zero.



Parallel Plate Capacitor

parallel plate capacitor

A parallel plate capacitor consists, parallel plates each of equal areas (A) separated by a small distance (d). Where uniform electric field E develops between the plates.

The capacitance of an air filled parallel plate is found as follows.

$$E = \frac{\sigma}{\epsilon_o} = \frac{Q}{A\epsilon_o}, \text{ Gauss's Law}$$

And we also know that

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$$E = \frac{V}{d}$$

So we have, $\frac{Q}{A\epsilon_o} = \frac{V}{d}$

$$Q = \frac{A\epsilon_o V}{d} \quad \text{.....(*)}$$

$$Q = CV \Rightarrow C = \frac{Q}{V} \quad \text{..... (**)}$$

Now, substituting for Q from equation (*) in to equation (**)

$$C = \frac{A\epsilon_o V}{\frac{d}{V}} = \frac{A\epsilon_o V}{dV}$$

$$\boxed{C = \epsilon_o \frac{A}{d}}$$ ←The air filled parallel plate capacitor.

Illustrative Examples

68. A 2μF capacitor is connected to a 1.5V battery. What is the charge stored on the capacitor?

Solution: $Q = CV = \left(2\times10^{-6}\right)(1.5) = 3\times10^{-6}C = 3\mu C$

69. What is the capacitance of an air filled parallel plate capacitor of plates area $100cm^2$ and plates separation 0.1mm?

Solution: $C = \frac{\varepsilon_o A}{d}$

$$C = \frac{\varepsilon_o A}{d} = (8.85\times10^{-12})\frac{(100\times10^{-4})}{(0.1\times10^{-3})} = 8.85\times10^{11}F$$

Capacitor with Dielectric

In reality, air filled parallel plate capacitors have very little importance since they are limited to capacitances of around $10^{-14}F$ which is too small for most electronic devices to operate well. Hence most capacitors are constructed with a dielectric material in between the plates.

Dielectrics: Are insulating materials such as glass, mica, paraffin, wood, plastic, rubber etc that maybe polarized by an electric field and which allows more charge to be stored.

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- **Remember!** The effect of inserting an insulator between the pl will enhance the capacitance of the capacitor by reducing potential difference between them. The capacitance of a capac with a dielectric is given by

$$C = \epsilon_r \epsilon_o \frac{A}{d}$$

Where, ϵ_r is relative permittivity of the material, also called *dielectric constant*.

- dielectric constant is the ratio of the permittivity of the dielec material to the permittivity of free space. $\epsilon_r = \frac{\epsilon}{\epsilon_o}$.

  For example $\epsilon_r = 2$ would mean twice as 'easy' for the electric fi to travel through the space between the plates than to travel throu air.

A dielectric blocks the flow of charges from one plate to the other. E when the electric field across a dielectric exceeds a certain valu becomes a conductor. The maximum value of the electric field tha dielectric withstand without breakdown is called *dielectric strength*.

Dielectric Strength: Is the maximum electric field strength tha material can withstand before breakdown or tear.

For example, if the electric field in air exceeds about 3000,000 dielectric breakdown will occur, leading to a spark.

The capacitance of a capacitor can be increased by

- Increasing plates area
- Decreasing plates separation, and
- Introducing dielectric between the plates.

- The effect of inserting a dielectric is that.

  (i) The capacitance increases from C_o to $C_o \epsilon_r$

$$C_o = \epsilon_o \frac{A}{d} \leftarrow \text{air filled}$$

$$C = \epsilon_r \epsilon_o \frac{A}{d} \leftarrow \text{with dielectric}$$

$$\Rightarrow C = \epsilon_r C_o$$

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(ii) The electric field decrease, from E_o to $\frac{E_o}{\epsilon_r}$

$$E_o = \frac{Q}{\epsilon_o A} \leftarrow \text{air filled}$$

$$E = \frac{Q}{\epsilon_r \epsilon_o A} \leftarrow \text{with dielectric}$$

$$\Rightarrow E = \frac{E_o}{\epsilon_r}$$

Conceptual Examples

70. When the separation between the plates of a capacitor increased, what will happen to the electric field between the plates? Explain

Explanation: The electric field in parallel plates is uniform.

71. If the separation between the plates of a capacitor is increased, what will happen to the charge stored on the capacitor. Explain

Explanation: Capacitance, $C = \epsilon_o \frac{A}{d}$

Charge stored, $Q = CV = \epsilon_o \frac{AV}{d}$ decreases.

72. If you were asked to design a capacitor that has small size and large capacitance, what will you do? Explain

Explanation: One possible way is a parallel plate capacitor with small plates separation and dielectric with large dielectric constant.

Illustrative Examples

73. A certain capacitor has a capacitance of 4μF when its plates are separated by 0.2mm in vacuum. A battery is used to charge the plates to a potential difference of 500v, and is then disconnected and sheet of mica with dielectric constant 5 is inserted between the plates. After the insertion of the dielectric. Calculate

a) The potential difference
b) The capacitance
c) The increase in charge assuming the source remain connected

Solution: a) $V = \frac{V_o}{\epsilon_r} = \frac{500}{5} = 100V$

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b) $C = \epsilon_r C_o = (5)(4) = 20\mu F$

c) $Q_o = C_o V_o = (4)(500) = 2000\mu C$

$Q = CV = (20)(500) = 10,000\mu C$

The increase in charge,

$\Delta Q = Q - Q_o = 10,000 - 2000 = 8000\mu C$

74. A capacitor with dielectric connected to a 24V power supply has a normal plate separation of 4mm and a plate area of $0.64cm^2$.
a) What is the dielectric constant if the capacitance is 1.2pF?
b) What charge stored if the plates spacing is compressed to 2mm?

Solution: A) $C_o = \epsilon_o \frac{A}{d} = \frac{(8.85\times10^{-12})(0.64\times10^{-4})}{4\times10^{-3}} = 1.4\times10^{-3}F$

$C = \epsilon_r C_o \Rightarrow \epsilon_r = \frac{C}{C_o} = \frac{1.2\times10^{-12}}{1.4\times10^{-3}} = 8.57$

B) $C^1 = \epsilon_r \frac{\epsilon_o A}{d} = \frac{(8.57)(8.85\times10^{-12})(0.64\times10^{-4})}{2\times10^{-3}}$

$C^1 = 2.4\times10^{-12}F$

$Q^1 = C^1V = (2.4\times10^{-12})(24) = 5.76\times10^{-11}C$

$Q^1 = 0.576pC$

75. An air filled parallel plate capacitor has a capacitance of 1pF. When the plate separation is doubled and filled with wax, the capacitance increases to 2pF. What is the dielectic constant of wax. (UEE 2002)
A) 2 B) 4 C) 6 D) 8

Solution: $C_1 = \epsilon_o \frac{A}{d}$ ...air filled

$C_2 = \epsilon_r \frac{\epsilon_o A}{2d_1}$ ...wax filled $\Rightarrow C_2 = \frac{\epsilon_r}{2} C_1$

$2 = \frac{\epsilon_r}{2}(1) \Rightarrow \epsilon_r = 4$

Answer: B

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76. When a p.d of 150V is applied to the plates of a parallel plate capacitor, the plates carry a surface charge density of $30nc/cm^2$. What is the spacing between the plates (UEE: 2004)

A) 2.24μm B) 5.32μm C) 12.24μm D) 4.42μm

Solution: $V = Ed,\ E = \frac{\sigma}{\epsilon_o} = \frac{Q}{A\epsilon_o}$

$$V = \frac{\sigma}{\epsilon_o} d$$

$$d = \frac{V\epsilon_o}{\sigma} = \frac{(150)(8.85\times10^{-12})}{30\times10^{-9}/10^{-4}} = 4.42\mu m$$ **Answer: D**

77. A parallel plate capacitor has a capacitance of 10μF. If the distance between the plates is doubled, keeping all other quantities constant, what is the new capacitance of the capacitor. (UEE. 2007)

A) 0.2μF B) 5μF C) 10μF D) 20μF

Solution: $C_1 = \epsilon_o \frac{A}{d_1}$ ..... (*)

$$C_2 = \epsilon_o \frac{A}{d_2} = \frac{\epsilon_o A}{2d_1}$$, from equation (*)

$$C_2 = \frac{C_1}{2} = \frac{10}{2} = 5\mu F$$ **Answer: B**

78. An air filled capacitor is charged to 12V. When dielectric is inserted the voltage is found to be 5V. What is the dielectric constant.

A) 2 B) 2.4 C) 4 D) 4.2

Solution: $\epsilon_r = \frac{V_o}{V} = \frac{12}{5} = 2.4$ **Answer: B**

Energy Stored in a Capacitor

When a capacitor is charged, the potential difference across the plates increases from an initial value of zero to a final value of V. Thus, the average potential is considered in the calculation of energy stored.

Energy stored = (charge stored) (average potential)

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$$U = QV_{av} = Q\left(\frac{V+0}{2}\right)$$

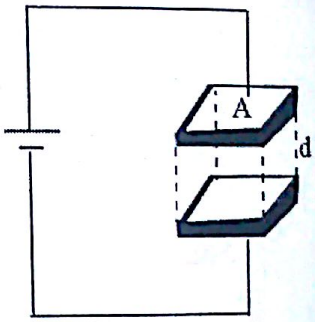
$\boxed{U = \frac{1}{2}QV}$, but since $Q = CV$

We can have, $U = \frac{1}{2}(CV)V \Rightarrow \boxed{U = \frac{1}{2}CV^2}$

Again, from $Q = CV$, $V = \frac{Q}{C}$

$$U = \frac{1}{2}Q\left(\frac{Q}{C}\right) \Rightarrow \boxed{U = \frac{Q^2}{2C}}$$

We may use one of the three equations depending on the given and required variables but all give us the same result.



Electric Energy Density

$$\text{Energy density} = \frac{\text{Energy stored}}{\text{volume of capacitor}}$$

$$\text{Energy density} = \frac{\frac{1}{2}CV^2}{Ad}, \text{ volume of rectangular prism.}$$

Again, $C = \epsilon_r\epsilon_o\frac{A}{d}$ and $V = Ed$

$$\text{Energy density} = \frac{\frac{1}{2}\left(\epsilon_r\epsilon_o\frac{A}{d}\right)(Ed)^2}{Ad}$$

$$\boxed{\text{Energy density} = \frac{1}{2}\epsilon_r\epsilon_o E^2}$$

Its unit is J/m^2.

Illustrative Examples

79. A 1.8μF capacitor is charged to 500V. Calculate the energy stored in the capacitor.

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Solution: $U=\frac{1}{2}CV^2=\frac{1}{2}\left(1.8\times10^{-6}\right)\left(500\right)^2=0.225J$

80. An air filled capacitor has plates 0.8mm apart. The charge on each plate is 0.04μC when the potential difference is 300V. Calculate
A) The energy stored
B) The area of plates
C) The energy density

Solution: A) $U=\frac{1}{2}QV=\frac{1}{2}\left(0.04\times10^{-6}\right)\left(300\right)=6\mu J$

B) $C=\epsilon_o\frac{A}{d}\Rightarrow\frac{Q}{V}=\frac{\epsilon_o A}{d}$

$$A=\frac{Qd}{V\epsilon_o}=\frac{\left(0.04\times10^{-6}\right)\left(0.8\times10^{-3}\right)}{\left(300\right)\left(8.85\times10^{-12}\right)}=1.2\times10^{-2}m^2$$

C) $\text{Energy density}=\frac{\text{energy}}{\text{volume}}$

$$=\frac{V}{Ad}=\frac{6\times10^{-6}}{\left(1.2\times10^{-2}\right)\left(0.8\times10^{-3}\right)}=0.625J/m^2$$

81. A parallel plate capacitor of 120μF is charged to 1500V. The space between plates is filled with a 0.02mm thick dielectric with ϵ_r =240. What is the charge density.

Solution: Charge density$=\frac{1}{2}\epsilon_o\epsilon_r E^2$, $E=\frac{V}{d}=\frac{1500}{2\times10^{-5}}=7.5\times10^7$

$$=\frac{1}{2}=\left(8.85\times10^{-12}\right)\left(240\right)\left(7.5\times10^7\right)=5.9\times10^6J/m^3$$

82. Which one of the following changes to a parallel plate capacitor would not increase energy stored: (UEE: 2005)
A) Increasing the area of the plates
B) Increasing the dielectric constant
C) Decreasing the charges on the plates
D) Increasing the distance between the plates

Explanation: $U=\frac{1}{2}CV^2$ and $C=\epsilon_r\epsilon_o\frac{A}{d}$ **Answer: D**

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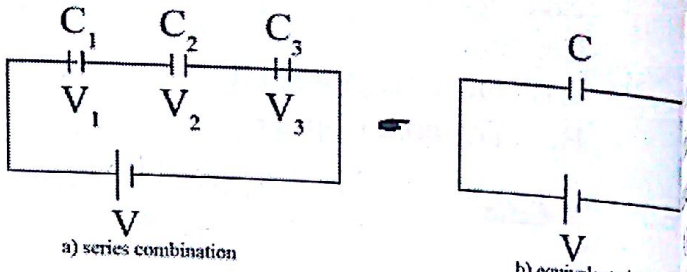
12.8 Capacitors Combination

Based on the desired value of capacitance in a circuit, capacitors can be combined in different ways. The most common forms are series and parallel combinations.

Capacitors in series

When capacitors are combined in series, the charge in the network is constant and the total potential drop is the sum of potential drops across each individual capacitors.



a) series combination b) equivalent circuit

That is, $Q_1 = Q_2 = Q_n = Q_T = Q$

$$V_T = V_1 + V_2 + \ldots\ldots + V_n$$

but we know that, $V = \frac{Q}{C}$

$$\Rightarrow \frac{Q_T}{C_T} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \ldots\ldots + \frac{Q_n}{C_n}$$

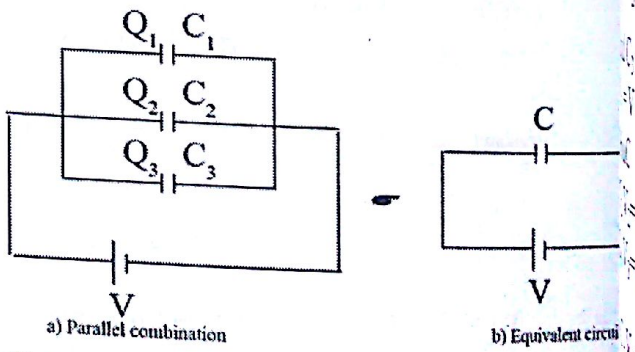
but for series combination, $Q_1 = Q_2 = Q_n = Q_T = Q$

$$\boxed{\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots\ldots + \frac{1}{C_n}} \leftarrow$$

total capacitance of n capacitors in series.

Capacitors in parallel

When capacitors are combined in parallel, the potential drop, in the system is constant and the total charge stored in the network is the sum of charges on the individual capacitors.



a) Parallel combination b) Equivalent circuit

That is, $V_1 = V_2 = V_n = V_T = V$

$$Q_T = Q_1 + Q_2 + \ldots\ldots\ldots + Q_n$$

But we know that, $Q = CV$

$$\Rightarrow C_T V_T = C_1 V_1 + C_2 V_2 + \ldots\ldots\ldots + C_n V_n$$

but for parallel combination $V_1 = V_2 = V_n = V_T = V$

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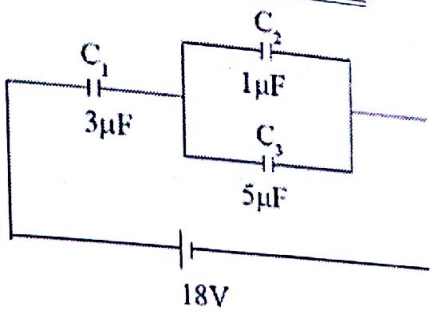
$C_T = C_1 + C_2 + \ldots\ldots + C_n$ ← total capacitance of n capacitors in parallel

Illustrative Examples

83. From the combination shown; Calculate

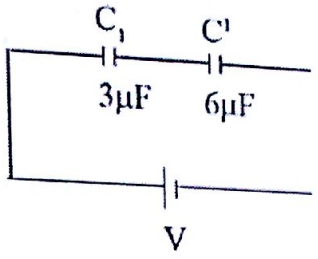
A) The equivalent capacitance
B) The potential drop across each capacitor
C) The charge on each capacitor
D) The energy stored in C_2



Solution: A) C_2 and C_3 are in parallel.

$C^1 = C_2 + C_3 = 1 + 5 = 6\mu F$

The circuit modified to,



Next, C_1 and C^1 are in series.

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C'} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6}$$

$C_T = 2\mu F$

B) $Q_T = CV = (2)(8) = 36\mu C$

$Q_1 = Q_T = 36\mu C$

$$V_1 = \frac{Q_1}{C_1} = \frac{36}{3} = 12V$$

Since C_2 and C_3 are in parallel, they have the same potential

$V_2 = V_3 = V - V_1 = 18 - 12 = 6V$

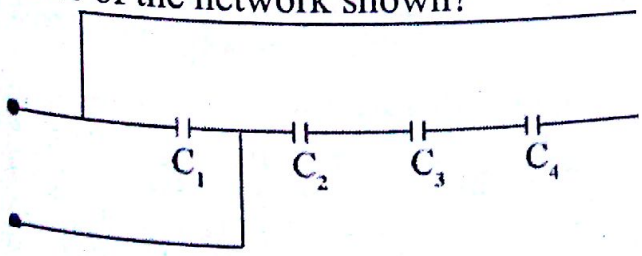
C) $Q_1 = 36\mu C$

$Q_2 = C_2 V_2 = (1)(6) = 6\mu C$

$Q_3 = C_3 V_3 = (5)(6) = 30\mu C$

D) $U = \frac{1}{2} C_2 V_2 = \frac{1}{2}\left(1\times10^{-6}\right)(6) = 3\times10^{-6} J.$

84. If each capacitor has a capacitance of 2μF, what is the equivalent capacitance of the network shown?



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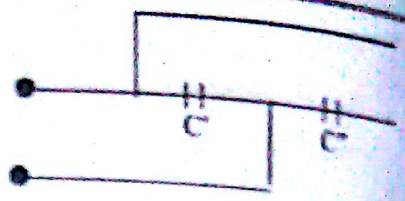
Solution: C_1 and C_2, C_3 and C_4 are in series



$$\Rightarrow \frac{1}{C^{1}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{2} + \frac{1}{2} = C^{1} = 1\mu F, \text{ also}$$

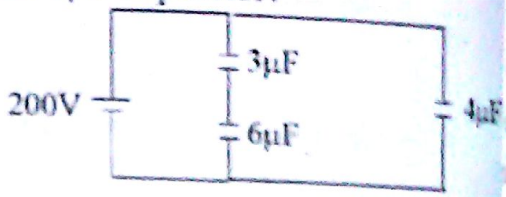
$$\frac{1}{C^{11}} = \frac{1}{C_3} + \frac{1}{C_4} = \frac{1}{2} + \frac{1}{2} = C^{11} = 1\mu F \text{ and } C^{1} \text{ and } C^{11} \text{ are in parallel.}$$

Thus, $C^{1} + C^{11} = 1 + 1 = 2\mu F$

85. A) What is the charge on the 4μF capacitor?

B) What is the voltage across the 6μF capacitor?

Solution: A) The 3μF and 6μF are in series



$$\frac{1}{C^{1}} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6} = \frac{1}{2}$$

$\Rightarrow C^{1} = 2\mu F$

now C^{1} and the 4μF are in parallel,

$C_T = C^{1} + 4\mu F = 2 + 4 = 6\mu F$

The voltage across C^{1} and the 4μF are same.

Thus, the charge on the 4μF is, $Q = CV = (4)(200) = 800\mu C$

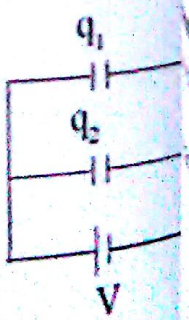
B) The voltage on the 6μF is $V = \frac{Q}{C} = \frac{400}{6} = \frac{200}{3} V$

86. Two capacitors $C_1 = 2\mu F$ and $C_2 = 6\mu F$ are separately charged across a 12V battery. After being disconnected from the battery they are connected together with the negative plate of one attached to positive plate of the other. Calculate the final charge on each capacitor.

Solution: $q_1 = C_1V_1 = (2)(12) = 24\mu C$ and $q_2 = C_2V_2 = (6)(12) = 72\mu C$. The charges will partly cancel when the capacitors are connected. Their final charges will be,

$$q_1^{1} + q_2^{1} = q_2 - q_1 = 72 - 24 = 48\mu C$$

The potential drops across q_1^{1} and q_2^{1} are same.



$$V_1^{1} = V_2^{1} \Rightarrow \frac{q_1^{1}}{C_1} = \frac{q_2^{1}}{C_2} \Rightarrow \frac{q_1^{1}}{2} = \frac{48 - q_1^{1}}{6}$$

$\Rightarrow q_1^{1} = 12\mu C$ and $q_2^{1} = 36\mu C$

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87. Two capacitors have a combined series capacitance of 4μF and a combined parallel capacitance of 18μF. What is the capacitance of each capacitor?

Solution: In series, $\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2}\Rightarrow C=\frac{C_1C_2}{C_1+C_2}$ .....(*)

In parallel, $C_P = C_1 + C_2$.....(**)

From equation (*) and (**),

$$4=\frac{(18-C_2)C_2}{18}$$

$$C_2^2 - 18C_2 + 72 = 0$$

$$(C_2 - 6)(C_2 - 12) = 0$$

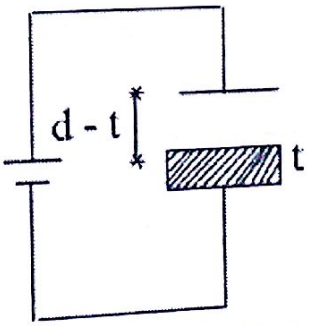
$C_2 = 6\mu F$ or $C_2 = 12\mu F$ hence, $C_1 = 12\mu F$ or $C_1 = 6\mu F$

88. A parallel plate capacitor of plates area A and separation d is partially filled with a dielectric material of thickness t and dielectric constant K. What is the effective capacitance of the system?

Explanation: The arrangement can be treated as if there are two capacitors C_1 and C_2 in series.

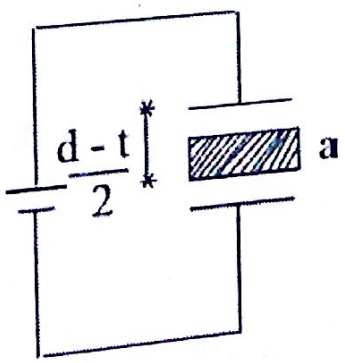
$$\frac{1}{C_T}=\frac{1}{C_1}+\frac{1}{C_2}\Rightarrow C_1=\frac{\epsilon_o A}{d-t} \text{ and } C_2=\frac{K\epsilon_o A}{t}$$

$$\frac{1}{C_T}=\frac{1}{\frac{\epsilon_o A}{d-t}}+\frac{1}{\frac{K\epsilon_o A}{t}}\Rightarrow C=\frac{K\epsilon_o A}{K(d-t)+t}$$



89. A parallel plate capacitor has a plate separation d and plate area A. An uncharged metal slab of thickness a is inserted midway between the plates. Find the capacitance of the system.

Explanation: The two capacitors are in series with effective plate separation of $\frac{d-a}{2}$ each.



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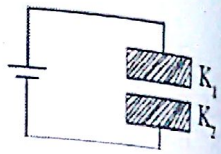
$$\frac{1}{C_T}=\frac{1}{C_1}+\frac{1}{C_2}=\frac{1}{\frac{\epsilon_o A}{\frac{d-a}{2}}}+\frac{1}{\frac{\epsilon_o A}{\frac{d-a}{2}}}\Rightarrow C=\frac{\epsilon_o A}{d-a}$$

90. Suppose a parallel plate capacitor with plate area A and separation d is filled half of its space with dielectric material of constant K_1 and the remaining half with dielectric material of constant K_2. Find the effective capacitance.

Explanation: C_1 and C_2 are in series.

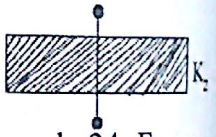
$$\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2},\ C_1=\frac{K_1\epsilon_o A}{\frac{d}{2}},\ C_2=\frac{K_2\epsilon_o A}{\frac{d}{2}}$$



91. Suppose a parallel plate capacitor with plate area A and separation d. The left half of the gap is filled with K_1 and the right half with another dielectric of constant K_2. Calculate the capacitance.

Explanation: C_1 and C_2 are in parallel. $d_1=d_2=\frac{d}{2}$

$$C=C_1+C_2=\frac{\epsilon_o A}{d}(K_1+K_2),\ A_1=A_2=\frac{A}{2}$$



92. Three capacitors of capacitance 8μF, 12μF and 24μF are connected in series and then in parallel. What is the ratio of equivalent capacitances in series and in parallel (UEE: 2001).

A) 1:11 B) 11:1 C) 1:1 D) 1:3

Solution:

$$\frac{1}{C_S}=\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}=\frac{1}{8}+\frac{1}{12}+\frac{1}{24}\Rightarrow C_S=4\mu F$$

$$C_P=C_1+C_2+C_3=8+12+24\Rightarrow C_P=44\mu F$$

$$\frac{C_S}{C_P}=\frac{4}{44}=\frac{1}{11}=1:11$$

Answer: A

93. A certain capacitor is charged to a potential V. If you wish to increase its stored energy by 10%, then by what percentage should you increase V? (UEE: 2001)

A) 4.88% B) 5.92% C) 6.98% D) 5.13%

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Solution: $U_1=\frac{1}{2}CV_1^2$ .....(*)

$U_2=\frac{1}{2}CV_2^2$, $U_2=U_1+10\%U_1=U_1+0.1U_1=1.1U_1$

$1.1U_1=\frac{1}{2}CV_2^2$ .....(**)

taking ratio, $\frac{1.1U_1}{U_1}=\frac{\frac{1}{2}CV_2^2}{\frac{1}{2}CV_1^2}$

$1.1=\frac{V_2^2}{V_1^2}=\frac{V_2}{V_1}=\sqrt{11.1}\Rightarrow V_2=1.0488V_1$

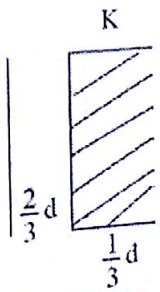
$\Rightarrow V_2=V_1+0.0488V_1\Rightarrow 4.88\%$ **Answer: A**

94. A parallel plate capacitor with plate separation d has a capacitance C in the absence of dielectric. What is the capacitance when a slab of dielectric material of constant K and thickness $\frac{1}{2}$d is inserted between the plates? (UEE: 2003)

A) $\frac{3K}{2K+1}C$ C) $\frac{3K}{2K-1}C$

B) $\frac{2K+1}{3K}C$ D) $\frac{2K-1}{3K}C$

Solution: C_1 and C_2 are in series.



$C_1=\frac{\epsilon_o A}{\frac{2}{3}d}$ and $C_2=\frac{K\epsilon_o A}{\frac{1}{3}d}=\frac{3K\epsilon_o A}{d}$

$C_T=\frac{3k}{2k+1}C$ **Answer: A**

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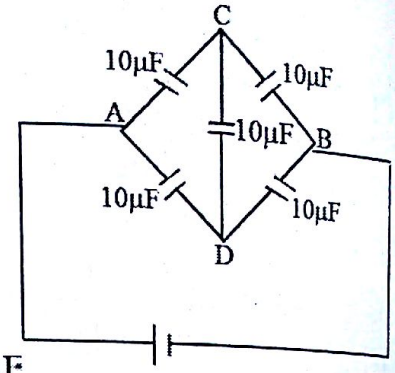
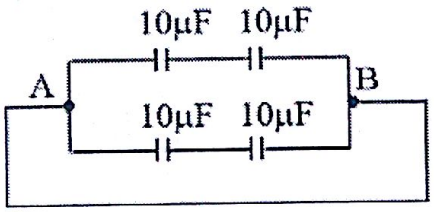
95. What is the equivalent capacitance between A and B? (UEE: 2004)

A) 10μF B) 40μF C) 30μF D) 60μF

Solution: The bridge circuit has equal potentials at C and D hence we can neglect the middle capacitor since it do not carry charge. And the circuit is modified as follows.





$\Rightarrow C_T = 10\mu F$

Answer: A

96. What is the capacitance of a capacitor that should be connected with 2 other capacitors having capacitance of 4μF and 8μF all in series to produce an effective capacitance of $\frac{24}{11}\mu F$?

(UEE: 2004)

A) 12μF B) 10μF C) 8μF D) 4μF

Solution: $\frac{1}{C_T}=\frac{1}{C}+\frac{1}{4}+\frac{1}{8}$

$$\frac{1}{\frac{24}{11}}=\frac{1}{C}+\frac{1}{4}+\frac{1}{8}\Rightarrow C=12\mu F\Rightarrow C=12\mu F$$

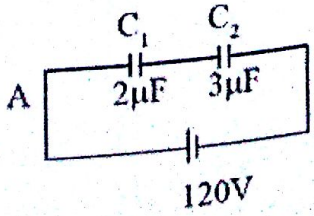
Answer: A

97. Two different capacitors of capacitance $C_1 = 2\mu F$ and $C_2 = 3\mu F$ are connected in series across a 120V supply line. The charged capacitors are disconnected from the line and from each other, and reconnected with terminals of like sign together. What will be the final charges on C1 and C2, respectively? (UEE: 2006).

A) 57.6μC and 86.4μC C) 144μC across each

B) 115.2μC and 127.8μC D) 288μC across each

Solution: initial condition:



$\Rightarrow C = \frac{6}{5}\mu F, Q = 144\mu C$

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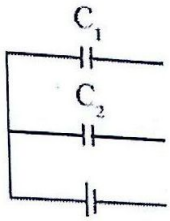
$Q_1 + Q_2 = Q = 144\mu C$



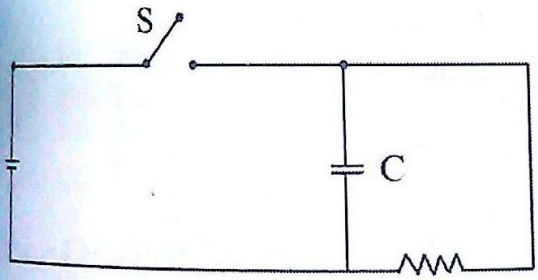
final condition

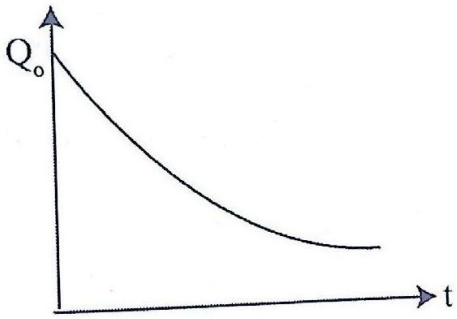
$$V_1 = V_2 = \frac{Q_1}{C_1} = \frac{Q_2}{C_2} = \frac{144 - Q_1}{C_2}$$

$$\frac{Q_1}{2} = \frac{144 - Q_1}{3} \Rightarrow Q_1 = 57.6\mu C,\ Q_2 = 86.4\mu C$$ **Answer: A**

12.9 Discharging a capacitor

When a capacitor is discharging, the charge on it decays exponentially and can be computed as follows.





Discharging

The current through the circuit at any time is given by

$$I = -\frac{\Delta Q}{\Delta t} \quad \text{..... (*)}$$

The negative sign shows that the charge on the capacitor decreases as time increases.

We also know that, $I = \frac{V}{R}$ ..... (**) and

$$V = \frac{Q}{C} \quad \text{..... (***)}$$

From the above three equations we can have,

$$-\frac{\Delta Q}{\Delta t} = \frac{Q}{RC}$$

For very small time interval,

$$\frac{dQ}{Q} = -\frac{dt}{RC}$$, integrating both sides,

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$$\int_{Q_o}^{Q} \frac{dQ}{Q} = -\int_{0}^{t} \frac{dt}{RC}$$

$$\ln Q\Big|_{Q_o}^{Q} = -\frac{t}{RC}\Big|_{o}^{t}$$

$$\ln Q - \ln Q_o = -\frac{t}{RC}, \text{ changing to exponential function.}$$

$$\frac{Q}{Q_o} = e^{-t/RC}$$

$$\Rightarrow \boxed{Q = Q_o e^{-t/RC}} \leftarrow \text{charge discharged at any time.}$$

Where: Q_o is the charge on the capacitor at t = o.
C is the capacitance of the capacitor.
R is the resistance if the resistor.
RC is called the *time constant* of the circuit.

☛ **Important points to Remember**

- The time taken for the capacitor to lose half its charge is know as the *half-life* $t_{1/2}$ of the decay process. In this case

$Q = \frac{Q_o}{2}$ when $t = t_{1/2}$ hence,

$$Q = Q_o e^{-t/RC}$$

$$\frac{Q_o}{2} = Qe^{-t/RC}$$

$$\frac{1}{2} = e^{-t/RC}$$

$$-\frac{t}{RC} = \ln 1/2 = 0.693$$

$$\boxed{t = 0.693RC} \leftarrow \text{half-life.}$$

- When t = the time constant, RC

$$Q = Q_o e^{-RC/RC}, \; t = RC$$

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$$\frac{Q_o}{2} = Q_o e^{-1}$$

$$Q = \frac{Q_o}{e}$$

- $\boxed{Q = 0.37Q_o}$ ← RC is the time for the charge on the capacitor to fall to 0.37times its initial value, Q_o.

Illustrative Examples

98. A 50μF capacitor has an initial charge of 60μC. It is then connected to a 15MΩ resistor and allowed to discharge through the resistor. Calculate
   A) the time constant
   B) the half-life
   C) the p. d at half life
   D) the charge at half life
   C) the current at half life

Solution: A) $RC = (15 \times 10^6)(50 \times 10^{-6}) = 750\text{sec}$

B) $t_{1/2} = 0.693RC = (750)(0.693) = 520\text{sec}$

C) $V_C = V_o e^{-t/RC} = \frac{Q_o}{C} e^{-1} = \left(\frac{6\times10^{-6}}{50\times10^{-6}}\right)\left(e^{-1}\right) = 0.044\text{V}$

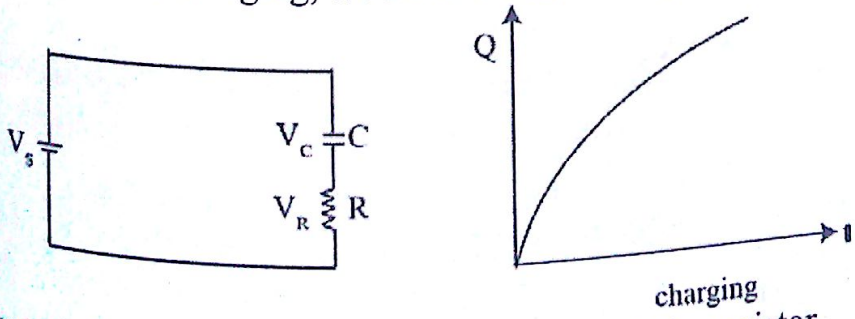
D) $Q = 0.37Q_o = (0.37)\left(6\times10^{-6}\right) = 2\mu\text{C}$

E) $I = I_o e^{-t/RC} = \frac{V}{R} e^{-t/RC} = \frac{Q}{RC} e^{-1}$

$$= \frac{6\times10^{-6}}{\left(15\times10^{6}\right)\left(50\times10^{-6}\right)} e^{-1} = 0.03\mu\text{A}$$

12.10 Charging a Capacitor

When a capacitor is charging, we have the following.



The source voltage is dropped in the capacitor and the resistor.

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$$V_S = V_R + V_C = IR + \frac{Q}{C}$$

As the capacitor charges, I and Q will change while V_S and C will not change, so that in a time interval Δt, we can have

$$\frac{\Delta V_S}{\Delta t} = 0 = \frac{\Delta IR}{\Delta t} + \frac{I}{C}\frac{\Delta Q}{\Delta t}, \text{ dividing by } \Delta t$$

Since $\frac{\Delta Q}{\Delta t} = I$, we get

$$0 = \frac{\Delta IR}{\Delta t} + \frac{I}{C}$$

$$\Rightarrow \frac{\Delta I}{\Delta t} = -\frac{I}{RC}, \text{ if } \Delta t \to 0, \text{ we assume differential equation.}$$

$$\frac{dI}{dt} = -\frac{I}{RC}$$

$$\int_{I_o}^{I} \frac{dI}{dt} = -\int_{O}^{t} \frac{dt}{RC}$$

$$\Rightarrow \boxed{I = I_o e^{-t/RC}},$$

At $t = 0$, $V_C = 0$ and $I = I_O = V_S/R = \frac{V_R}{R}$

$$I = \frac{V_R}{R} = \frac{V_S}{R} e^{-t/RC}$$

$\boxed{V_R = V_S e^{-t/RC}}$, the p.d across the capacitor is then

$$V_C = V_S - V_R = V_S - V_S e^{-t/RC}$$

$$\boxed{V_C = V_S\left(1 - e^{-t/RC}\right)}$$

And, the charge stored on the capacitor at any time is

$$\boxed{Q = CV_S\left(1 - e^{-t/RC}\right)}$$

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Illustrative Examples

99. Uncharged capacitor of capacitance 4μF is connected in series with a 0.6MΩ resistor and a 10V battery calculate
   a) The maximum charge on the capacitor
   b) The maximum current in the circuit
   c) The time taken by the capacitor to be charged 80%

Solution: a) $Q_o = C_v = (4)(10) = 40\mu C$

b) $I_o = \frac{V}{R} = \frac{10}{0.6\times10^6} = 16\mu A$

c) $Q = Q_o[1 - e^{-t/RC}]$

$0.8Q_o = Q_o[1 - e^{-t/RC}]$, solving this,

$t \simeq 0.1$ sec

100. A charged capacitor discharges through a 3MΩ resistor. If the voltage in the capacitor at time t = 0.5RC is 18V. Calculate
   a) The potential difference between the plates at time t = 0
   b) The current through the resistor at i) t = 0, ii) t = RC

Solution: a) $V_C = V_o e^{-t/RC} \Rightarrow 18 = V_o e^{-\frac{0.5RC}{RC}} = 0.6V_o$

$$V_o = \frac{1.8}{0.6} = 30V$$

b) i) $I_o = \frac{V_o}{R} = \frac{30}{3\times10^6} = 0.3\mu A$

ii) $I = I_o e^{-t/RC} = I_o e^{-RC/RC} = \frac{I_o}{e} = 0.11\mu A$

101. Uncharged capacitor of 6μF capacitance is connected to a 0.25MΩ resistor and a 12V battery. Calculate
   a) The p.d cross the capacitor in one time constant
   b) The p.d across the capacitor and the charge on it at t = 5sec
   c) Once the capacitor is fully discharged through a 1MΩ resistor. How long does it take for half of its charge to discharge?

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Solution: a) at $t = RC$, $V_C = 0.63V_S = (0.63)(12) = 7.6V$

b) $V_C = V_S(1 - e^{-t/RC}) = 12[1 - e^{-5/1.5}] = 11.6V$

$Q = CV = (6 \times 10^{-6})(11.6) = 7 \times 10^{-5}C$

c) $Q = Q_o e^{-t/RC} \Rightarrow \frac{Q_o}{2} = Q_o e^{-t/RC} = t = 0.693RC$

$t = 4.2sec$

Additional Examples

Coulomb's Law and Electric field.

102. Two identical metal balls with charges 2Q and –Q are separated by some distance and exert a force F on each other. They are joined by a conducting wire which is then removed. The force between them will now be

A) F B) F/2 C) F/4 D) F/8

Solution: $F = \frac{K(2Q)(Q)}{r^2} \Rightarrow F = \frac{2KQ^2}{r^2}$ .....(*)

after contact, the net charge is $2Q - Q = Q$

and, there will be $\frac{Q}{2}$ on each.

$$F^1 = \frac{K\left(\frac{Q}{2}\right)\left(\frac{Q}{2}\right)}{r^2} = \frac{KQ^2}{4r^2} \quad \text{.....(**)}$$

Talking ratio of (**) to (*), $\frac{F^1}{F} = \frac{1}{8} \Rightarrow F^1 = \frac{F}{8}$ **Answer: D**

103. A cube of side (b) has charge (Q) at each of its vertices. The field at the center of the cube will be

A) $\frac{Q}{b^2}$ B) $\frac{Q}{2b^2}$ C) $32\frac{Q}{b^2}$ D) zero

Explanation: From symmetry, the field cancel out. **Answer: D**

104. Charge Q_1 exerts a force on charge Q_2. If another charge Q_3 is brought near, the force of Q_1 on Q_2 will

A) increase
B) decrease
C) remain the same
D) unknown

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Explanation: Force between two charges does not depend on the presence or absence of another charge. **Answer: C**

105. The ratio of the forces between a 1μC and 5μC charges will be
A) 1:1 B) 1:25 C) 1:5 D) 5:1
Explanation: From Newton's third law, the forces between two point charges are equal. **Answer: A**

106. An electreic field line is leaving a charged sphere. What is the angle between the surface of the sphere and the field line.
A) 0° B) 30° C) 45° D) 90° **Answer: D**

107. Two plates are 1cm apart and the potential difference between them is 10V. What is the electric field between the plates.
A) 10N/C B) 500N/C C) 1000N/C D) 250N/C
Solution: $E = \frac{V}{d} = \frac{10}{0.01} = 1000 N/C$ **Answer: C**

108. A Spherical charged conductor of charge density σ has an electric field of E. If the radius of the sphere is doubled keeping the charge density constant, the electric field will be
A) $E/4$ B) $E/2$ C) E D) 4E
Solution: $E = \sigma/\epsilon_o$, if the charge density σ is constant, then the electric field is also constant. **Answer: C**

109. Charge Q is divided into two parts which are then kept some distance apart. The force between them will be maximum if the two parts are
A) $\frac{Q}{2}$ each C) $\frac{Q}{3}$ and $\frac{2}{3}Q$
B) $\frac{Q}{4}$ and $\frac{3}{4}Q$ D) unknown
Solution: $F \sim q(Q-q)$. The force is maximum when,
$q = Q - q \Rightarrow 2q = Q, q = \frac{Q}{2}$ **Answer: A**

110. When 10^{19} electrons are removed from a neutral metal plate, the electric change on it is?
A) $10^{19}C$ B) 1.6C C) −1.6C D) $10^{-19}C$

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Solution: $Q = ne \Rightarrow Q = (10^{19})(1.6\times10^{-19}) = 1.6C$ **Answer: B**

111. Two parallel oppositely charged plates that are at some separation are dipped into an oil tank. If the oil is pumped out, the electric field will

A) increase
B) decrease
C) remain the same
D) become zero

Explanation: In the absence of dielectric electric field increases.

Answer: A

112. A charge is uniformly distributed over an iron plate and the electric field at some point is found to be 40v/m. If the iron plate is replaced by a copper plate of the same dimention and carrying the same charge, then the electric field at the same point will be?

A) 1v/m B) 10v/m C) 40v/m D) 80v/m

Explanation: The nature of the material is not important.

Answer: C

113. A proton and an electron are placed in a uniform electric field. Which of the following is correct?

A) The electric forces acting on them will be equal
B) The magnitude of the forces will be equal
C) Their accelerations will be equal
D) The magnitude of their acceleration will be equal

Answer: B

114. Proton is nearly 1836 times more massive than the electron. The coulomb force of repusion between two protons separated by a finite distance is F. The electrostatic force between two electrons separated by the same distance is

A) F B) $F/1836$ C) 1837 D) $\frac{1837}{4}F$

Explanation: Electrostatic force does not depend on mass.

Answer: A

115. Which of the following graphs represent the relationship between the force F between two point charges and the distance r between them?

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D) oscillate between the centre and the nearer wall

Explanation: There is no field inside the sphere.

Answer: C

119. Three point charges are placed at the corners of an equilateral triangle. Assuming only electrostatic force is acting

A) The system can never be in equilibrium

B) The system will be in equilibrium if the charges rotate about the centre

C) The system will be in equilibrium if the charges have different magnitude and different sings

D) The system will be in equilibrium if the charges have the same magnitude but different signs

Explanation: No system can be in equilibrium under the influence of electrostatic forces only.

Answer: A

120. The ratio of the forces between two small spheres of charges Q_1 and Q_2 when they are in air and then in a medium of dielectric constant K is,

A) 1:K B) K:1 C) $1:K^2$ D) $K^2:1$

Explanation: in air, $F_1 = \dfrac{Q_1Q_2}{4\pi \in_o r^2}$ ..... (*)

in a medium, $F_2 = \dfrac{Q_1Q_2}{4\pi \in_o kr^2}$ .....(**)

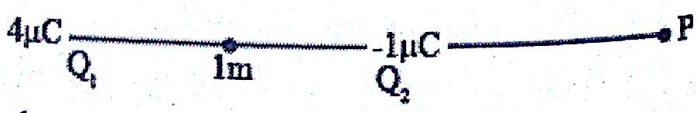
talking ratio $\dfrac{F_1}{F_2} = \dfrac{1}{K}$ **Answer: A**

Note: When a dielectric is introduced between point charges, the electrostatic force between them decreases by a factor of K.

121. Two point charges $4\mu C$ and $-1\mu C$ are 1m a part. At what distance from the $-\mu C$ is the net electric field become zero to the right of it?

A) 1.5m B) 0.5m C) 1m D) 2m

Solution:



at point P, the electric field is assumed to be zero.

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